FORTIGATE FIREWALL LAB

A PRACTICAL HANDS-ON GUIDE

VOLUME 1

Step-by-Step Lab Exercises for FortiGate Firewall

POINTS COVERED

- VDOM
- Packet Flow
- Interfaces
- Routing
- Policy & Objects
- NAT
- HA
- VPN
- Authentication

LAB-BASED LEARNING | REAL-WORLD SCENARIOS | BEGINNER TO ADVANCED

PREPARED BY
ASIF PATEL

TECHNICAL REFERENCE

FortiGate Firewall Lab

Hands-On-Guide Volume-1

Lab ID	: LAB-2026-002
Version	: v1.0
Platform	: FortiGate
Audience	: Network engineers familiar with Firewall technologies
Prerequisites	: Routing and Switching Knowledge, Firewall, GNS3 or EVE-NG Network Emulator
Prepared By	: Asif Patel

DISCLAIMER

This guide is published for educational and reference purposes only. All topologies, IP addresses, host names, and credentials shown here are fictitious and intended solely for use in an isolated lab environment such as GNS3 or EVE-NG.

Contents

1 Prerequisites for FortiGate Firewall Lab in EVE-NG 4

2 FortiGate Hardware Architecture and Packet Flow 5
2.1 FortiGate Hardware Architecture . . . 5
2.2 FortiGate Packet Flow . . . 6
2.3 FortiGate Slow-Path and Fast-Path . . . 8

3 Initial Configuration Branch-1 9
3.1 Branch-1 Topology and IP Addressing Table . . . 9
3.2 Default parameters for a FortiGate firewall . . . 9
3.3 Management Interface Configuration for lab topology . . . 10

4 VDOM (Virtual Domain) 14

5 Zones in FortiGate 17

6 FortiGate Firewall Interface Types 19
6.1 Physical Interface . . . 19
6.2 Aggregate Interface (LAG) . . . 21
6.3 VLAN interface . . . 25
6.4 Loopback interface . . . 26

7 DHCP Server 29

8 Supporting Lab Infrastructure - Cisco IOS 34
8.1 ISP Topology . . . 34
8.2 ISP IP Addressing Table . . . 35

9 Branch-1 WAN Connectivity 39
9.1 BGP . . . 39
9.2 Routing Objects . . . 43

10 Policy and Object 47
10.1 FortiGate Feature Visibility . . . 47
10.2 Firewall Objects . . . 47
10.3 Firewall Policies . . . 48
10.4 FortiGate Firewall Inspection Modes . . . 49
10.5 How Policy Evaluation Works . . . 49
10.6 Firewall Policy Creation . . . 50
10.7 Verification of Firewall Policy . . . 51
10.7.1 Policy Verification . . . 51
10.7.2 Traffic Flow Verification . . . 52
10.7.3 Log Verification . . . 53
10.8 Policy and Object Troubleshooting Scenario . . . 53
10.8.1 Traffic Not Matching Firewall Policy . . . 53
10.8.2 LAN Users Can Ping Internet but Cannot Browse Webpages . . . 55

11 Source NAT 57
11.1 IP Pool . . . 57
11.2 SNAT (Source NAT) . . . 58
11.2.1 Overload (PAT - Port Address Translation) . . . 58
11.2.2 One-to-One IP Pool . . . 61
11.2.3 Fixed Port Range . . . 62

11.2.4 Port Block Allocation (PBA) . . . 64
11.3 Central SNAT . . . 66

12 Initial Configuration Data-Center 69
12.1 Data Center Topology and IP Addressing Table . . . 69
12.2 Data Center WAN Connectivity – Static Route . . . 72

13 DNAT (Destination NAT) 74
13.1 Virtual IP (Static DNAT) . . . 74
13.2 Virtual IP DNAT with Port Forwarding . . . 77
13.3 DNAT Troubleshooting Scenarios . . . 80
13.3.1 DNAT – External Users Cannot Reach DMZ Server . . . 80
13.3.2 DNAT – Asymmetric Routing / Missing Default Gateway . . . 82

14 FortiGate Local User Authentication 84

15 VPN (Virtual Private Network) 87
15.1 Cryptography . . . 87
15.2 IPSec Tunnel . . . 88
15.2.1 IPsec Security Building Blocks . . . 88
15.2.2 Ports and IP Protocols . . . 89
15.2.3 IPsec Tunnel Lifecycle with IKEv1 . . . 89
15.2.4 Black Hole Route . . . 91
15.2.5 Dead Peer Detection . . . 91
15.3 Site-To-Site VPN . . . 92
15.3.1 Site-to-Site VPN using Template / Wizard . . . 93
15.3.2 Site-to-Site VPN using custom method . . . 99
15.4 Remote Access VPN . . . 106
15.4.1 Split and Full Tunnel mode . . . 106
15.4.2 SSL VPN Portals . . . 106
15.4.3 SSL VPN Web Mode (Client-less) . . . 107
15.4.4 SSL VPN Tunnel mode (Client-based) . . . 109
15.4.5 Remote Access IPsec VPN . . . 111
15.5 VPN Troubleshooting Scenario . . . 116
15.5.1 VPN Tunnel Down Phase-1 Fails . . . 116
15.5.2 VPN Tunnel Phase-1 Up, Phase 2 Down . . . 118
15.5.3 VPN Tunnel Up but No Traffic . . . 120

16 HA (High Availability) 123
16.1 FGCP (FortiGate Clustering Protocol) . . . 123
16.2 HA Link . . . 124
16.3 A-P HA (Active-Passive HA) . . . 124
16.4 A-A HA (Active-Active HA) . . . 130
16.4.1 Behavior of set override disable in Active-Active HA . . . 133
16.4.2 Effect of set load-balance-all enable in Active-Active HA . . . 134
16.5 Troubleshooting Scenario . . . 136
16.5.1 HA Cluster Won't Form / Peer Not Detected . . . 136
16.5.2 Constant Failover / Flapping . . . 138
16.5.3 Failover Not Happening When It Should . . . 138
16.5.4 Split-Brain (Both Units Think They're Primary) . . . 139

17 Daily FortiGate Firewall Commands 141

18 Conclusion 143

1 Prerequisites for FortiGate Firewall Lab in EVE-NG

- **System Requirements**
  - Host Machine: Minimum 16 GB RAM, Quad-Core CPU, 200 GB storage. Recommended: 32 GB RAM, 8-Core CPU, SSD storage.
  - Virtualization Support: BIOS/UEFI must have Intel VT-x or AMD-V enabled.
  - Installation: Deploy EVE-NG Community/Pro Edition on bare-metal or VM.
  - Networking: Configure bridged networking for external connectivity.
  - Access: Verify management access via HTTPS and SSH.
- **FortiGate Image Preparation**
  - Image Source: Obtain FortiGate VM (KVM/Qcow2) from Fortinet support portal.
  - Conversion: Rename image as per EVE-NG standards (virtioa.qcow2).
  - Placement: Store under /opt/unetlab/addons/qemu/fortinet/.
- **FortiGate 15-Day Trial vs Permanent Evaluation License**
  - **VDOMs:** – Both 15-Day Trial (Pre-7.2.0) and Permanent Trial (7.2.1+) : Up to 2 VDOMs (root + traffic VDOM)
  - **Encryption:** – Both 15-Day Trial (Pre-7.2.0) and Permanent Trial (7.2.1+) : HTTPS admin disabled, Low encryption only (DES/weak ciphers)
  - **Security Features:** – Both 15-Day Trial (Pre-7.2.0) and Permanent Trial (7.2.1+) : Limited built-in IPS/AV signatures; no FortiGuard services (Web Filter, Antispam, Security Rating)
  - **Support:** – Both 15-Day Trial (Pre-7.2.0) and Permanent Trial (7.2.1+) : No technical support; firmware upgrades locked until permanent license applied
  - **Resource Limitations:**
    - – 15-Day Trial (Pre-7.2.0) : 1 CPU, 2 GB RAM
    - – Permanent Trial (7.2.1+) : 1 CPU, 2 GB RAM; max 3 interfaces, 3 firewall policies, 3 routes

> ☞ **Note**
> This lab demonstrated using FortiGate Firewall version 7.2.0, build1157.

- **Networking Prerequisites**
  - Management Network: Define a dedicated management bridge.
  - Lab Topology: Plan VLANs, WAN uplinks, and internal LAN segments.
  - Client Access: Web browser (Chrome/Edge) for GUI; SSH client (PuTTY, SecureCRT) for CLI.

2 FortiGate Hardware Architecture and Packet Flow

2.1 FortiGate Hardware Architecture

FortiGate's architecture is built around hardware acceleration via proprietary ASICs (Application-Specific Integrated Circuits) working in tandem with its custom operating system, FortiOS.

> ☞ **Note**
> FortiGate-VM running in EVE-NG does not utilize NP7, CP9 or SP5 hardware acceleration. All traffic is processed in software.

Instead of relying solely on general-purpose CPUs to perform compute-heavy networking, cryptography, and pattern matching, FortiGate separates tasks across dedicated hardware processing engines.

- **Network Processor (NP7)**: The NP7 accelerates Layer 3 and Layer 4 traffic at wire speed. It offloads firewall functions such as stateful inspection, NAT translations, routing lookups, and IPsec encapsulation/decapsulation. By handling these tasks in hardware, NP7 ensures ultra-low latency and deterministic performance, ideal for high-bandwidth WAN, SD-WAN, and data center deployments.
- **Content Processor (CP9)**: The CP9 is Fortinet's cryptographic and deep inspection engine. It specializes in SSL/TLS decryption and re-encryption, IPsec VPN encryption/decryption. With most modern traffic encrypted, CP9 prevents performance collapse when enabling full UTM features, allowing FortiGate firewalls to inspect encrypted traffic at scale without exhausting CPU resources.
- **Security Processor (SP5)**: The SP5 consolidates both network and content security functions into a single chip, combining NP and CP capabilities. It provides application identification, Layer 7 firewalling, and overlay security features such as DLP and web filtering. This integration makes SP5 highly efficient for edge deployments, offering high compute density and lower power consumption while maintaining strong security performance.
- **CPU Role**: While Forti ASIC/SP5 processors handle the heavy lifting, the general-purpose CPU remains responsible for control plane tasks, session setup, and traffic that cannot be offloaded. This division of labor ensures FortiGate devices can scale securely, balancing hardware acceleration with flexible software-based processing.

Below table summarizes different FortiGate firewall processor architectures.

Table 1: FortiGate Hardware Processor Architecture

Processor	Layer Focus	Key Functions	Deployment Benefit
NP7	L3/L4	Firewall, NAT, Routing, IPsec Encapsulation	Wire-speed forwarding, ultra-low latency
CP9	Crypto/L7	SSL/TLS Decryption, VPN, IPS, AV	reduce the performance impact of security inspection by offloading supported processing from the general-purpose CPU
SP5	Consolidated	App ID, Layer 7 Firewall, Combined NP + CP	Efficient edge security, high compute density

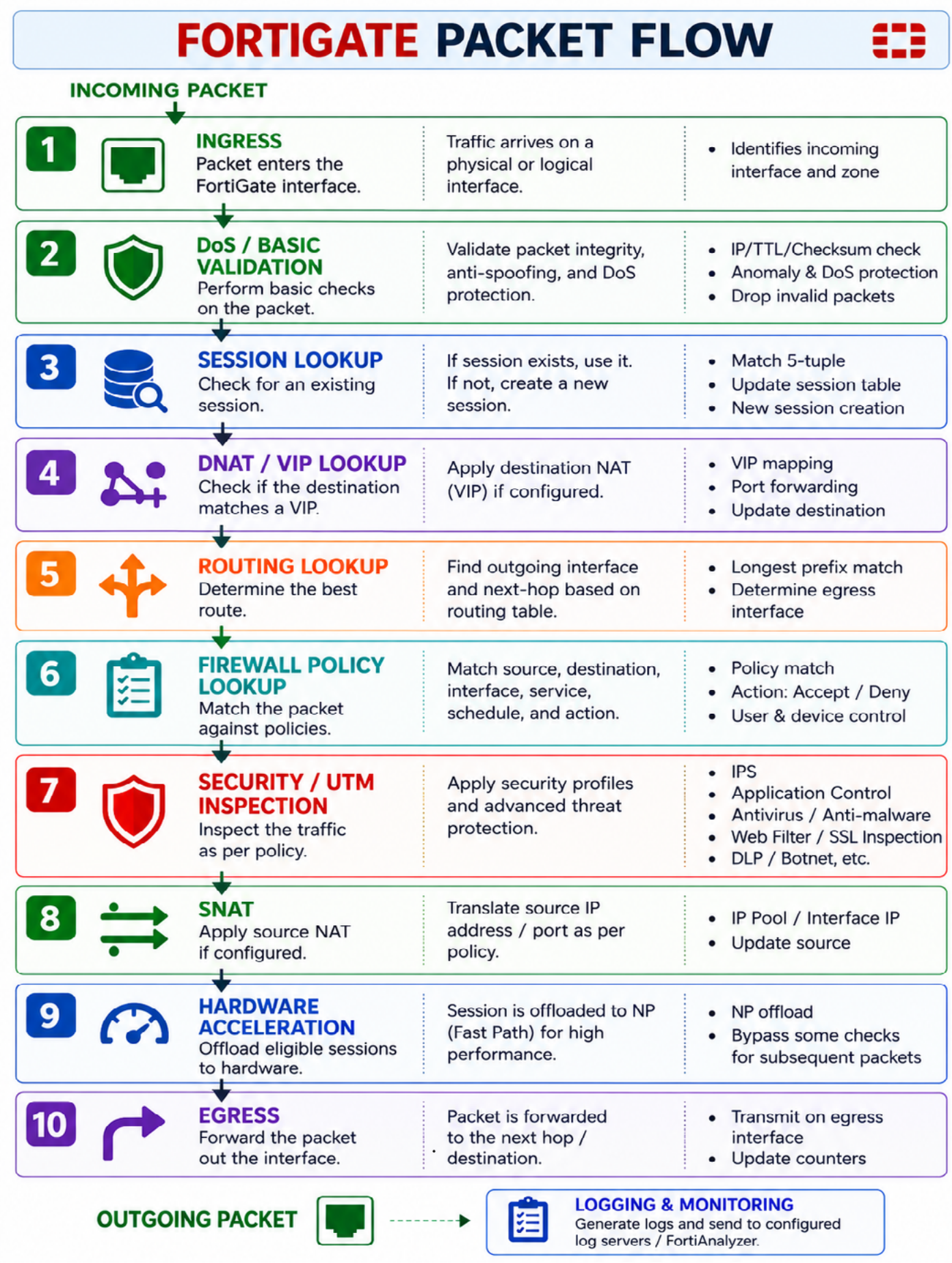
2.2 FortiGate Packet Flow

Traffic through a FortiGate firewall follows a multi-stage pipeline designed to balance performance and security. Below is the step-by-step packet flow.

- **Ingress**: Packets arrive on a physical or logical interface. FortiGate identifies the incoming interface and zone, starting the inspection process.
- **DoS / Basic Validation**: The firewall performs integrity checks – IP, TTL, and checksum – and applies anti-spoofing and DoS protection. Invalid or malformed packets are dropped immediately.
- **Session Lookup**: FortiGate checks if the packet belongs to an existing session (using the 5-tuple: source IP, destination IP, source port, destination port, protocol). If found, it uses the existing session; otherwise, it creates a new one.
- **DNAT / VIP Lookup**: If the destination matches a Virtual IP (VIP), FortiGate applies destination NAT. This enables port forwarding and updates the destination address.
- **Routing Lookup**: The firewall determines the best route using the routing table, selecting the outgoing interface and next hop based on longest prefix match.
- **Firewall Policy Lookup**: FortiGate matches the packet against configured policies – source, destination, service, schedule, and action (allow or deny). It also applies user and device identity controls.
- **Security / UTM Inspection**: Traffic is inspected according to policy. Security profiles such as IPS, antivirus, application control, web filtering, SSL inspection, DLP, and botnet protection are applied here.
- **SNAT**: If source NAT is configured, FortiGate translates the source IP or port using an IP pool or interface IP before forwarding.
- **Hardware Acceleration**: Eligible sessions are offloaded to FortiASIC/SPU hardware (NP7, CP9, SP5) for fast-path processing. Subsequent packets bypass some software checks for higher performance.
- **Egress**: The packet is transmitted through the egress interface toward its destination. Counters are updated to track throughput and session statistics.
- **Logging & Monitoring**: Finally, FortiGate generates logs and sends them to FortiAnalyzer or configured log servers for analysis and reporting.

> ☞ **Note**
> Conceptual packet flow: Actual processing order can vary depending on session state, NAT type, inspection mode, local-in traffic, IPsec, hardware offload, and FortiOS implementation.

Figure on next page shows all these steps for packet life cycle inside FortiGate firewall.


FORTIGATE PACKET FLOW
INCOMING PACKET
1
INGRESS
Packet enters the FortiGate interface.
Traffic arrives on a physical or logical interface.
Identifies incoming interface and zone
2
DoS / BASIC VALIDATION
Perform basic checks on the packet.
Validate packet integrity, anti-spoofing, and DoS protection.
IP/TTL/Checksum check
Anomaly & DoS protection
Drop invalid packets
3
SESSION LOOKUP
Check for an existing session.
If session exists, use it. If not, create a new session.
Match 5-tuple
Update session table
New session creation
4
DNAT / VIP LOOKUP
Check if the destination matches a VIP.
Apply destination NAT (VIP) if configured.
VIP mapping
Port forwarding
Update destination
5
ROUTING LOOKUP
Determine the best route.
Find outgoing interface and next-hop based on routing table.
Longest prefix match
Determine egress interface
6
FIREWALL POLICY LOOKUP
Match the packet against policies.
Match source, destination, interface, service, schedule, and action.
Policy match
Action: Accept / Deny
User & device control
7
SECURITY / UTM INSPECTION
Inspect the traffic as per policy.
Apply security profiles and advanced threat protection.
IPS
Application Control
Antivirus / Anti-malware
Web Filter / SSL Inspection
DLP / Botnet, etc.
8
SNAT
Apply source NAT if configured.
Translate source IP address / port as per policy.
IP Pool / Interface IP
Update source
9
HARDWARE ACCELERATION
Offload eligible sessions to hardware.
Session is offloaded to NP (Fast Path) for high performance.
NP offload
Bypass some checks for subsequent packets
10
EGRESS
Forward the packet out the interface.
Packet is forwarded to the next hop / destination.
Transmit on egress interface
Update counters
OUTGOING PACKET
LOGGING & MONITORING
Generate logs and send to configured log servers / FortiAnalyzer.

2.3 FortiGate Slow-Path and Fast-Path

FortiGate can process traffic through different paths depending on the session state, configured security features, and whether the traffic is eligible for hardware acceleration.

- **Slow-Path / Main CPU Processing:**
  - New sessions initially require FortiOS to perform session setup and determine how the traffic should be handled.
  - Processing can include routing lookup, firewall policy lookup, NAT decisions, session creation, and other required security checks.
  - Traffic that is not eligible for hardware offloading continues to be processed by the FortiGate's main processing resources.
  - Certain features, traffic types, or proxy-based security functions can prevent a session from being offloaded to the Network Processor.
- **Fast-Path / Hardware-Accelerated Processing:**
  - After a session is established, eligible sessions can be offloaded to a FortiASIC Network Processor (NP).
  - The Network Processor uses the session information created by FortiOS to process subsequent packets efficiently.
  - Fast-path processing reduces CPU utilization and improves throughput and latency.
  - TCP, UDP, ICMP, and other supported traffic can be hardware accelerated when the session meets the required offload conditions.
  - Some flow-based security profiles can work with hardware acceleration, whereas several proxy-based inspection features can prevent session offloading.

> ☞ **Note**
> Fast-path processing does not mean that security controls are bypassed. The session has already been evaluated by FortiOS, and supported security processing may continue while the traffic is hardware accelerated. Sessions that are not eligible for hardware offloading remain processed by the main CPU.

3 Initial Configuration Branch-1

In this section, we will cover the following objectives.

Objective

- Topology and IP address for Branch-1 firewall.
- Access a FortiGate firewall using factory default settings.
- Configure hostname and management interfaces, IP addressing.
- Verify management connectivity using GUI and CLI tools.

3.1 Branch-1 Topology and IP Addressing Table

We will use the following topology in this lab demonstration.

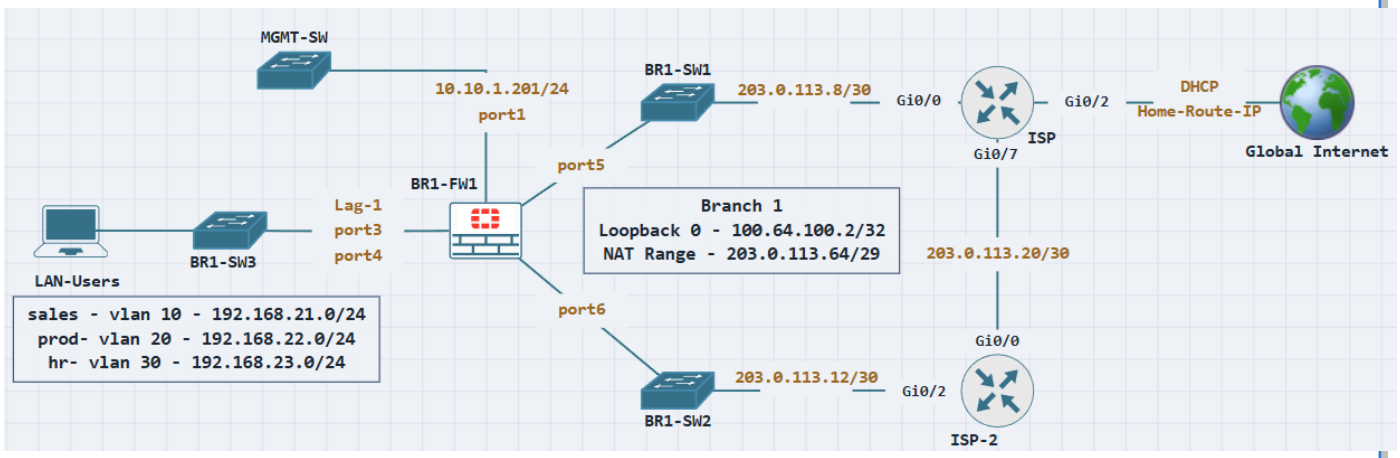


Figure: Branch-1 Topology

Device	Interface	IP Address / mask	Description
BR1-FW1	port1	10.10.1.201/24	Management Interface
BR1-FW1	port5	203.0.113.10/30	WAN link to ISP-Gi0/0
BR1-FW1	port6	203.0.113.14/30	WAN link to ISP-2-Gi0/2
BR1-FW1	port3-4 (Lag-1)	–	Po1 to BR1-SW3
	VLAN 10	192.168.21.1/24	Sales LAN
	VLAN 20	192.168.22.1/24	Prod LAN
	VLAN 30	192.168.23.1/24	HR LAN
BR1-FW1	Loopback 0	100.64.100.2/32	Loopback Interface
BR1-FW1	NAT Range	203.0.113.64/29	Public NAT Pool

Table 2: Branch-1 IP Addressing Summary

3.2 Default parameters for a FortiGate firewall

The factory default parameters for a FortiGate firewall are:

- **Management IP Address:** Typical default management IP on many model is 192.168.1.99 (Subnet mask: 255.255.255.0) on port1, but this may vary by model and FortiOS version.
- **Default Interface:** port1 (or the dedicated MGMT interface on enterprise/high-end models)

- **Default Username:** admin
- **Default Password:** Blank (no password required on first login)
- **Operation Mode:** NAT
- **Protocols Allowed on Management Interface:** HTTPS, HTTP, SSH, and Ping (ICMP)
- **DHCP Server:** Enabled by default on entry-level models (assigns IP addresses within the $192.168.1.X$ range to connected hosts)

Following is the details for how to Access the FortiGate firewall with factory default settings.

- **Connect your PC:** Plug an Ethernet cable from your computer into port1 (or MGMT). Set your computer's IP manually to something like $192.168.1.110$ with a subnet mask of $255.255.255.0$ if it doesn't pull an IP via DHCP.
- **Access the GUI:** Open a web browser and go to [https://192.168.1.99]. (Note: Accept any self-signed SSL certificate warnings).
- **Log In:** Enter admin as the username and leave the password field completely blank. You will immediately be prompted to create a new secure password.

3.3 Management Interface Configuration for lab topology

As per topology above, port1 is used as the management interface. In this EVE-NG lab demonstration, port1 receives an IP address through DHCP by default. For the lab demo, we can log in to the GUI with the assigned DHCP IP address, but we need to change it to a static IP address so the management address is predictable across reboots (a DHCP lease can also change and may install a default gateway that interferes with the lab routing).

> ☞ **Note**
> Management IP Addresses are taken from lab environment. Adjust them according to your local IP addressing scheme.

Site-Name	Hostname	MGMT IP
Branch 1	BR1-FW1	10.10.1.201/24

Table 3: Hostname and MGMT Interface IP Plan

Before starting the management interface configuration, please note the following points regarding allowed accesses.

> ☞ **Note**
> HTTP management access should only be enabled in the lab environment. Production deployment should use HTTPS and SSH only.

In this lab guide, both GUI and CLI methods are demonstrated for configuring hostname and management interface IP address on BR1-FW1.

Configuration Field Breakdown

- **Host name:** Uniquely identifies the firewall device on the network, in logs, and within management systems.

- **Idle timeout:** Automatically terminates inactive GUI/CLI administrative sessions after 60 minutes to prevent unauthorized access.
- **Alias:** Provides an easily recognizable descriptive name for physical interface port1.
- **VRF ID:** Assigns the interface to a Virtual Routing and Forwarding instance, isolating management traffic into a dedicated routing domain.
- **Virtual Domain (VDOM):** assigns the interface to the VDOM context, controlling which VDOM owns and manages it.
- **IP/Netmask:** Defines the static IPv4 management address and the subnet mask for the interface.
- **Administrative Access:**
  - **HTTPS:** Grants secure, encrypted web GUI access.
  - **HTTP:** Grants unencrypted web GUI access (only for the lab Demo).
  - **PING:** Responds to ICMP echo requests for connectivity diagnostics.
  - **SSH:** Enables encrypted Command Line Interface (CLI) administration.
- **LLDP Settings:** Receive / Transmit LLDP. Set to Use VDOM Setting, meaning LLDP behavior follows the global or VDOM-specific configuration.

Why keep Management and Data Networks Separate? (Management VRF 31)

- **Security:** Prevents exposure of device management interfaces to user or Internet traffic.
- **Traffic Segregation:** Keeps control-plane traffic (SSH, HTTPS, SNMP) apart from data-plane flows.
- **Compliance:** Aligns with industry standards (PCI-DSS, ISO 27001, NIST).
- **Operational Stability:** Ensures administrators retain access even if LAN/WAN policies fail.

The following listing shows the CLI configuration for the management interface.

Hostname & MGMT Interface

```
config system global
  set admintimeout 60
  set hostname BR1-FW1
 end
!
 config system interface
  edit port1
   set vdom root
   set mode static
   set vrf 31
   set ip 10.10.1.201 255.255.255.0
   set allowaccess ping https ssh http
   set description management-interface
   set alias management-interface
  end
 end
```

The screenshot below shows the hostname, NTP, and idle-timeout configuration for the FortiGate firewall.

GUI PATH: **System > Settings**

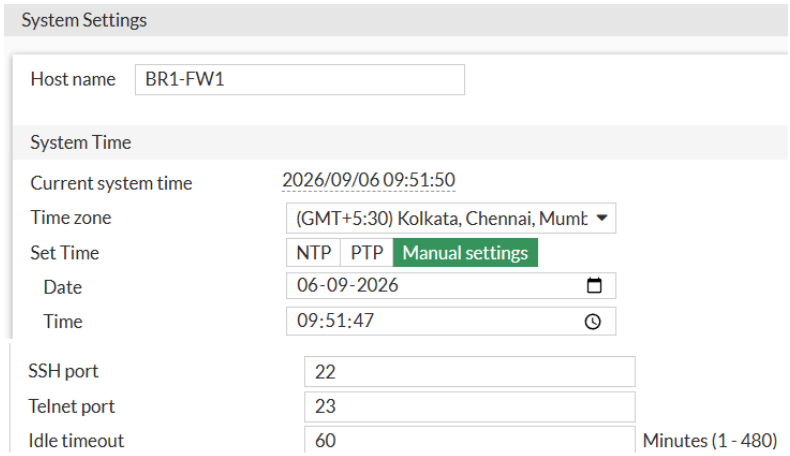


Figure: Hostname and Idle timeout value Configure

The screenshot below shows the management interface configuration for Fortigate firewall.

GUI PATH: **Interfaces --> port1**

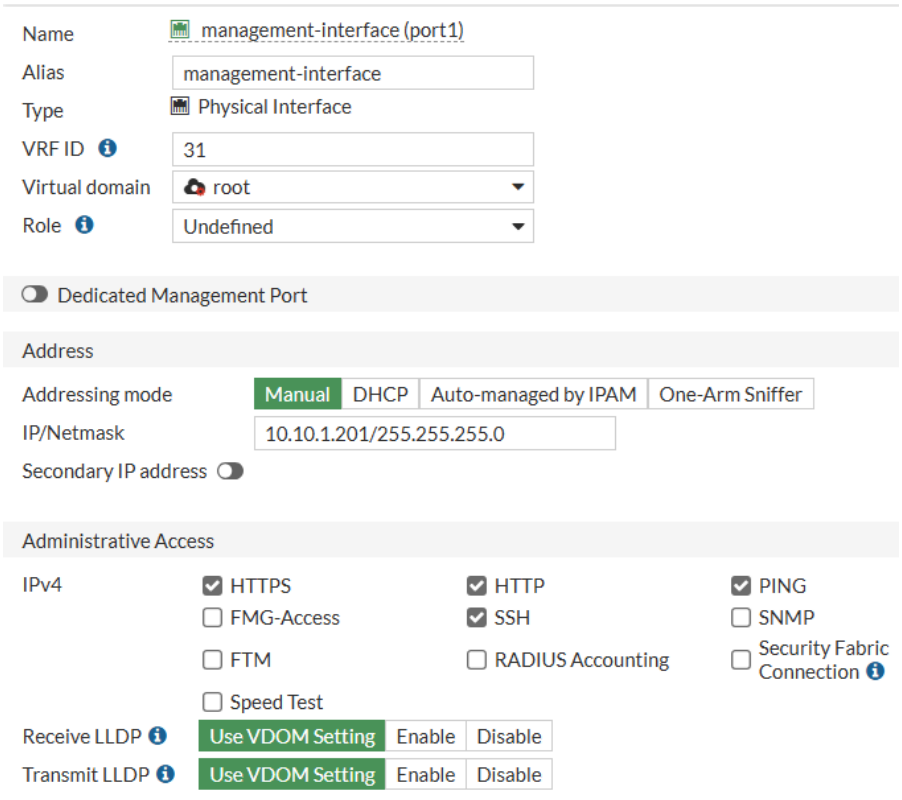


Figure: Management Interface Configuration

We can verify management interface details as per following.

MGMT Interface Verification

```
BR1-FW1 # get system interface physical port1
        ==[port1]
```

```
mode: static
ip: 10.10.1.201 255.255.255.0
ipv6: ::/0
status: up
speed: 10000Mbps (Duplex: full)
FEC: none
FEC_cap: none
```

The following image illustrates the initial FortiGate deployment and GUI access – a step-by-step workflow covering console connection, power setup, management IP configuration, and GUI login.

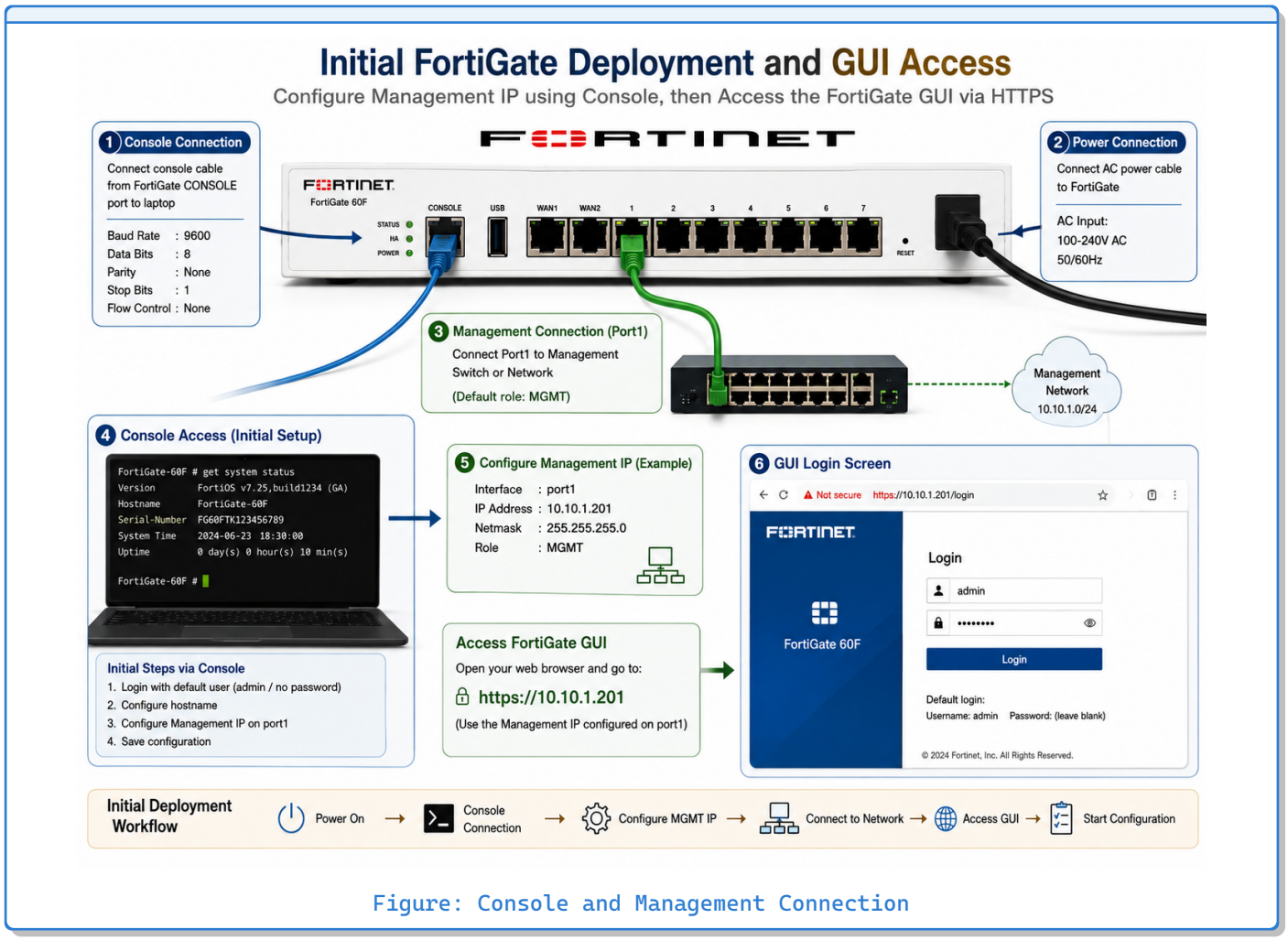


Figure: Console and Management Connection

4 VDOM (Virtual Domain)

In this section, we will cover the following objectives.

> **Objective**
>
> - Understand Virtual Domain concepts.
> - Differentiate between Single VDOM and Multi VDOM.
> - Create and manage VDOMs.

In FortiGate, a VDOM (Virtual Domain) is a way to split one physical firewall into multiple virtual firewalls. Each VDOM operates independently, with its own configuration, routing, policies, and administrators – but all share the same hardware resources.

FortiGate supports two VDOM modes: Multi-VDOM (multiple independent virtual firewalls on one device) and Split-Task VDOM (one management VDOM + one traffic VDOM). Multi-VDOM is ideal for multi-tenant or complex enterprise segmentation, while Split-Task is simpler, used when you only need a dedicated management plane separate from traffic handling.

In FortiGate, the root VDOM is the default Virtual Domain that exists when you first configure the firewall. It acts as the primary VDOM unless you enable multi-VDOM mode. The root VDOM cannot be deleted because it is the foundation of the system. Below are some key characteristics of VDOMs:

- **Segmentation** – Divide a single FortiGate into multiple logical firewalls.
- **Isolation** – Each VDOM has its own interfaces, routing table, policies, and objects.
- **Multi-Tenancy** – Useful for service providers or enterprises hosting different departments/customers on the same hardware.
- **Independent Administration** – Different admins can manage different VDOMs without interfering with each other.
- **Resource Sharing** – CPU and memory are shared, but traffic and policies remain isolated.

By default, most FortiGate models support up to 10 VDOMs, but higher-end appliances and licensed upgrades can expand this limit to 25, 50, 100, 250, or even 500 VDOMs depending on the model and purchased license. On lower-end FortiGate models, VDOMs can be enabled only through the CLI, while on higher-end devices they can also be enabled through the GUI. Global Setting like Interface, HA Configuration are configured outside VDOM.

We can enable multi-vdom via CLI with the following command:

Enable multi-vdom mode

```
config system global
 set vdom-mode multi-vdom
end
```

> **Note**
>
> FortiGate 15-day trial only allowed one traffic VDOM. So before creating a new VDOM we will change root vdom mode to admin only as per below GUI Screenshot.

GUI PATH: **System --> VDOM --> Edit --> root**

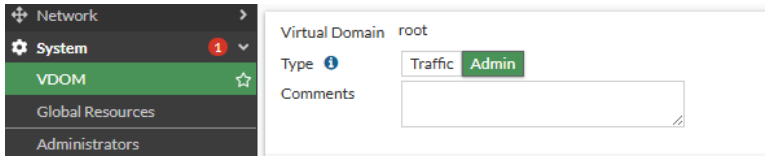


Figure: Make root vdom admin only

We can create new VDOM from CLI using the configuration below.

Create VDOM

```
conf vdom
edit FG-traffic
end
```

Below are the roles of VDOM types on a FortiGate firewall:

- **management (root)** – Handles system traffic such as DNS, NTP, FortiGuard updates, and logging.
- **traffic** (FG-traffic) – Handles user traffic, firewall policies, and routing.

The following figure shows Created VDOM features.

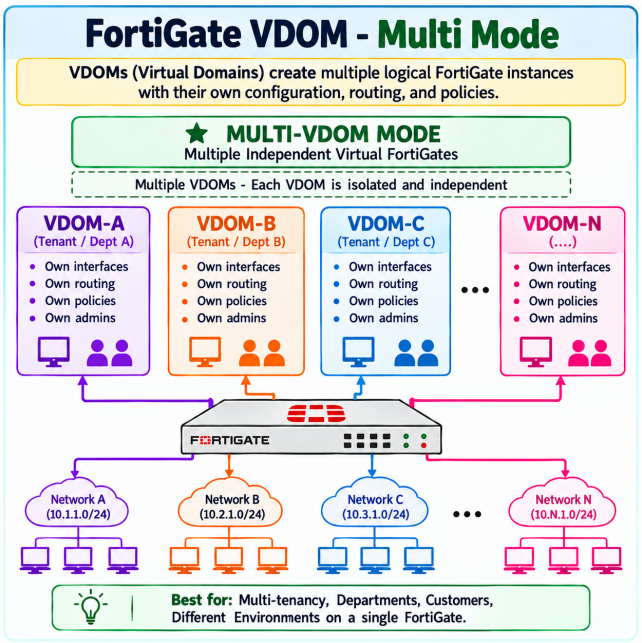


Figure: VDOM-Multi Mode

Below GUI screenshot showing created multi-vdoms.

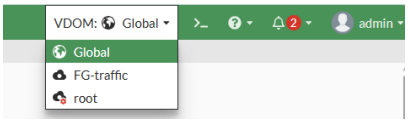


Figure: BR1-FW1 multi-vdom

On a FortiGate firewall, we can check the list of VDOMs directly from the CLI also as per below.

➟ **get system status**

The command **get system status** is used on a FortiGate firewall to quickly display device-wide information.

- Shows system information, including the current VDOM mode (single, multi-VDOM, or split-VDOM). In multi-VDOM mode, it also displays the number of VDOMs allowed and currently configured.
- **Firmware details:** Version, build number, and release date.
- **Serial number & hardware model:** Useful for inventory and licensing.
- **System time & uptime:** Confirms clock synchronization and stability.
- **License information:** Shows FortiCare registration, support contract, and feature entitlements.
- **Operational limits:** Maximum sessions, memory usage, and other capacity indicators.

BR1-FW1 system status

```
BR1-FW1 # get system status
Version: FortiGate-VM64-KVM v7.2.0,build1157,220331 (GA.F)
!
Serial-Number: FGVMEVAXXXXXX27
License Status: Valid
Evaluation License Expires: Sun Sep 20 21:49:45 2026
VM Resources: 1 CPU/1 allowed, 2007 MB RAM/2048 MB allowed
!
Hostname: BR1-FW1
Operation Mode: NAT
Max number of virtual domains: 2
Virtual domains status: 2 in NAT mode, 0 in TP mode
```

We can verify the created VDOMs from the vdom configuration context or with the show command, as follows.

VDOM context

```
BR1-FW1 # conf vdom
BR1-FW1 (vdom) # edit
<vdom>    Virtual Domain Name
FG-traffic
root
```

VDOM Verification – show system vdom-property

```
BR1-FW1 (global) # show system vdom-property
config system vdom-property
    edit "root"
    edit "FG-traffic"
end
```

5 Zones in FortiGate

In this section, we will cover the following objectives.

Objective

- Understand security zones and their benefits.
- Create LAN and WAN zones.

In FortiGate firewalls, a Zone is a logical grouping of multiple interfaces into a single security boundary, allowing you to apply firewall policies, security profiles, and routing rules consistently across all interfaces in that zone.

- Grouping multiple LAN and WAN interfaces into a single **LAN Zone** and **WAN Zone**.
- Simplifying firewall rules when multiple interfaces share identical security requirements.

We can configure the zone as shown below with GUI and CLI. At branch sites, we can configure two zones: LAN and WAN. We will add member interface later in the Interfaces section.

Configuration Field Breakdown

- **Name:** Defines the logical name of the zone (e.g. WAN). Used to group multiple interfaces under a single security zone for simplified policy management.
- **Block intra-zone traffic:** When enabled, prevents traffic between interfaces within the same zone. Useful for isolating subnets between members of the same zone.
- **Interface members:** Lists the physical or virtual interfaces assigned to this zone. These interfaces inherit the zone's security policies and behave as a unified group.
- **Comments:** Descriptive note for administrative clarity (e.g. wan-towards-isp). Helps identify the zone's purpose or connectivity direction in complex deployments.

We will configure the zones on the firewall using both GUI and CLI methods, as shown below. Please select the FG-traffic VDOM from right side panel in the GUI and in CLI configuration also.

GUI PATH: **Network -> Interfaces -> Create New -> Zone.**

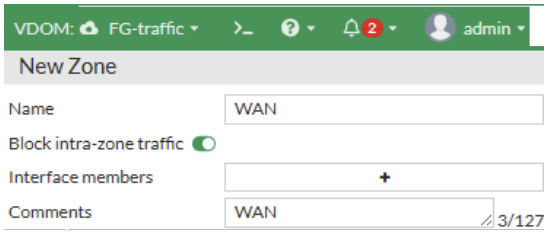


Figure: BR1-FW1 Zone Configuration

✓ Important Considerations

- All interfaces in a zone must share the same security requirements; otherwise, grouping them may cause unintended access.
- Zones cannot overlap; an interface can belong to only one zone.
- Routing behavior: Zones simplify firewall policies but do not replace routing

configuration—we still need to define static or dynamic routes.

BR1-FW1 Zone Configuration

```
config vdom
 edit FG-traffic
  config system zone
   edit WAN
    set description WAN
    set intrazone deny
  next
   edit LAN
    set description LAN
    set intrazone deny
  end
 end
```

We will verify the created zones using the following CLI command.

Zone Verification – show system zone

```
BR1-FW1 (FG-traffic) # show system zone
config system zone
    edit "WAN"
        set description "WAN"
        set intrazone deny
    next
    edit "LAN"
        set description "LAN"
        set intrazone deny
    next
end
```

6 FortiGate Firewall Interface Types

In this section, we will cover the following objectives.

> **Objective**
>
> • Configure and verification for Physical, Aggregate, VLAN, Loopback Interface.

FortiGate firewalls support multiple interface types, ranging from physical ports to advanced logical constructs like VLANs, software switches, and VXLAN. Choosing the right type depends on performance needs, flexibility, and topology design.

- **Physical Interface**: Hardware port (LAN/WAN/DMZ), fastest with ASIC acceleration.
- **VLAN Interface**: Logical sub-interface with VLAN ID, used for segmentation.
- **Aggregate Interface (LAG)**: Bundled ports for redundancy and throughput.
- **Software Switch**: CPU-based grouping of interfaces, flexible but less efficient.
- **Hardware Switch**: Integrated Switch Fabric, hardware-accelerated LAN switching.
- **Loopback Interface**: Virtual interface for testing/management, CPU-based.
- **Virtual Wire Pair**: Transparent inline inspection between paired interfaces.

6.1 Physical Interface

For the WAN Interface we will use Physical port on all Firewall. We will configure the WAN Interface of Firewall using the configuration below. For the Demonstration we shown GUI Configuration screenshot for WAN Interface Port5 of BR1-FW1. Below table shows Firewall WAN Interface Configuration Summary.

Site-Name	Device-ID	Interface	Description	IP Address
Branch-1	BR1-FW1	port5	wan-to-isp-Gi0/0	203.0.113.10/30
		port6	wan-to-isp-2-Gi0/2	203.0.113.14/30

Table 4: BR1-FW1 WAN Interface Configuration Summary

Configuration Field Breakdown

- **Name/Alias:** Identifies the interface or zone (e.g. port5, Lag-1, vlan10, WAN); used for internal reference and CLI commands.
- **Type:** Defines interface type – *Physical*, *802.3ad Aggregate*, or *VLAN*.
- **Role:** Assigns functional category (LAN, WAN); determines policy grouping.
- **Interface Members:** Lists physical ports or sub-interfaces participating in a zone or aggregate (e.g. port3, port4).
- **VLAN Protocol & ID:** Specifies tagging standard (802.1Q or 802.1AD) and VLAN identifier (e.g. 10) for logical segmentation.
- **VRF ID:** Defines routing instance; 0 indicates default routing table.
- **Addressing Mode:** Determines IP assignment – Manual (static), DHCP (dynamic), or IPAM-managed.

- **IP/Netmask:** Sets network identity (e.g. 203.0.113.10/30, 192.168.31.1/24); used for routing and connectivity.
- **Administrative Access:** Controls management protocols (HTTPS, SSH, SNMP, PING, etc.); enabling PING allows reachability tests.
- **LLDP Settings:** Manages link-layer discovery; can inherit VDOM settings or be manually enabled/disabled.
- **Comments:** Notes describing interface or zone purpose (e.g. wan-towards-isp); useful for documentation.

GUI PATH: **Interfaces --> port5**

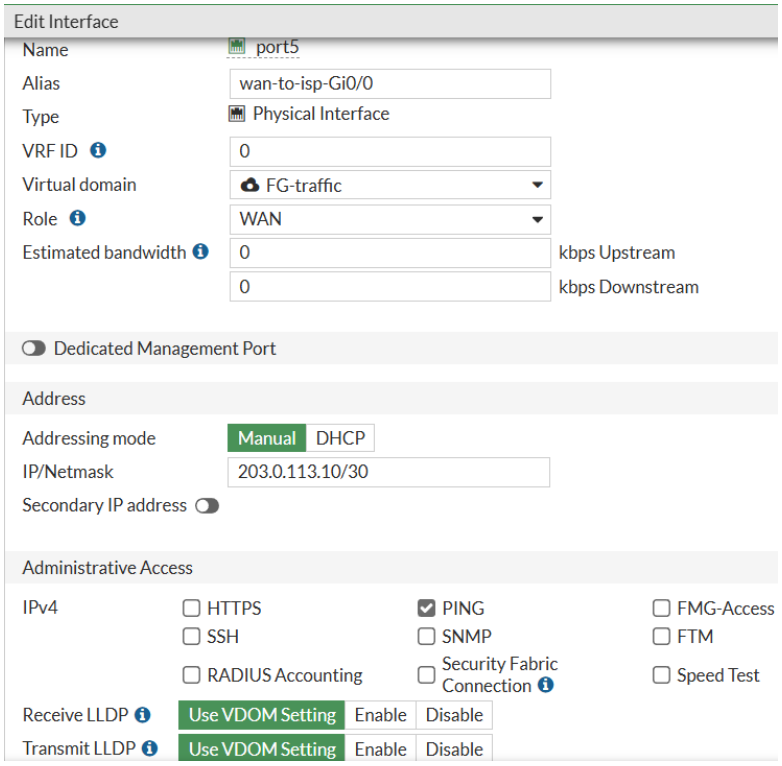


Figure: BR1-FW1 Configure Physical Interface Port5

Now add the WAN interface Port5 as a member of the WAN zone.

GUI PATH: **Network -> Interfaces -> Zone -> Edit**

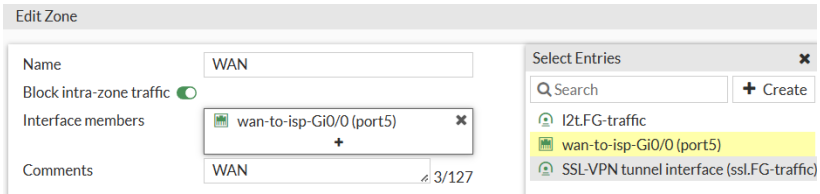


Figure: BR1-FW1 add member in Zone

Now we will configure BR1-FW1 WAN Interface Port5/Port6 using the CLI method as per below.

BR1-FW1 WAN Interface Port5/Port6 and Add Member to Zone

```
config vdom
 edit FG-traffic
  config system interface
  edit port5
   set vdom FG-traffic
   set ip 203.0.113.10 255.255.255.252
   set allowaccess ping
   set description wan-to-isp-Gi0/0
   set alias wan-to-isp-Gi0/0
   set role wan
  next
  edit port6
   set vdom FG-traffic
   set ip 203.0.113.14 255.255.255.252
   set allowaccess ping
   set description wan-to-isp-2-Gi0/2
   set alias wan-to-isp-2-Gi0/2
   set role wan
  end
  !
  config system zone
   edit WAN
    set interface port5 port6
  end
```

We can verify the firewall's connectivity with ISP Gateway using ping command as per follows.

Ping to ISP from BR1-FW1

```
BR1-FW1 (FG-traffic) # execute ping 203.0.113.9
PING 203.0.113.9 (203.0.113.9): 56 data bytes
64 bytes from 203.0.113.9: icmp_seq=0 ttl=255 time=3.6 ms
.
64 bytes from 203.0.113.9: icmp_seq=4 ttl=255 time=1.0 ms
--- 203.0.113.9 ping statistics ---
5 packets transmitted, 5 packets received, 0\% packet loss
round-trip min/avg/max = 0.9/1.5/3.6 ms
!
BR1-FW1 (FG-traffic) # execute ping 203.0.113.13
PING 203.0.113.13 (203.0.113.13): 56 data bytes
64 bytes from 203.0.113.13: icmp_seq=0 ttl=255 time=4.5 ms
.
64 bytes from 203.0.113.13: icmp_seq=4 ttl=255 time=1.3 ms
--- 203.0.113.13 ping statistics ---
5 packets transmitted, 5 packets received, 0\% packet loss
round-trip min/avg/max = 1.2/2.0/4.5 ms
```

6.2 Aggregate Interface (LAG)

FortiGate Aggregate Interface (LAG) is a logical interface created by bundling two or more physical interfaces. It uses IEEE 802.3ad LACP (Link Aggregation Control Protocol). Some benefits of an aggregate interface are:

- Increased throughput (bandwidth scaling).
- Load balancing across member links.

- Redundancy and failover protection.
- Simplified management (single logical interface).

In this lab, the aggregate interface is configured between BR1-FW1 and BR1-SW3 as per the following table.

Table 5: BR1-FW1 Aggregate Interface Member Mapping

Device-ID	Aggregate:Member-Interface	Switch-ID	Port-channel:Member-interface
BR1-FW1	Lag-1:port3,port4	BR1-SW3	Po1:Gi0/0,Gi0/1

The GUI screenshot below for BR1-FW1 shows the fields which we can select for the Aggregate Interface configuration. Before start Aggregate Interface configuration we need to change VDOM for physical port from root to FG-traffic VDOM.

GUI PATH: **Interfaces --> port3/port4**

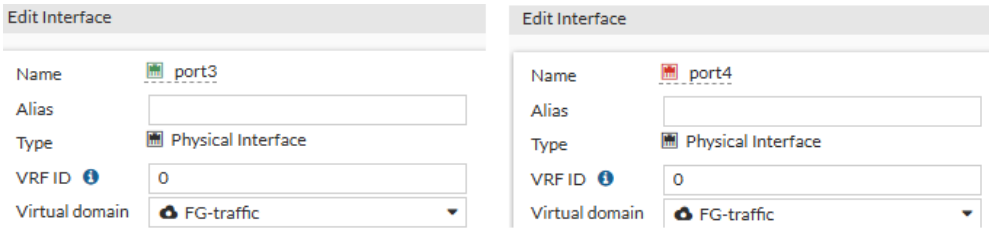


Figure: BR1-FW1 Configure physical Interface

GUI PATH: **Interfaces --> Create New --> Interface**

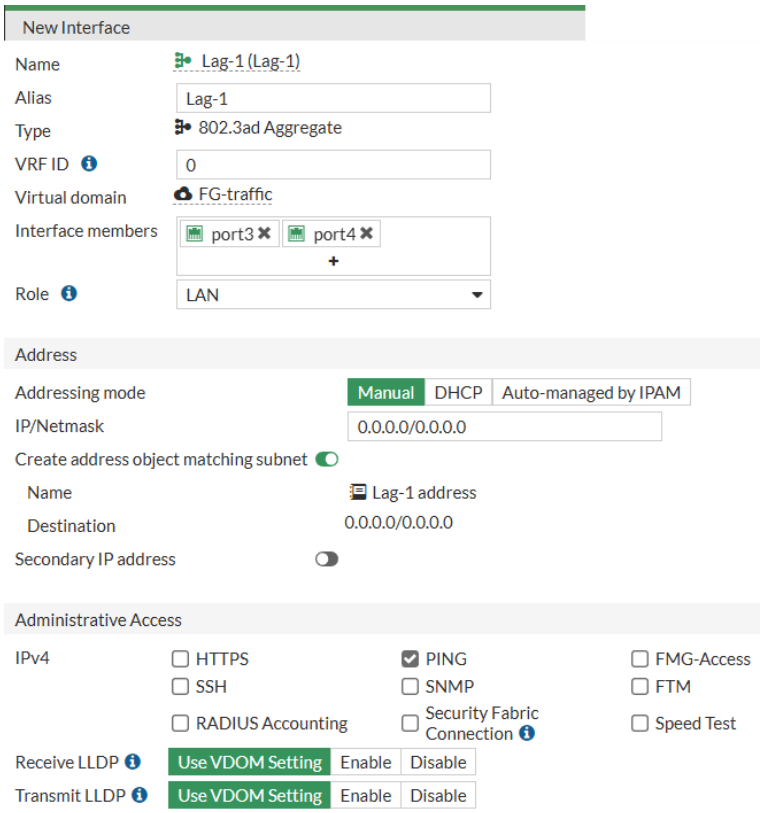


Figure: BR1-FW1 Configure Aggregate Interface

We can configure Aggregate Interface with CLI Commands as per below.

BR1-FW1 Aggregate Interface Configuration

```
config vdom
 edit FG-traffic
  config system interface
   edit port3
    set vdom FG-traffic
   next
   edit port4
    set vdom FG-traffic
   next
   edit Lag-1
    set vdom FG-traffic
    set allowaccess ping
    set type aggregate
    set member port3 port4
    set description Lag-1
    set alias Lag-1
    set device-identification enable
    set device-user-identification enable
    set lldp-transmission enable
    set role lan
   end
  config system zone
   edit LAN
    set interface Lag-1
   end
  end
```

In the above configuration why device-identification, lldp-transmission enable and set device-user-identification enable.

- **set device-identification enable** Activates passive device profiling on the interface. FortiGate inspects traffic (DHCP, HTTP headers, MAC OUI) to classify connected endpoints.
- **set device-user-identification enable** Enables user-based identification on the interface. FortiGate attempts to associate traffic with specific user identities for visibility and policy enforcement.
- **set lldp-transmission enable** Turns on LLDP advertisement. FortiGate actively sends LLDP packets to neighbors, announcing its identity, interface name, and capabilities for topology discovery.

For the Aggregate Interface Validation we required to configure Channel group on switch BR1-SW3. Also allow the vlan on Port channel interface as per the below configuration to enable link aggregation on switch.

BR1-SW3 Channel-Group Po1 Configuration

```
config t
 int range Gi0/0-1
  description to-primary-firewall
  channel-group 1 mode active
```

BR1-SW3 Po1 allowed VLAN Configuration

```
config t
 int Po1
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30
```

We can verify Aggregation link of FortiGate Firewall with following commands.

BR1-FW1 aggregate interface configuration - show system interface Lag-1

```
BR1-FW1 (FG-traffic) # show system interface Lag-1
config system interface
    edit "Lag-1"
        set vdom "FG-traffic"
        set allowaccess ping
        set type aggregate
        set member "port3" "port4"
        set description "Lag-1"
        set device-identification enable
        set role lan
    next
end
```

BR1-FW1 - diagnose netlink aggregate name Lag-1

```
BR1-FW1 (FG-traffic) # diagnose netlink aggregate name Lag-1
LACP flags: (A|P)(S|F)(A|I)(I|O)(E|D)(E|D)
status: up
LACP mode: active
LACP HA: enable
actor MAC address: 50:4d:b9:00:4f:02
partner MAC address: 50:39:a6:00:80:00
slave: port3
  index: 0
  link status: up
  permanent MAC addr: 50:4d:b9:00:4f:02
  LACP state: established
  partner system: 28672 50:39:a6:00:80:00
slave: port4
  index: 1
  link status: up
  permanent MAC addr: 50:4d:b9:00:4f:03
  LACP state: established
  partner system: 28672 50:39:a6:00:80:00
```

BR1-FW1 - diagnose netlink aggregate list

```
BR1-FW1 (FG-traffic) # diagnose netlink aggregate list
List of 802.3ad link aggregation interfaces:
 2  name Lag-1             status up      algorithm L4  lacp-mode active
```

BR1-SW3 - show etherchannel summary

```
BR1-SW3#show etherchannel summary
Number of channel-groups in use: 1
Number of aggregators:           1
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)         LACP      Gi0/0(P)    Gi0/1(P)
```

6.3 VLAN interface

On a FortiGate firewall, a VLAN interface is a logical sub-interface created on top of a physical port or aggregate interface with a VLAN ID. It's how we segment traffic into multiple broadcast domains over the same physical link. We will allocate VLAN in our demonstration as per below table and use vlan for ip address assignment by DHCP.

Table 6: LAN VLAN Allocation with Names and IPs

Site	VLAN ID	Subnet Range	VLAN Name
Branch 1	10	192.168.21.0/24	SALES
	20	192.168.22.0/24	PROD
	30	192.168.23.0/24	HR

Now we will configure VLAN interface on BR1-FW1 as per below GUI screenshot. Before starting the vlan configuration select FG-traffic VDOM for GUI.

GUI PATH: **Interfaces --> Create New --> Interface**

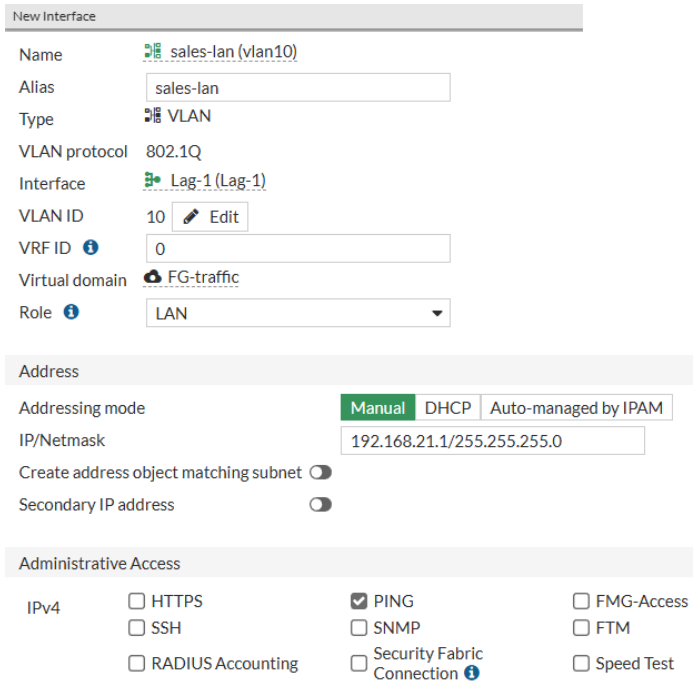


Figure: BR1-FW1 Configure VLAN Interface

We can configure Vlan Interface as per the below CLI Configuration and also map the vlan under LAN zone as discussed in Zones in FortiGate section.

BR1-FW1 VLAN Interface Configuration

```
config vdom
edit FG-traffic
 config system interface
  edit vlan10
   set vdom FG-traffic
   set ip 192.168.21.1 255.255.255.0
   set allowaccess ping
   set description sales-lan
   set alias sales-lan
   set role lan
   set interface Lag-1
```

```
    set vlanid 10
   next
    edit vlan20
     set vdom FG-traffic
     set ip 192.168.22.1 255.255.255.0
     set allowaccess ping
     set description prod-lan
     set alias prod-lan
     set role lan
     set interface Lag-1
     set vlanid 20
   next
    edit vlan30
     set vdom FG-traffic
     set ip 192.168.23.1 255.255.255.0
     set allowaccess ping
     set description hr-lan
     set alias hr-lan
     set role lan
     set interface Lag-1
     set vlanid 30
    end
    config system zone
     edit LAN
      append interface vlan10 vlan20 vlan30
     end
    end
```

We can perform the system Interfaces and Routing Table verification as shown in the following output.

BR1-FW1 VLAN Interface Verification

```
BR1-FW1 (FG-traffic) # show system interface
vlan10   static   0.0.0.0 0.0.0.0  192.168.21.1 255.255.255.0  up   disable   vlan  enable
vlan20   static   0.0.0.0 0.0.0.0  192.168.22.1 255.255.255.0  up   disable   vlan  enable
vlan30   static   0.0.0.0 0.0.0.0  192.168.23.1 255.255.255.0  up   disable   vlan  enable
!
BR1-FW1 (FG-traffic) # get router info routing-table all
C       192.168.21.0/24 is directly connected, vlan10
C       192.168.22.0/24 is directly connected, vlan20
C       192.168.23.0/24 is directly connected, vlan30
```

6.4 Loopback interface

A Loopback Interface on a FortiGate firewall is a virtual interface that doesn't correspond to any physical port. It's extremely useful in routing and management scenarios, especially when working with protocols like BGP, OSPF, or IPsec tunnels.

Key Characteristics of Loopback Interfaces

- **No physical port**: Purely logical, exists in software.
- **Stable endpoint**: Remains up as long as the firewall is running, independent of physical link status.
- **Used for router ID**: Often assigned as the BGP/OSPF router ID.
- **Management IP**: Can host services like SNMP, syslog, or monitoring.

- **VPN/IPsec termination**: Provides a consistent endpoint for tunnels.

Device	Interface	IPv4 Address
BR1-FW1	Loopback0	100.64.100.2/32

Table 7: Device and Loopback Interface IP Address Assignment

We can configure the loopback interface on BR1-FW1 as shown in the GUI screenshot. Before Loopback interface configuration select FG-traffic from GUI.

GUI PATH: **Interfaces --> Create New --> Interface**

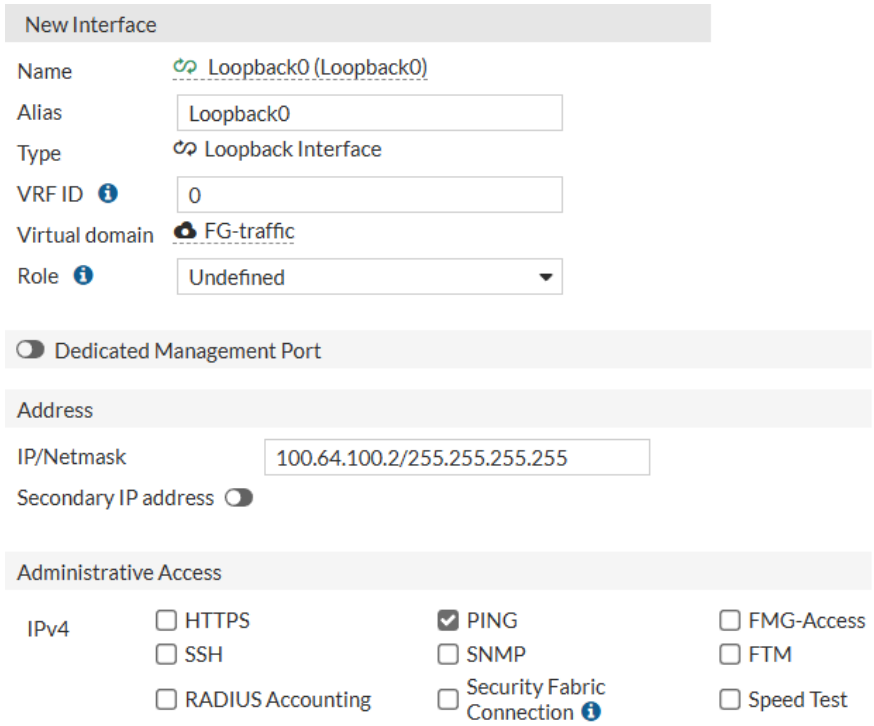


Figure: BR1-FW1 Configure Loopback Interface

We can configure Loopback Interface as per the below CLI Configuration and use the IP address shown in the table above.

BR1-FW1 Loopback0 Interface Configuration

```
config vdom
edit FG-traffic
 config system interface
  edit Loopback0
   set vdom FG-traffic
   set ip 100.64.100.2 255.255.255.255
   set allowaccess ping
   set type loopback
   set description Loopback0
   set alias Loopback0
 end
```

To verify a loopback interface on a FortiGate firewall, check its configuration and confirm that it is present in the routing table.

BR1-FW1 Loopback0 Verification

```
BR1-FW1 (FG-traffic) # show system interface Loopback0
config system interface
    edit "Loopback0"
        set vdom "FG-traffic"
        set ip 100.64.100.2 255.255.255.255
        set allowaccess ping
        set type loopback
        set description "Loopback0"
        set alias "Loopback0"
        set snmp-index 19
    next
end
!
BR1-FW1 (FG-traffic) # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
Routing table for VRF=0
C       100.64.100.2/32 is directly connected, Loopback0
!
BR1-FW1 (global) # sh system interface
Loopback0   static   0.0.0.0 0.0.0.0  100.64.100.2 255.255.255.255  up   disable   loopback
    enable
```

7 DHCP Server

In this section, we will cover the following objectives.

> **Objective**
>
> - Understand DHCP Operations and DORA process.
> - Configuration and verification of FortiGate as a DHCP server.

DHCP (Dynamic Host Configuration Protocol) is classified as an application layer protocol in the OSI model. It uses UDP transport (ports 67 for server, 68 for client) to exchange messages. DHCP uses the DORA process for Dynamic IP address allocation (DISCOVER, OFFER, REQUEST, ACKNOWLEDGMENT).

- **DHCP Discover**: A client (e.g. PC, phone) broadcasts a request on the network asking for an IP address. This is a “Who can give me an IP?” message.
- **DHCP Offer**: The DHCP server (FortiGate firewall in lab) responds with an available IP address, subnet mask, gateway, DNS, and lease time. This is a “Here’s an IP you can use” message.
- **DHCP Request**: The client replies, requesting the offered IP address. This confirms “I’d like to use the IP you suggested.”
- **DHCP Acknowledgment (ACK)**: The server acknowledges and finalizes the lease. The client can now use the IP address and other configuration details.

The following figure shows the workflow of FortiGate firewall as DHCP Server.

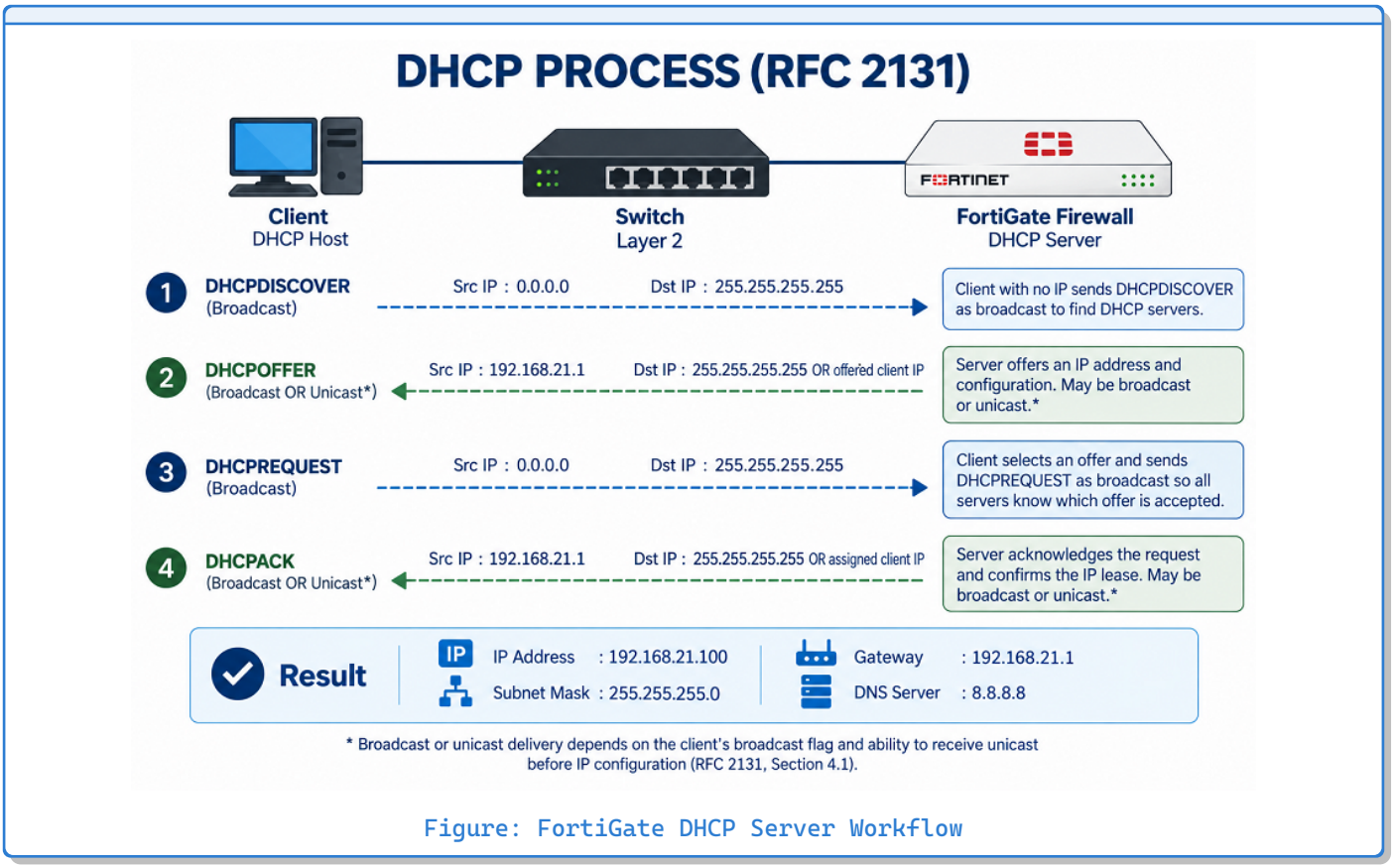


Figure: FortiGate DHCP Server Workflow

In a small or medium scale network we can use FortiGate as DHCP Server. Following are some key points regarding centralized DHCP server.

- **Centralized Control**: The firewall manages IP allocation and security together.
- **Ease of Management**: No need for a separate DHCP appliance in small/medium networks.
- **Integration & Flexibility**: DHCP options (gateway, DNS, lease time) align directly with firewall interface configuration and supports multiple scopes per VLAN/interface.

We will use a public DNS server on all firewalls in our lab. GUI and CLI Configuration are shown below.

> ☞ **Note**
>
> - Enterprise environments should use internal DNS resolver whenever possible.
> - Public DNS Services are used in this lab for simplicity only.

GUI PATH: **Network --> DNS**

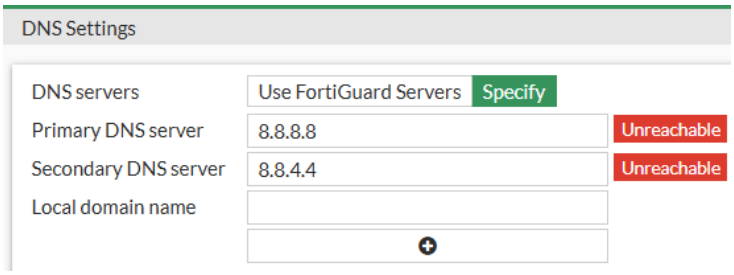


Figure: FortiGate DNS Server Configuration

FortiGate DNS Server Configuration

```
config global
 config system dns
  set primary 8.8.8.8
  set secondary 8.8.4.4
 end
```

The DHCP server is configured on the FortiGate firewall using the GUI method, as shown below.

Configuration Field Breakdown

- **DHCP Status:** Enables or disables the DHCP service on the selected interface. When Enabled, FortiGate automatically assigns IP addresses to connected hosts.
- **Address Range:** Defines the pool of IP addresses that clients can receive (e.g. 192.168.21.2–192.168.21.254). Ensures devices on the VLAN or subnet get unique IPs within this range.
- **Netmask:** Specifies the subnet mask (e.g. 255.255.255.0) to define the network boundary. Determines how many IPs are available in the subnet.
- **Default Gateway:** Sets the gateway IP that clients use to reach other networks. Same as Interface IP means the interface's own IP acts as the gateway.

- **DNS Server:** Defines how clients resolve domain names. Options include System DNS (uses FortiGate's global DNS), Interface IP, or a custom DNS server.
- **Lease Time:** Determines how long an IP address is assigned to a client. (e.g. 604800 seconds ≈ 7 days). Longer lease times reduce DHCP traffic; shorter times suit dynamic environments.

We can configure DHCP Server from GUI as shown in the screenshot below.

GUI PATH: **Interface --> Lag-1 --> vlan10/20/30 --> Edit**

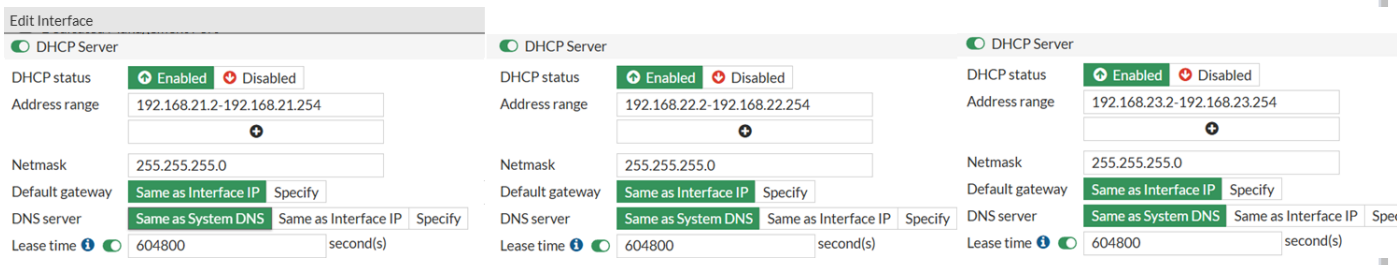
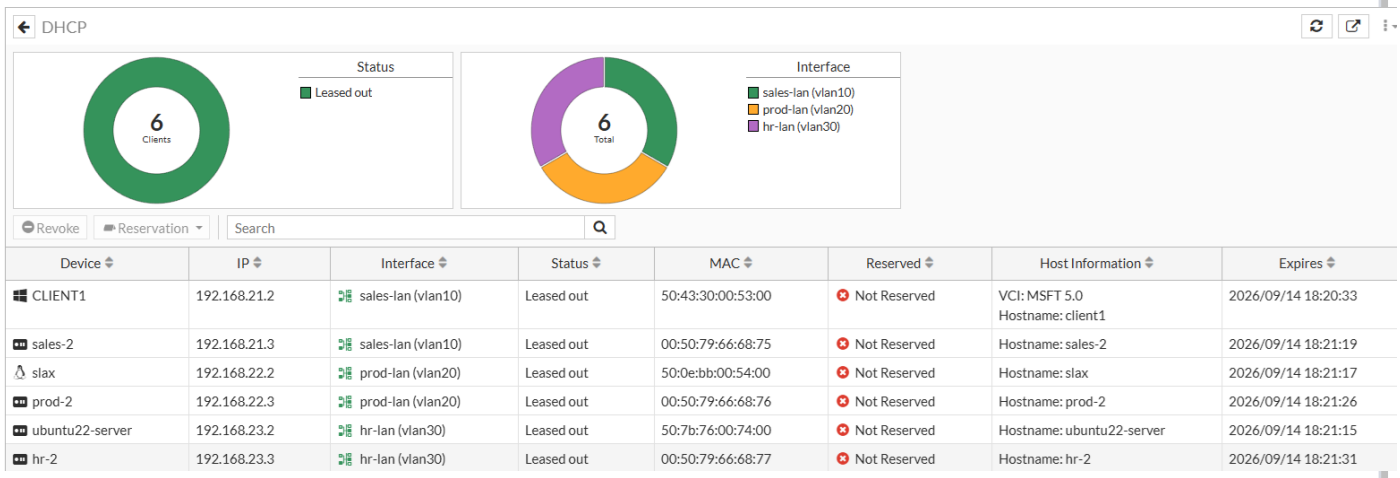


Figure: BR1-FW1 DHCP Server Configuration

We can verify DHCP IP address lease as per below GUI.

GUI PATH: **Network --> DHCP**



Device	IP	Interface	Status	MAC	Reserved	Host Information	Expires
CLIENT1	192.168.21.2	sales-lan (vlan10)	Leased out	50:43:30:00:53:00	Not Reserved	VCI: MSFT 5.0 Hostname: client1	2026/09/14 18:20:33
sales-2	192.168.21.3	sales-lan (vlan10)	Leased out	00:50:79:66:68:75	Not Reserved	Hostname: sales-2	2026/09/14 18:21:19
slax	192.168.22.2	prod-lan (vlan20)	Leased out	50:0e:bb:00:54:00	Not Reserved	Hostname: slax	2026/09/14 18:21:17
prod-2	192.168.22.3	prod-lan (vlan20)	Leased out	00:50:79:66:68:76	Not Reserved	Hostname: prod-2	2026/09/14 18:21:26
ubuntu22-server	192.168.23.2	hr-lan (vlan30)	Leased out	50:7b:76:00:74:00	Not Reserved	Hostname: ubuntu22-server	2026/09/14 18:21:15
hr-2	192.168.23.3	hr-lan (vlan30)	Leased out	00:50:79:66:68:77	Not Reserved	Hostname: hr-2	2026/09/14 18:21:31

Figure: BR1-FW1 DHCP Dashboard

Before configuring the DHCP server using the CLI check current created DHCP server with **show system dhcp server** command.

BR1-FW1 DHCP Server Configuration

```
config vdom
 edit FG-traffic
  config system dhcp server
   edit 2
    set dns-service default
    set default-gateway 192.168.21.1
    set netmask 255.255.255.0
    set interface vlan10
    config ip-range
     edit 1
      set start-ip 192.168.21.2
```

```
      set end-ip 192.168.21.254
     next
    end
   next
   edit 3
    set dns-service default
    set default-gateway 192.168.23.1
    set netmask 255.255.255.0
    set interface vlan30
    config ip-range
     edit 1
      set start-ip 192.168.23.2
      set end-ip 192.168.23.254
     next
    end
   next
   edit 4
    set dns-service default
    set default-gateway 192.168.22.1
    set netmask 255.255.255.0
    set interface vlan20
    config ip-range
     edit 1
      set start-ip 192.168.22.2
      set end-ip 192.168.22.254
     next
    end
   next
  end
```

We can confirm both configuration and operational status of FortiGate DHCP server with these commands.

BR1-FW1 Verify DHCP Configuration

```
BR1-FW1 (FG-traffic) # sh system dhcp server
config system dhcp server
        set dns-service default
        set default-gateway 192.168.21.1
        set netmask 255.255.255.0
        set interface "vlan10"
        config ip-range
            edit 1
                set start-ip 192.168.21.2
                set end-ip 192.168.21.254
            next
        end
output omitted
```

BR1-FW1 Current leased IPs with MAC, hostname, expiry

```
BR1-FW1 (FG-traffic) # execute dhcp lease-list vlan10
  IP            MAC-Address             Hostname
  192.168.21.2  50:43:30:00:53:00     client1
  192.168.21.3  00:50:79:66:68:75  sales-2
BR1-FW1 (FG-traffic) # execute dhcp lease-list vlan20
  IP            MAC-Address             Hostname
```

```
  192.168.22.2  50:0e:bb:00:54:00       slax
  192.168.22.3  00:50:79:66:68:76       prod-2
BR1-FW1 (FG-traffic) # execute dhcp lease-list vlan30
  IP            MAC-Address             Hostname
  192.168.23.2  50:7b:76:00:74:00       ubuntu22-server
  192.168.23.3  00:50:79:66:68:77       hr-2
```

➟ **execute dhcp lease-clear (dhcp-leased-ip)**

The command **execute dhcp lease-clear 192.168.23.2** manually removes the DHCP lease entry for the specified IP address from the FortiGate DHCP server.

- **Removes the server-side lease entry** FortiGate deletes the lease from its DHCP lease table. However, this does not automatically force the client to immediately restart the DHCP DORA process. The client may continue using its currently configured IP address until it renews the lease, the DHCP client is restarted, or the network interface is disconnected/reconnected.
- **Troubleshooting address-allocation issues** Clearing a lease can be useful when troubleshooting stale DHCP entries, address-allocation problems, or when a client needs to obtain a new lease.
- **After DHCP scope changes** If the DHCP pool, gateway, DNS, or other options have been modified, clear the server-side lease if required and then renew/restart DHCP on the client so that the client requests updated DHCP information.

➲ **BR1-FW1 Clear/reserve leases (for troubleshooting)**

```
BR1-FW1 (FG-traffic) # execute dhcp lease-list vlan30
vlan30
  IP            MAC-Address             Hostname
  192.168.23.2  50:7b:76:00:74:00       ubuntu22-server
!
BR1-FW1 (FG-traffic) # execute dhcp lease-clear 192.168.23.2
!
BR1-FW1 (FG-traffic) # execute dhcp lease-list vlan30
#### The client has not yet requested/renewed its lease. ####
```

8 Supporting Lab Infrastructure – Cisco IOS

In this section, we will cover the following objectives.

> **Objective**
>
> - Understand the ISP and ISP-2 supporting network topology and IP addressing.
> - Configure Cisco ISP router interfaces and BGP connectivity.
> - Configure static routes toward the Global Internet and lab networks.
> - Configure NAT overload on the ISP router for Internet access.
> - Configure ISP-2 as a DHCP server and NAT gateway for remote users.
> - Verify end-to-end connectivity across the supporting ISP infrastructure.

8.1 ISP Topology

In this demonstration we consider two cisco router ISP and ISP-2 as follow topology.

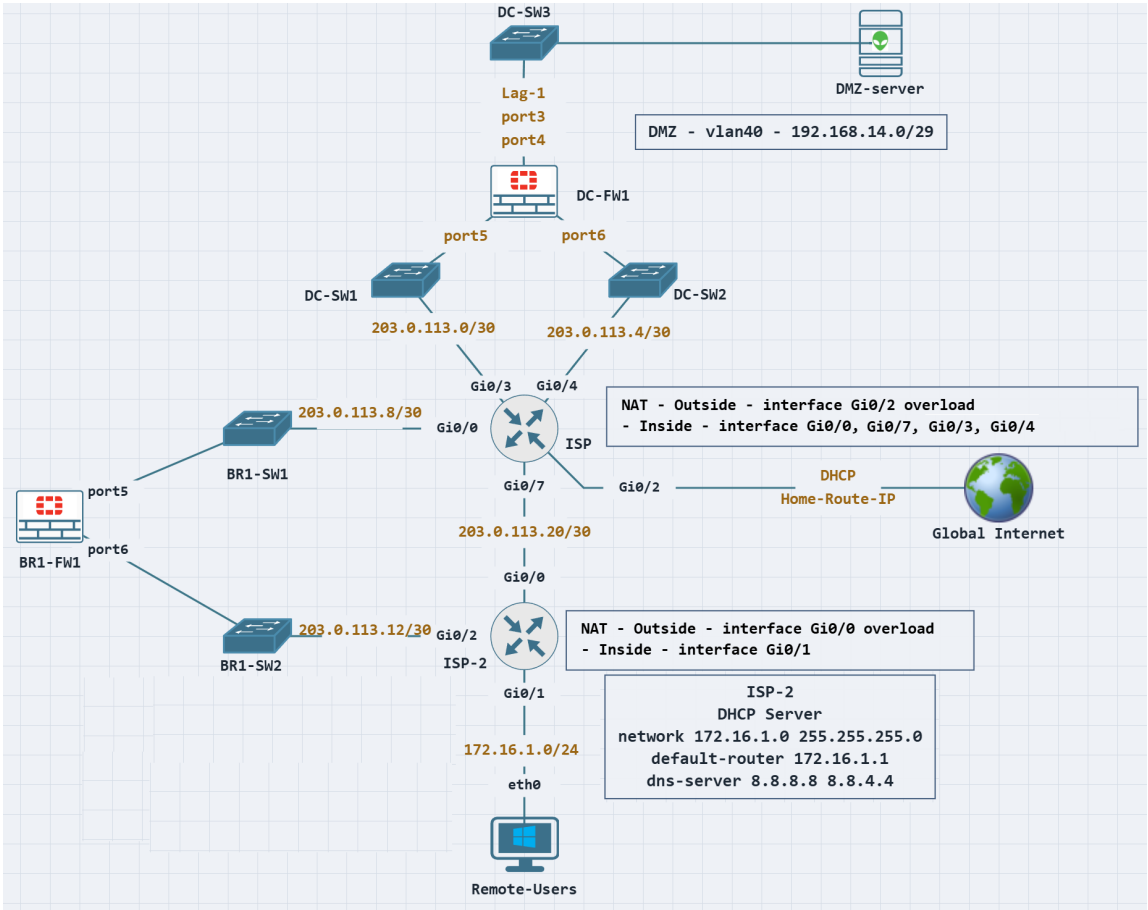


Figure: ISP Topology

In the following section we will see the ISP Route (ISP, ISP-2) Role and Function Notes.

- **ISP Router:** Acts as the primary NAT gateway to access the Global Internet for entire topology.

  - NAT configuration: Outside interface Gi0/2 (overload).
  - Inside interfaces on ISP Router: Gi0/0, Gi0/1, Gi0/7, Gi0/3, Gi0/4.
  - Provides DHCP Home-Route-IP for Internet connectivity.
- **ISP-2 Router:** Serves as both NAT and DHCP Server for local user connectivity.
  - NAT configuration: Outside interface Gi0/0 (overload).
  - Inside interface: Gi0/1.
  - DHCP scope: 172.16.1.0/24, default gateway 172.16.1.1, DNS servers 8.8.8.8 and 8.8.4.4.

8.2 ISP IP Addressing Table

Table 8: ISP IP Address Summary

Segment / Device	Interface	IP Address / Subnet	Notes
ISP Router (Core)	Gi0/3	203.0.113.1/30	To-DC-SW1-port5
	Gi0/4	203.0.113.5/30	To-DC-SW2-port6
	Gi0/0	203.0.113.9/30	To BR1-FW1-port5
	Gi0/7	203.0.113.21/30	To ISP-2
	Gi0/2	DHCP	Internet uplink (Home-Router-IP)
ISP-2 Router	Gi0/0	203.0.113.22/30	To ISP router
	Gi0/2	203.0.113.13/30	To BR1-FW1-port6
	Gi0/1	172.16.1.1/24	Internal LAN
Remote Users	eth0	DHCP from ISP-2	172.16.1.0/24 network

Configure cisco router using the configuration below. BR1-FW1 and DC-FW1 using ASN 65010 and Internet-cloud use ASN 65000.

ISP Router Interface Configuration

```
config t
hostname ISP
 interface GigabitEthernet0/0
  description to-br1-fw1-port5
  ip address 203.0.113.9 255.255.255.252
  no shutdown
!
interface GigabitEthernet0/2
 description to-home-router-Global-Internet
 ip address dhcp
 no shutdown
!
interface GigabitEthernet0/3
 description to-dc-fw1-port5
 ip address 203.0.113.1 255.255.255.252
 no shutdown
!
interface GigabitEthernet0/4
 description to-dc-fw1-port6
 ip address 203.0.113.5 255.255.255.252
 no shutdown
!
interface GigabitEthernet0/7
 description to-isp-2-router-gi0/0
 ip address 203.0.113.21 255.255.255.252
 no shutdown
```

ISP Router BGP Configuration

```
config t
 router bgp 65000
  bgp router-id 203.0.113.9
  no bgp default ipv4-unicast
  neighbor 203.0.113.10 remote-as 65010
  neighbor 203.0.113.22 remote-as 65000
 !
  address-family ipv4
   neighbor 203.0.113.10 activate
   neighbor 203.0.113.10 default-originate
   neighbor 203.0.113.22 activate
   neighbor 203.0.113.22 next-hop-self
 exit-address-family
```

ISP Router Static Route Configuration

```
config t
 ip route 100.64.100.1 255.255.255.255 203.0.113.2
 ip route 100.64.100.1 255.255.255.255 203.0.113.6
 ip route 203.0.113.64 255.255.255.248 203.0.113.10
 ip route 203.0.113.96 255.255.255.252 203.0.113.2
 ip route 203.0.113.100 255.255.255.252 203.0.113.6
```

ISP Router NAT Configuration

```
config t
 interface GigabitEthernet0/0
  ip nat inside
 interface GigabitEthernet0/7
  ip nat inside
 interface GigabitEthernet0/3
  ip nat inside
 interface GigabitEthernet0/4
  ip nat inside
!
 interface GigabitEthernet0/2
  ip nat outside
!
 access-list 1 permit 203.0.113.0 0.0.0.255
 ip nat inside source list 1 interface GigabitEthernet0/2 overload
```

ISP-2 Router we will going to use for DNAT and Remote SSL and IPsec VPN.

ISP-2 Router Interface and NAT Configuration

```
config t
hostname ISP-2
 access-list 1 permit 172.16.1.0 0.0.0.255
 ip nat inside source list 1 interface GigabitEthernet0/0 overload
!
 interface GigabitEthernet0/0
  description to-isp-router-gi0/7
  ip address 203.0.113.22 255.255.255.252
  ip nat outside
  no shutdown
!
 interface GigabitEthernet0/1
  description to-remote-users
  ip address 172.16.1.1 255.255.255.0
  ip nat inside
  no shutdown
!
 interface GigabitEthernet0/2
  description to-br1-fw1-port6
  ip address 203.0.113.13 255.255.255.252
  no shutdown
```

ISP-2 Router Static Route, bgp and DHCP Server Configuration

```
config t
 ip route 0.0.0.0 0.0.0.0 203.0.113.21
 ip route 203.0.113.64 255.255.255.248 203.0.113.14
!
 ip dhcp excluded-address 172.16.1.1 172.16.1.10
!
 ip dhcp pool LAN-POOL
  network 172.16.1.0 255.255.255.0
  default-router 172.16.1.1
  dns-server 8.8.8.8 8.8.4.4
!
 router bgp 65000
  bgp router-id 203.0.113.22
  no bgp default ipv4-unicast
  neighbor 203.0.113.14 remote-as 65010
   neighbor 203.0.113.21 remote-as 65000
 !
  address-family ipv4
   network 203.0.113.12 mask 255.255.255.252
   neighbor 203.0.113.14 activate
   neighbor 203.0.113.14 default-originate
   neighbor 203.0.113.21 activate
   neighbor 203.0.113.21 next-hop-self
 exit-address-family
```

We can verify ISP and ISP-2 routes as follows. After all above configuration we can see the route table as shown following cli output. Also the IBGP peering will be formed between ISP and ISP-2 routers.

ISP Router show ip route

```
ISP#show ip route
S*    0.0.0.0/0 [254/0] via 10.10.1.1
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.10.1.0/24 is directly connected, GigabitEthernet0/2
L        10.10.1.110/32 is directly connected, GigabitEthernet0/2
      203.0.113.0/24 is variably subnetted, 11 subnets, 3 masks
C        203.0.113.0/30 is directly connected, GigabitEthernet0/3
L        203.0.113.1/32 is directly connected, GigabitEthernet0/3
C        203.0.113.4/30 is directly connected, GigabitEthernet0/4
L        203.0.113.5/32 is directly connected, GigabitEthernet0/4
C        203.0.113.8/30 is directly connected, GigabitEthernet0/0
L        203.0.113.9/32 is directly connected, GigabitEthernet0/0
C        203.0.113.20/30 is directly connected, GigabitEthernet0/7
L        203.0.113.21/32 is directly connected, GigabitEthernet0/7
S        203.0.113.64/29 [1/0] via 203.0.113.10
S        203.0.113.96/30 [1/0] via 203.0.113.2
S        203.0.113.100/30 [1/0] via 203.0.113.6
```

ISP-2 Router show ip route

```
isp-2#sh ip route
S*    0.0.0.0/0 [1/0] via 203.0.113.21
```

```
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.16.1.0/24 is directly connected, GigabitEthernet0/1
L        172.16.1.1/32 is directly connected, GigabitEthernet0/1
      203.0.113.0/24 is variably subnetted, 4 subnets, 2 masks
C        203.0.113.12/30 is directly connected, GigabitEthernet0/2
L        203.0.113.13/32 is directly connected, GigabitEthernet0/2
C        203.0.113.20/30 is directly connected, GigabitEthernet0/0
L        203.0.113.22/32 is directly connected, GigabitEthernet0/0
S        203.0.113.64/29 [1/0] via 203.0.113.14
```

ISP and ISP-2 Routers IBGP Peering

```
ISP#sh ip bgp summary
BGP router identifier 203.0.113.9, local AS number 65000

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
203.0.113.10    4        65010      49      50       19    0    0 00:39:49        1
203.0.113.22    4        65000      18      16        5    0    0 00:11:17        0
!
isp-2#sh ip bgp summary
BGP router identifier 203.0.113.22, local AS number 65000

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
203.0.113.14    4        65010      49      51       19    0    0 00:40:02        1
203.0.113.21    4        65000      16      18        6    0    0 00:11:14        0
```

ISP-2 Routers DHCP pool

```
isp-2# sh ip dhcp pool
Pool LAN-POOL :
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 172.16.1.1           172.16.1.1       - 172.16.1.254      2
 !
 isp-2#sh ip dhcp binding assigned
 IP address          Client-ID/              Lease expiration        Type
                     Hardware address/
                     User name
 172.16.1.12         0100.5079.6668.49       Sep 8 2026 02:25 AM    Automatic
 172.16.1.25         0150.639d.004d.00       Sep 8 2026 02:27 AM    Automatic
```

9 Branch-1 WAN Connectivity

In this section we will see how we can configure BGP Routing on firewall for Branch-1.

9.1 BGP

For this lab demonstration firewall WAN connectivity, we will configure Branch-1 with BGP Routing with ASN 65010. Below are GUI method to configure BGP Peer on FortiGate Firewall also advertise the Loopback0 within BGP Network Advertisement.

Configuration Field Breakdown

- **Local AS:** Defines organization's Autonomous System Number (ASN). Used to identify network in BGP routing; here, 65010 represents internal AS.
- **Router ID:** Unique identifier for the BGP process, typically an interface or loopback IP (e.g. 100.64.100.2). It ensures deterministic route selection and must be unique per router.
- **Neighbors:** Lists external peers (ISPs or other AS routers) that exchange routing information. Each neighbor entry includes its IP and remote AS.
- **IP (Neighbor):** Specifies the peer's IP address (e.g. 203.0.113.9). This is the remote router's BGP endpoint.
- **Remote AS:** Defines the peer's Autonomous System Number (e.g. 65000). Establishes an eBGP session between AS and the ISP's AS.
- **Password:** Optional MD5 authentication for BGP sessions. Ensures secure communication between peers.
- **Interface:** Specifies the outgoing interface used to reach the neighbor. Useful when multiple WAN links exist.
- **Update Source:** Defines which local interface IP is used as the source for BGP updates. Commonly set to a loopback interface for stability.
- **Graceful Restart Time:** Determines how long the router maintains routes during a restart. Helps preserve routing stability during short outages.
- **Activate IPv4:** Enables IPv4 address family for the BGP session. Must be turned on for IPv4 route exchange.
- **IPv4 Filtering Options:** Control which routes are accepted or advertised using lists and maps:
  - **Filter list in/out:** Applies AS-path filters to incoming/outgoing routes.
  - **Distribute list in/out:** Filters routes based on access lists.
  - **Prefix list in/out:** Filters routes by prefix (network) criteria.
  - **Route map in/out:** Applies complex policy rules for route manipulation.
- **Allow AS in:** Permits own AS number in received routes (used in multi-AS designs).
- **Max Prefix:** Limits the number of prefixes accepted from a neighbor. Prevents route table overflow due to misconfiguration or route leaks.

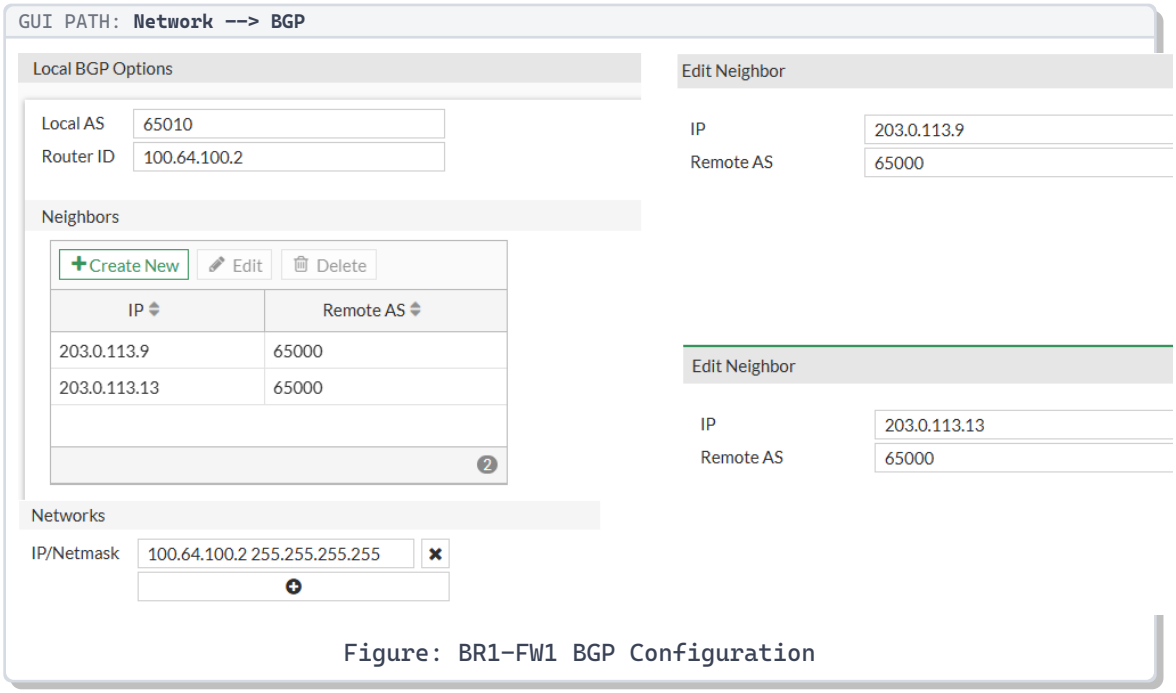


Figure: BR1-FW1 BGP Configuration

We can configure the BGP neighbor and advertise the network using the CLI configuration below. The network configuration advertises the loopback address 100.64.100.2/32, which will be used as the tunnel source in the future IPsec site-to-site VPN lab.

BR1-FW1 BGP Peer with ISP

```
config vdom
 edit FG-traffic
  config router bgp
   set as 65010
   set router-id 100.64.100.2
   config neighbor
    edit 203.0.113.9
     set remote-as 65000
    next
    edit 203.0.113.13
     set remote-as 65000
    next
   end
   config network
    edit 1
     set prefix 100.64.100.2 255.255.255.255
    next
   end
```

Following set of FortiGate BGP **show** commands we'll find useful for monitoring and troubleshooting neighbor sessions.

★ **Verification Command**

- **get router info bgp summary**
  - Provides a high-level overview of the BGP process.
  - Confirms the router ID and local AS number.
  - Shows the number of neighbors, their state, uptime, and how many prefixes are received.

- Useful for quickly checking if sessions are Established and how many routes are being exchanged.

BGP Summary on BR1-FW1

```
BR1-FW1 (FG-traffic) # get router info bgp summary
VRF 0 BGP router identifier 100.64.100.2, local AS number 65010
BGP table version is 6
5 BGP AS-PATH entries
0 BGP community entries
Neighbor     V         AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
203.0.113.9  4      65000      20      18        6    0    0 00:08:13        1
203.0.113.13 4      65000      21      16        5    0    0 00:08:11        1
Total number of neighbors 2
```

★ Verification Command

- **get router info bgp neighbor**
  - Displays detailed information about each BGP peer.
  - Confirms peer AS numbers, session type (internal vs external), and router IDs.
  - Shows session uptime, hold timers, and keepalive intervals.
  - Helps troubleshoot if a neighbor is flapping or stuck in Idle/Active states.

BGP Neighbor details on BR1-FW1

```
BR1-FW1 (FG-traffic) # get router info bgp neighbor
VRF 0 neighbor table:
BGP neighbor is 203.0.113.9, remote AS 65000, local AS 65010, external link
  BGP version 4, remote router ID 203.0.113.9
  BGP state = Established, up for 00:09:10
  Last read 00:00:32, hold time is 180, keepalive interval is 60 seconds
  Configured hold time is 180, keepalive interval is 60 seconds
.
OUTPUT OMITTED
.
BGP neighbor is 203.0.113.13, remote AS 65000, local AS 65010, external link
  BGP version 4, remote router ID 203.0.113.22
  BGP state = Established, up for 00:09:08
  Last read 00:00:41, hold time is 180, keepalive interval is 60 seconds
  Configured hold time is 180, keepalive interval is 60 seconds
```

★ Verification Command

- **get router info bgp network**
  - Dumps the BGP routing table entries.
  - Verifies which prefixes are advertised and received.
  - Shows next hop, origin code, path attributes, and which route is selected as best (>).
  - Critical for confirming default routes, loopbacks, or customer prefixes are properly exchanged.

BGP network on BR1-FW1

```
BR1-FW1 (FG-traffic) # get router info bgp network
VRF 0 BGP table version is 3, local router ID is 100.64.100.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete
   Network          Next Hop            Metric     LocPrf Weight RouteTag Path
*>  0.0.0.0/0        203.0.113.13    0                      0        0 65000 i <-/->
*                    203.0.113.9     0                      0        0 65000 i <-/1>
*> 100.64.100.2/32  0.0.0.0                          100  32768        0 i <-/1>
Total number of prefixes 2
```

BGP routes on BR1-FW1

```
BR1-FW1 (FG-traffic) # get router info routing-table bgp
Routing table for VRF=0
B*      0.0.0.0/0 [20/0] via 203.0.113.13 (recursive is directly connected, port6), 00:02:04,
     [1/0]
```

We can do the verification of the bgp neighbor and received-routes on ISP router from BR1-FW1 as shown below.

ISP and ISP-2 BGP Verification

```
ISP#sh bgp ipv4 unicast summary
ISP#sh bgp ipv4 unicast summary
BGP router identifier 203.0.113.9, local AS number 65000
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
203.0.113.10    4        65010      29      26        5    0    0 00:20:02        1
203.0.113.22    4        65000      33      28        5    0    0 00:20:29        2
!
ISP#sh bgp ipv4 unicast
BGP table version is 11, local router ID is 203.0.113.9
     Network          Next Hop            Metric LocPrf Weight Path
      0.0.0.0          0.0.0.0                              0 i
 * i  100.64.100.2/32  203.0.113.22             0    100      0 65010 i
 *>                    203.0.113.10                         0 65010 i
 *>i  203.0.113.12/30  203.0.113.22             0    100      0 i
!
isp-2#sh bgp ipv4 unicast summary
BGP router identifier 203.0.113.22, local AS number 65000
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
203.0.113.14    4        65010      32      34        7    0    0 00:21:20        1
203.0.113.21    4        65000      29      34        7    0    0 00:21:15        1
!
isp-2#sh bgp ipv4 unicast
BGP table version is 7, local router ID is 203.0.113.22
     Network          Next Hop            Metric LocPrf Weight Path
      0.0.0.0          0.0.0.0                              0 i
 * i  100.64.100.2/32  203.0.113.21             0    100      0 65010 i
 *>                    203.0.113.14                         0 65010 i
 *>   203.0.113.12/30  0.0.0.0                  0         32768 i
```

Below are the observations from above outputs.

- **BGP Summary** Two neighbors up and stable, prefixes received correctly.

- **Neighbor Details** External peers in AS 65000. Sessions stable, timers consistent, no queue backlog.
- **Network Table** Default route learned from both peers on BR1-FW1, best via 203.0.113.13.
- Loopback 100.64.100.2/32 originated with local default LP and weight 32768.

In the below section we utilize the different Routing Objects for BGP path manipulation.

9.2 Routing Objects

In the FortiGate firewall we are able to configure below objects:-

- **Route Map** Controls redistribution between routing protocols. Can modify attributes like local preference or filter prefixes during import/export.
- **Access List** Defines permit/deny rules for specific networks. Used to filter routes before redistribution or advertisement.
- **Prefix List** Matches IP prefixes with sequence numbers. Provides fine-grained control over which routes are accepted or advertised.
- **AS Path List** Filters BGP routes based on Autonomous System path patterns. Useful for preferring or blocking routes from certain AS sequences.
- **Community List** Matches BGP communities to apply routing policies. Enables tagging, prioritization, or suppression of route advertisements.

To demonstrate this we will use the Routing object as per below for BGP path selection using Local Preference and AS-PATH Attributes. BGP path selection summary with local preference and AS-PATH prepend shown in below table, we will use this for configuration.

Path	Egress Interface	BGP Peer IP (ISP Router)	Local Preference	AS-Path Prepend
Primary	port5	203.0.113.9 (ISP)	200	None
Secondary	port6	203.0.113.13 (ISP-2)	100	Yes (2× AS 65010)

Table 9: BGP Path Selection Parameters

As the ISP advertise default route 0.0.0.0/0 to BR1-FW1, so we will create prefix list 0.0.0.0/0 as per below GUI Screenshot.

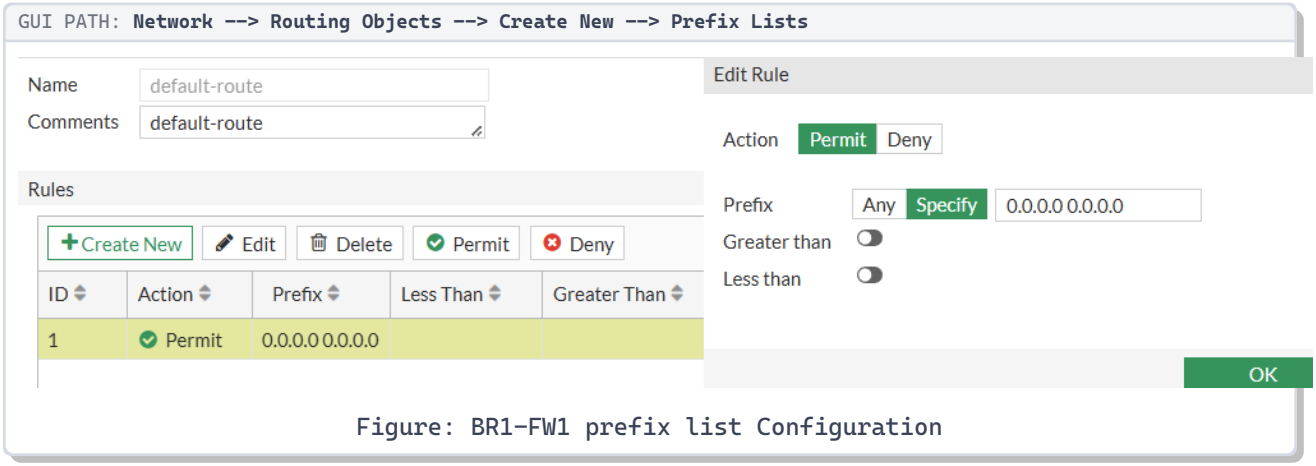


Figure: BR1-FW1 prefix list Configuration

BR1-FW1 peers with two routers in AS 65000: ISP (203.0.113.9, reachable via port5 on the 203.0.113.8/30 link) and ISP-2 (203.0.113.13, reachable via port6 on the 203.0.113.12/30 link). To make the ISP path primary, the route-map local-pref-200 sets local preference

200 on the default route learned from 203.0.113.9, while local-pref-100 keeps the ISP-2 path at 100. In the outbound direction, route-map as-path-2 prepends AS 65010 twice to advertisements toward ISP-2 (203.0.113.13), so ISP-2 is less preferred for return traffic.

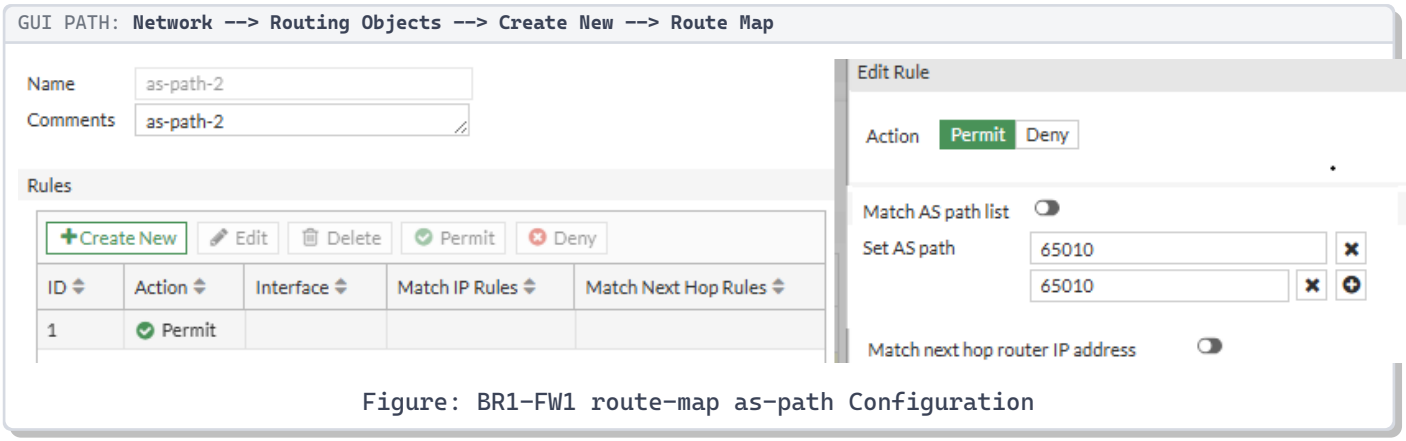


Figure: BR1-FW1 route-map as-path Configuration

Now create and apply local preference to default route as shown in the screenshot below.

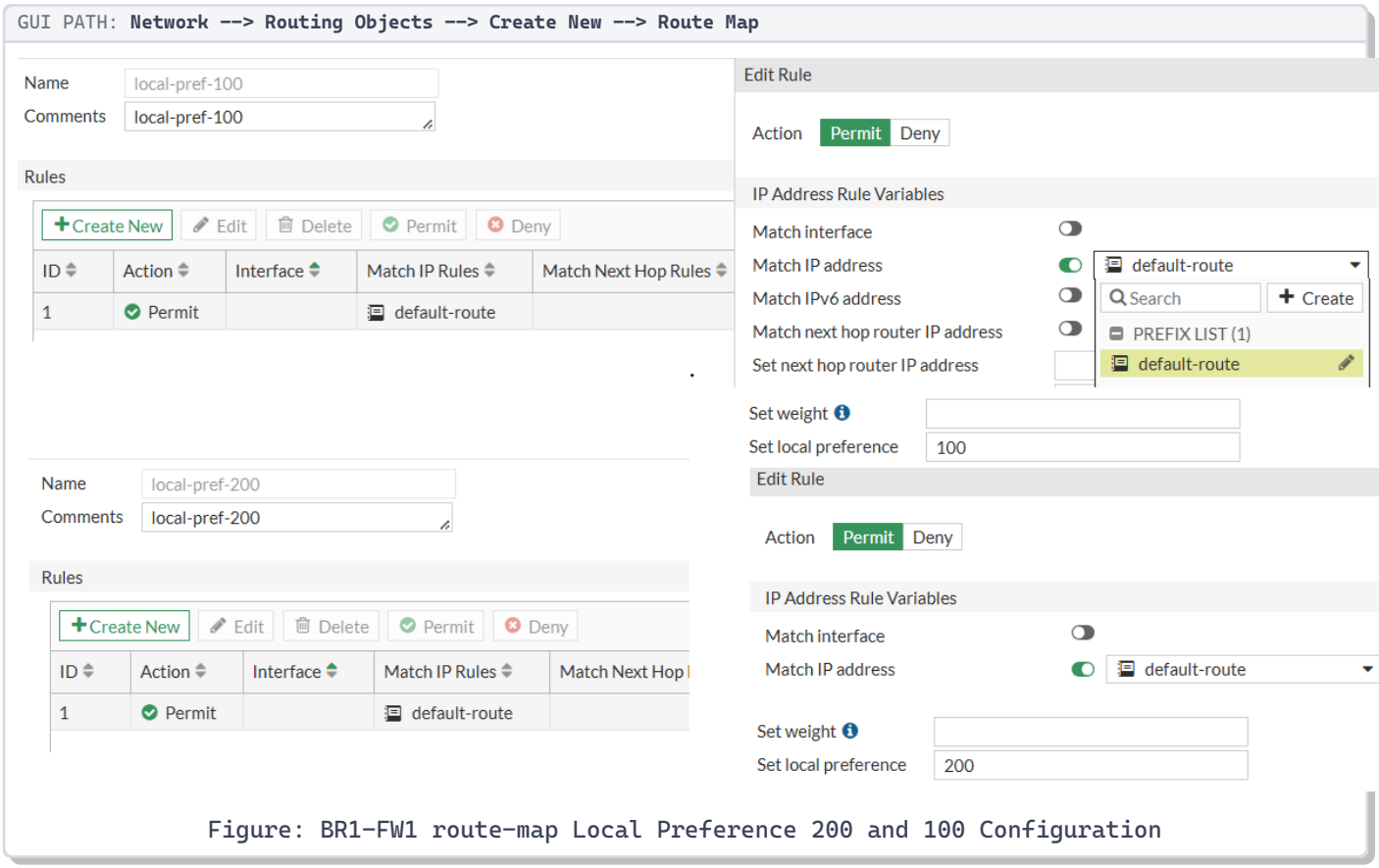


Figure: BR1-FW1 route-map Local Preference 200 and 100 Configuration

We can configure the prefix list, route-map for local-preference, as-path using the CLI configuration as shown below.

BR1-FW1 route-map

```
config vdom
 edit FG-traffic
  config router prefix-list
   edit default-route
    config rule
     edit 1
```

```
      set prefix 0.0.0.0 0.0.0.0
      unset ge
      unset le
     next
    end
    next
end
  config router route-map
   edit local-pref-200
    set comments local-pref-200
    config rule
     edit 1
      set match-ip-address default-route
      set set-local-preference 200
     next
    end
   next
   edit local-pref-100
    set comments local-pref-100
    config rule
     edit 1
      set match-ip-address default-route
      set set-local-preference 100
     next
    end
   next
   edit as-path-2
    set comments as-path-2
    config rule
     edit 1
      set set-aspath 65010 65010
     next
    end
   next
  end
```

Now apply these Route-map to ISP peer on BR1-FW1, as following GUI and CLI Configurations.

BR1-FW1 route-map apply to peer

```
config vdom
 edit FG-traffic
  config router bgp
   config neighbor
    edit 203.0.113.9
     set route-map-in local-pref-200
    next
    edit 203.0.113.13
     set route-map-in local-pref-100
     set route-map-out as-path-2
    next
   end
```

GUI PATH: **Network --> BGP --> Neighbors --> Route Map in/out**

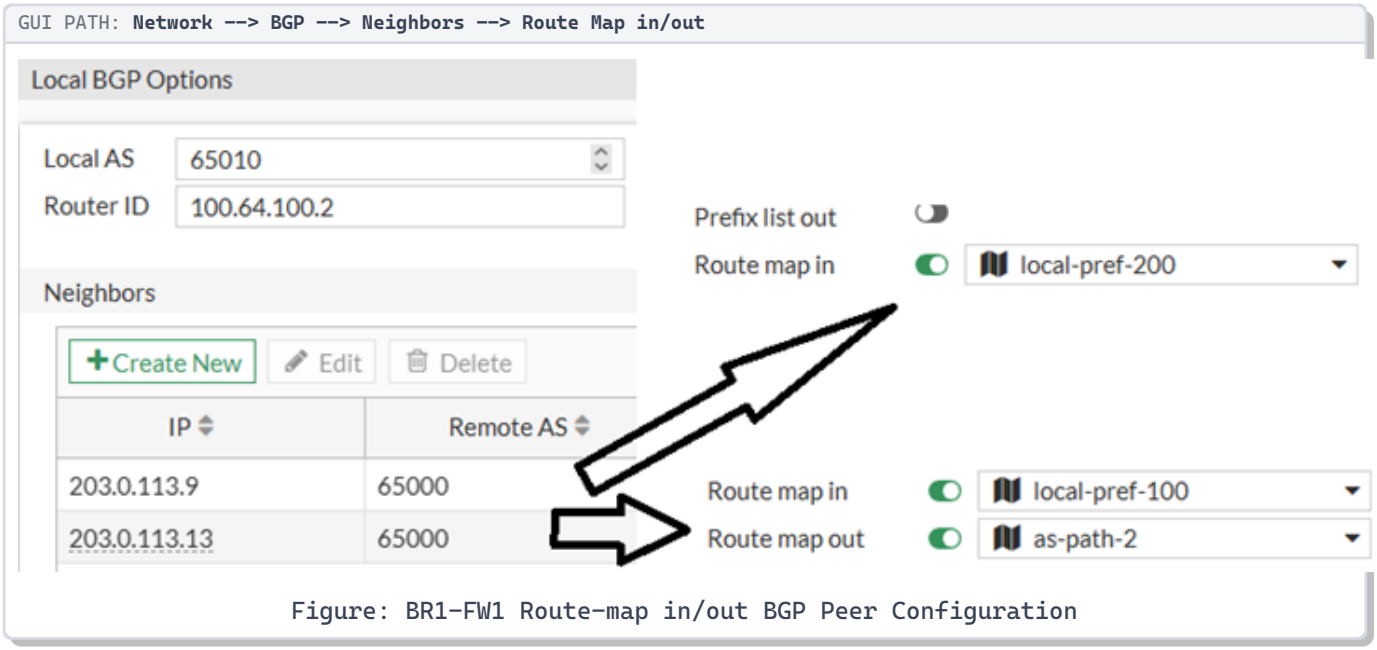


Figure: BR1-FW1 Route-map in/out BGP Peer Configuration

In the following output we can see the local preference 200 applied to the peer 203.0.113.9 cause this best path and path from peer 203.0.113.13 less preferred.

BGP network on BR1-FW1

```
BR1-FW1 (FG-traffic) # get router info bgp network
VRF 0 BGP table version is 4, local router ID is 100.64.100.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete
   Network          Next Hop            Metric     LocPrf Weight RouteTag Path
*> 0.0.0.0/0        203.0.113.9     0              200      0        0 65000 i <-/1>
*                   203.0.113.13    0              100      0        0 65000 i <-/->
*> 100.64.100.2/32  0.0.0.0                        100  32768        0 i <-/1>
Total number of prefixes 2
```

We can do the verification of bgp neighbor and received-routes on ISP router from BR1-FW1 as shown below which shows AS path prepending for peer 203.0.113.14 which cause it less preferred and peer 203.0.113.10 becomes best route.

ISP BGP Verification

```
ISP#sh bgp ipv4 unicast
BGP table version is 5, local router ID is 203.0.113.9
     Network          Next Hop            Metric LocPrf Weight Path
      0.0.0.0          0.0.0.0                              0 i
 *>   100.64.100.2/32  203.0.113.10                         0 65010 i
 *>i  203.0.113.12/30  203.0.113.22             0    100      0 i
!
isp-2#sh bgp ipv4 unicast
BGP table version is 8, local router ID is 203.0.113.22
     Network          Next Hop            Metric LocPrf Weight Path
      0.0.0.0          0.0.0.0                              0 i
 *>i  100.64.100.2/32  203.0.113.21             0    100      0 65010 i
 *                    203.0.113.14                         0 65010 65010 i
 *>   203.0.113.12/30  0.0.0.0                  0         32768 i
```

10 Policy and Object

In this section, we will cover the following objectives.

> **Objective**
>
> - Configure firewall address objects and groups.
> - Configure firewall security policy, verify traffic matches policy.
> - Troubleshoot policy related connectivity issues.

10.1 FortiGate Feature Visibility

The Feature Visibility menu allows administrators to control which features appear in the FortiGate GUI. Enabling or disabling modules like **VPN**, **Application Control**, **Web Filter**, and **SD-WAN Interface** helps tailor the interface to specific deployment needs. This improves usability and ensures only relevant configuration options are displayed.
Before configuring new policies and objects, it is essential to enable the **Multiple Interface Policies** option under **Feature Visibility**. This ensures that the FortiGate system interface supports advanced policy configurations across multiple interfaces, as illustrated in the screenshot below.

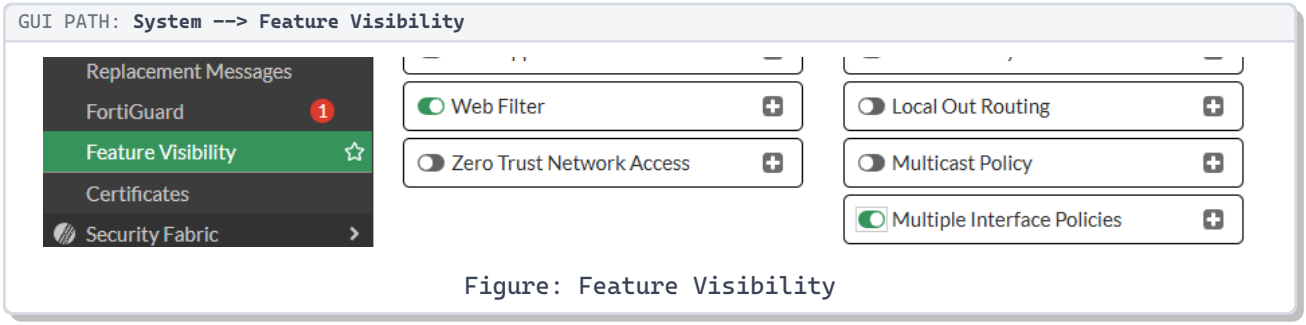


Figure: Feature Visibility

10.2 Firewall Objects

- **Address Objects**
  - Subnets (e.g., 192.168.1.0/24), Single IPs (e.g., 10.10.10.5/32). Ranges (e.g., 192.168.1.10–192.168.1.50), FQDNs (e.g., server.example.com)
- **Service Objects**
  - TCP/UDP ports (e.g., TCP/80, TCP/443, UDP/53)
  - Predefined services (HTTP, HTTPS, DNS, SSH, etc.)
- **Groups**
  - Address Groups (combine multiple address objects)
  - Service Groups (combine multiple service objects)

Next, we proceed with the configuration of a new **Address Object**. The following GUI screenshot illustrates the process.

GUI PATH: **Policy & Object --> Addresses --> Create New --> Address or Address Group**

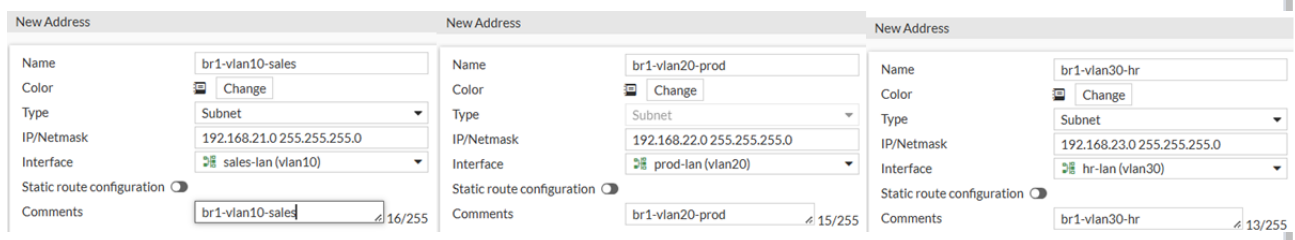


Figure: BR1-FW1 Configure Address Object

As per following CLI we can configure Address Objects on BR1-FW1.

BR1-FW1 Address Object

```
config vdom
 edit FG-traffic
  config firewall address
   edit br1-vlan10-sales
    set comment br1-vlan10-sales
    set subnet 192.168.21.0 255.255.255.0
   next
   edit br1-vlan20-prod
    set comment br1-vlan20-prod
    set subnet 192.168.22.0 255.255.255.0
   next
   edit br1-vlan30-hr
    set comment br1-vlan30-hr
    set subnet 192.168.23.0 255.255.255.0
   next
  end
 next
end
```

BR1-FW1 Address Group

```
config vdom
 edit FG-traffic
  config firewall addrgrp
   edit br1-lan-address
    set member br1-vlan10-sales br1-vlan20-prod br1-vlan30-hr
    set comment br1-lan-address
   next
  end
 next
end
```

10.3 Firewall Policies

- **Source**: Address object or group (e.g. LAN_Subnet)
- **Destination**: Address object or group (e.g. WAN_ISP, DMZ_Server)
- **Service**: Service object or group (e.g. Web_Service, Admin_Services)
- **Action**: Allow or Deny
- **NAT**: Enable/Disable (for internet access or IP translation)

- **Logging**: Enable/Disable (for monitoring traffic)
- **Schedule**: Always or custom time windows

10.4 FortiGate Firewall Inspection Modes

In FortiGate, we can choose between Flow-based and Proxy-based inspection modes. These define how the firewall analyzes and processes traffic:

Aspect	Flow-based Inspection	Proxy-based Inspection
Method	Inspects traffic as it passes through the FortiGate using flow/stream-based security engines without fully buffering the complete content.	Acts as a proxy for supported traffic, reconstructing/buffering content as required before forwarding it.
Performance	Generally requires fewer processing resources and provides lower latency.	Generally requires more processing resources and may introduce additional latency because more content can be buffered and reconstructed.
Security Features	Supports many FortiGate security profiles and performs threat inspection in real time.	Supports proxy-specific inspection features and processing that require full content reconstruction or proxy handling.
Best Use Case	Recommended for most deployments where good security inspection and performance are required.	Use when a required security feature specifically depends on proxy-based inspection or content reconstruction.

Table 10: Comparison of Flow-based vs Proxy-based Inspection

10.5 How Policy Evaluation Works

- FortiGate checks traffic against the policy list in sequence (from the first rule down).
- The first matching policy (based on Source, Destination, Service, Schedule, and Interface) is applied.
- Once a match is found, FortiGate does not continue checking further policies.
- If no policy matches, traffic is denied by the implicit **deny all** rule at the bottom.

Key Points

- **Order matters**: Place more specific rules above general ones.
- **Top-down evaluation**: FortiGate does not scan all rules; it stops at the first match.
- **Best practice**:
  - Put restrictive policies (e.g., deny rules) above broad allow rules.
  - Group related rules logically (LAN→WAN, VPN→LAN, DMZ→WAN).
  - Control policy evaluation order by arranging policies in the policy list. Policy IDs uniquely identify policies but do not determine their matching order. Place more specific policies above broader policies so the intended rule is matched first.

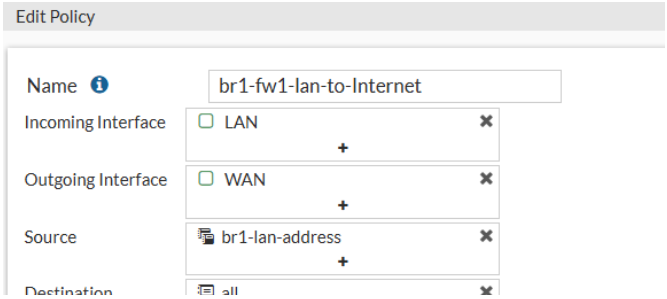
10.6 Firewall Policy Creation

The following steps and GUI screenshot illustrate the security policy configuration process. Shown only for BR1-FW1.

Configuration Field Breakdown

- **Name:** Identifies the policy (e.g., br1-fw1-lan-to-Internet); used for administrative clarity and log tracking.
- **Incoming Interface:** Specifies where traffic originates (e.g. sales-lan, prod-lan, hr-lan). These VLANs represent internal networks that will access the Internet.
- **Outgoing Interface:** Defines where traffic exits (e.g. WAN). This is the external interface connected to the ISP or upstream router.
- **Source:** Identifies the internal address object or subnet (e.g. br1-lan-address). Determines which internal hosts are allowed to use this policy.
- **Destination:** Specifies the target network or address object (e.g. all). Here, "all" means unrestricted Internet access.
- **Schedule:** Defines when the policy is active (e.g. always). Useful for time-based access control.
- **Service:** Determines which protocols or ports are allowed (e.g. ALL). Can be restricted to HTTP, HTTPS, DNS, etc., for tighter control.
- **Action:** Sets the traffic behavior – ACCEPT to permit or DENY to block. Here, traffic is allowed from LAN to Internet.
- **Inspection Mode:** Chooses how traffic is analyzed: Flow-based for faster performance, or Proxy-based for deeper content inspection.
- **NAT:** Enables Network Address Translation so internal private IPs are converted to public IPs before leaving the firewall.
- **IP Pool Configuration:** Defines how NAT uses external IPs: Use Outgoing Interface Address → single public IP; Use Dynamic IP Pool → multiple public IPs (e.g. br1-fw1-ip-pool-snat) for load distribution.
- **Preserve Source Port:** When disabled, FortiGate can remap source ports for efficiency; enabling it keeps original ports for compatibility.
- **Protocol Options:** Applies predefined inspection profiles (e.g. default) for handling specific protocols like FTP, SIP, or DNS.

GUI PATH: **Policy & Object --> Firewall Policy --> Create New**



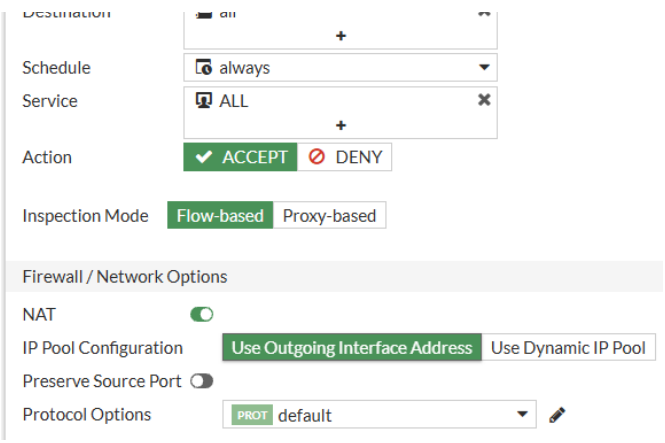


Figure: BR1-FW1 Configure Policy (continued)

We can configure Policy by CLI using the configuration below. (Note-We will use outgoing interface for SNAT, in next section we will see SNAT in detail)

BR1-FW1 Firewall Policy

```
config vdom
 edit FG-traffic
  config firewall policy
   edit 1
    set name br1-fw1-lan-to-Internet
    set srcintf LAN
    set dstintf WAN
    set action accept
    set srcaddr br1-lan-address
    set dstaddr all
    set schedule always
    set service ALL
    set logtraffic all
    set nat enable
   next
  end
 next
end
```

10.7 Verification of Firewall Policy

10.7.1 Policy Verification

- Run show firewall policy (Policy-ID) to confirm the policy exists.

BR1-FW1 Policy Verification

```
BR1-FW1 (FG-traffic) # show firewall policy 1
config firewall policy
        set name "br1-fw1-lan-to-Internet"
        set srcintf "LAN"
        set dstintf "WAN"
        set action accept
        set srcaddr "br1-lan-address"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
```

```
        set logtraffic all
        set nat enable
    next
end
```

- Ensure correct source/destination interfaces, addresses, and services.
- Verify that set nat enable is configured.

10.7.2 Traffic Flow Verification

- Use diagnose firewall iprope lookup <src-ip> <src-port> <dst-ip> <dst-port> <protocol> <interface> to check policy match.

BR1-FW1 matching Policy

```
BR1-FW1 (FG-traffic) # diagnose firewall iprope lookup 192.168.21.3 80 8.8.8.8 80 6 vlan10
<src [192.168.21.3-80] dst [8.8.8.8-80] proto 6 dev vlan10> matches policy id: 1
```

- Run diagnose sys session list to confirm active sessions and policy ID.

BR1-FW1 session list

```
BR1-FW1 (FG-traffic) # diagnose sys session list
session info: proto=6 proto_state=01 duration=16 expire=3583 timeout=3600 flags
=00000000 socktype=0 sockport=0 av_idx=0 use=3
tx speed(Bps/kbps): 809/6 rx speed(Bps/kbps): 24722/197
orgin->sink: org pre->post, reply pre->post dev=28->7/7->28 gwy
=203.0.113.9/192.168.21.3
hook=post dir=org act=snat 192.168.21.3:42414->151.101.131.19:443(203.0.113.10:42414)
hook=pre dir=reply act=dnat 151.101.131.19:443->203.0.113.10:42414(192.168.21.3:42414)
!
output omitted
!
```

Connectivity Testing

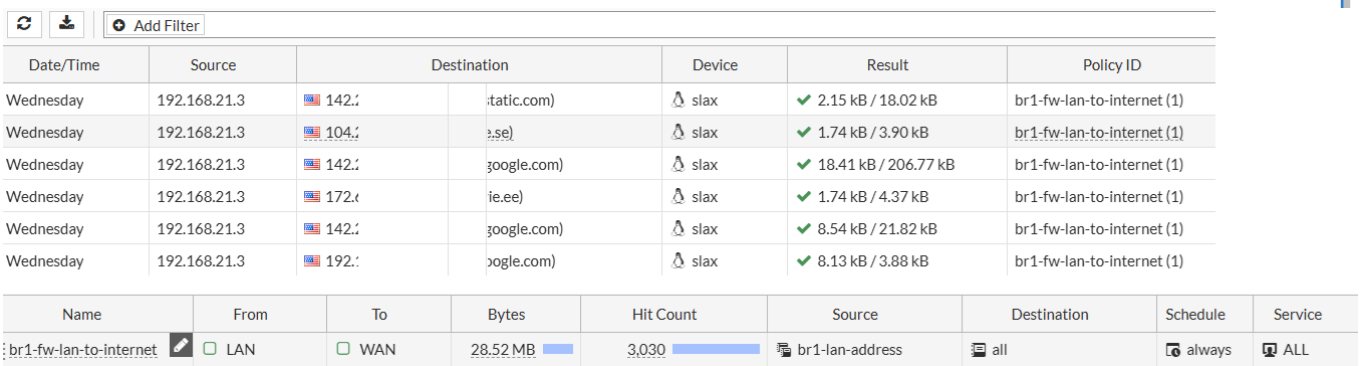
- From a LAN host, test with ping 8.8.8.8.
- Access a website
- On FortiGate, capture traffic with diagnose sniffer packet any 'host 8.8.8.8' 4.

BR1-FW1 capture traffic

```
BR1-FW1 (FG-traffic) # diagnose sniffer packet any 'host 8.8.8.8' 4
Using Original Sniffing Mode
interfaces=[any]
filters=[host 8.8.8.8]
15.393292 vlan10 in 192.168.21.3 -> 8.8.8.8: icmp: echo request
15.393455 port5 out 203.0.113.10 -> 8.8.8.8: icmp: echo request
15.402882 port5 in 8.8.8.8 -> 203.0.113.10: icmp: echo reply
15.402928 vlan10 out 8.8.8.8 -> 192.168.21.3: icmp: echo reply
15.402929 Lag-1 out 8.8.8.8 -> 192.168.21.3: icmp: echo reply
15.402931 port3 out 8.8.8.8 -> 192.168.21.3: icmp: echo reply
```

10.7.3 Log Verification

- Navigate to **Log & Report → Forward Traffic**.
- Confirm that traffic is hitting the intended policy.



Add Filter

Date/Time	Source	Destination	Device	Result	Policy ID
Wednesday	192.168.21.3	142.: :tatic.com)	slax	2.15 kB / 18.02 kB	br1-fw-lan-to-internet (1)
Wednesday	192.168.21.3	104.: :.se)	slax	1.74 kB / 3.90 kB	br1-fw-lan-to-internet (1)
Wednesday	192.168.21.3	142.: :google.com)	slax	18.41 kB / 206.77 kB	br1-fw-lan-to-internet (1)
Wednesday	192.168.21.3	172.(:ie.ee)	slax	1.74 kB / 4.37 kB	br1-fw-lan-to-internet (1)
Wednesday	192.168.21.3	142.: :google.com)	slax	8.54 kB / 21.82 kB	br1-fw-lan-to-internet (1)
Wednesday	192.168.21.3	192.: :ogle.com)	slax	8.13 kB / 3.88 kB	br1-fw-lan-to-internet (1)

Name	From	To	Bytes	Hit Count	Source	Destination	Schedule	Service
br1-fw-lan-to-internet	LAN	WAN	28.52 MB	3,030	br1-lan-address	all	always	ALL

Figure: BR1-FW1 Policy Verification

10.8 Policy and Object Troubleshooting Scenario

10.8.1 Traffic Not Matching Firewall Policy

- **Symptoms:** Users cannot access the internet even though a policy exists. They report no internet connectivity.
- **Troubleshooting Focus:**
  - Verify **source/destination interfaces** in the LAN→WAN policy are correct.
  - Confirm the **policy is enabled** and placed above any deny rules.
  - Check **address objects** – the LAN subnet must match actual client IPs; the destination should be all.
  - Ensure **service objects** include the required protocols (DNS, HTTP, HTTPS).
  - Validate that **NAT** is enabled in the policy (SNAT or IP pool).
  - Confirm the **default route (0.0.0.0/0)** points to the WAN gateway.
  - Check that the **WAN interface** has a valid IP address and upstream connectivity.
  - Verify **DNS resolution** works from the FortiGate CLI.
  - Ensure **no overlapping policies** are shadowing the LAN→WAN rule.

★ **Verification Command**

- show firewall policy
- show firewall address/addrgrp
- get router info routing-table all
- diagnose debug flow filter addr <lan-ip>
- diagnose debug flow trace start 50

Observation

```
BR1-FW1 (FG-traffic) # id=20085 trace_id=144 func=print_pkt_detail line=5824 msg="
vd-FG-traffic:0 received a packet(proto=1, 192.168.21.3:14167->8.8.8.8:2048) tun_id
```

```
=0.0.0.0 from vlan10. type=8, code=0, id=14167, seq=1."
id=20085 trace_id=144 func=init_ip_session_common line=6003 msg="allocate a new
session-00005ac4, tun_id=0.0.0.0"
id=20085 trace_id=144 func=vf_ip_route_input_common line=2604 msg="find a route: flag
=04000000 gw-203.0.113.9 via port5"
id=20085 trace_id=144 func=fw_forward_handler line=874 msg="Allowed by Policy-1: SNAT"
!
id=20085 trace_id=143 func=print_pkt_detail line=5824 msg="vd-FG-traffic:0 received a
packet(proto=6, 192.168.21.3:53136->142.251.153.119:443) tun_id=0.0.0.0 from vlan10.
flag [S], seq 216015635, ack 0, win 29200"
id=20085 trace_id=143 func=init_ip_session_common line=6003 msg="allocate a new
session-00005a76, tun_id=0.0.0.0"
id=20085 trace_id=143 func=vf_ip_route_input_common line=2604 msg="find a route: flag
=04000000 gw-203.0.113.9 via port5"
id=20085 trace_id=143 func=fw_forward_handler line=712 msg="Denied by forward policy
check (policy 0)"
!
BR1-FW1 (FG-traffic) # sh firewall policy 1
    set name "br1-fw1-lan-to-internet"
    set srcintf "LAN"
    set dstintf "WAN"
    set service "ALL_ICMP" "FTP" "DNS"
    !
    BR1-FW1 (FG-traffic) # sh firewall addrgrp br1-lan-address
        edit "br1-lan-address"
            set member br1-vlan20-prod br1-vlan30-hr
            set comment br1-lan-address
        next
    end
```

- **Root Cause:**
  - The firewall policy **br1-fw1-lan-to-internet** references the source object **br1-lan-address**.
  - That object is an **address group** containing only:
    - 192.168.22.0/24 (Prod VLAN)
    - 192.168.23.0/24 (HR VLAN)
  - The actual traffic source is **192.168.21.3 (VLAN10)**.
  - Since **192.168.21.0/24** is not included in the source object, the packet does not match the policy.
  - Debug confirms: *Denied by forward policy check (policy 0)* → no matching policy found.
- **Fix Note:** Update the source object so it includes the VLAN10 subnet.

Observation

```
BR1-FW1 (FG-traffic) # diagnose debug flow filter addr 192.168.21.3
BR1-FW1 (FG-traffic) # diagnose debug flow trace start 50
id=20085 trace_id=18 func=print_pkt_detail line=5824 msg="vd-FG-traffic:0 received a
packet(proto=1, 192.168.21.3:13843->8.8.8.8:2048) tun_id=0.0.0.0 from vlan10. type=8,
code=0, id=13843, seq=1."
id=20085 trace_id=18 func=init_ip_session_common line=6003 msg="allocate a new
session-00005164, tun_id=0.0.0.0"
id=20085 trace_id=18 func=vf_ip_route_input_common line=2604 msg="find a route: flag
=04000000 gw-203.0.113.9 via port5"
id=20085 trace_id=18 func=fw_forward_handler line=874 msg="Allowed by Policy-1: SNAT"
```

10.8.2 LAN Users Can Ping Internet but Cannot Browse Webpages

- **Symptoms:** LAN clients can successfully ping external IPs (e.g., 8.8.8.8). Web browsing (HTTP/HTTPS) fails – pages do not load.
- **Troubleshooting Focus:**
  - Confirm service objects include DNS, HTTP, and HTTPS.
  - Verify NAT configuration – ensure SNAT or IP pool is correctly applied.
  - Validate default route (0.0.0.0/0) points to the WAN gateway.
  - Ensure WAN interface has a valid public IP and upstream connectivity.

> ★ **Verification Command**
>
> - show firewall policy
> - diagnose debug flow filter addr <lan-ip>
> - diagnose debug flow trace start 50

Observation

```
BR1-FW1 (FG-traffic) # show firewall policy 1
config firewall policy
    edit 1
        set name "br1-fw1-lan-to-Internet"
        set srcintf "LAN"
        set dstintf "WAN"
        set service "ALL_ICMP" "DNS" "FTP"
        set logtraffic all
!
BR1-FW1 (FG-traffic) # id=65308 trace_id=1 func=print_pkt_detail line=5895 msg="
    vd-FG-traffic:0 received a packet(proto=1, 192.168.21.2:512->8.8.8.8:2048) tun_id
    =0.0.0.0 from vlan10. type=8, code=0, id=512, seq=256."
id=65308 trace_id=1 func=init_ip_session_common line=6076 msg="allocate a new
    session-00000b62, tun_id=0.0.0.0"
id=65308 trace_id=1 func=vf_ip_route_input_common line=2605 msg="find a route: flag
    =04000000 gw-203.0.113.9 via port5"
id=65308 trace_id=1 func=fw_forward_handler line=903 msg="Allowed by Policy-1: SNAT"
id=65308 trace_id=1 func=__ip_session_run_tuple line=3502 msg="SNAT 192.168.21.2-
    >203.0.113.65:5120"
id=65308 trace_id=2 func=print_pkt_detail line=5895 msg="vd-FG-traffic:0 received a packet
    (proto=1, 8.8.8.8:5120->203.0.113.65:0) tun_id=0.0.0.0 from port5. type=0, code=0, id
    =5120, seq=256."
!
id=65308 trace_id=19 func=print_pkt_detail line=5895 msg="vd-FG-traffic:0 received a
    packet(proto=6, 192.168.21.2:1054->151.101.209.91:443) tun_id=0.0.0.0 from vlan10.
    flag [S], seq 3932854848, ack 0, win 65535"
id=65308 trace_id=19 func=init_ip_session_common line=6076 msg="allocate a new
    session-00000b73, tun_id=0.0.0.0"
id=65308 trace_id=19 func=vf_ip_route_input_common line=2605 msg="find a route: flag
    =04000000 gw-203.0.113.9 via port5"
id=65308 trace_id=19 func=fw_forward_handler line=738 msg="Denied by forward policy check
    (policy 0)"
```

- **Root Cause:** Policy service list missing HTTP/HTTPS. ICMP and DNS were allowed, but web traffic was denied by forward policy check.

- **Fix Note:** Add HTTP/HTTPS services to the LAN→WAN policy.

Observation

```
BR1-FW1 (FG-traffic) # diagnose debug flow filter addr 192.168.21.3
BR1-FW1 (FG-traffic) # diagnose debug flow trace start 50
id=20085 trace_id=204 func=print_pkt_detail line=5824 msg="vd-FG-traffic:0 received a
packet(proto=6, 192.168.21.3:37004->142.251.150.119:443) tun_id=0.0.0.0 from vlan10.
flag [S], seq 2497089478, ack 0, win 29200"
id=20085 trace_id=204 func=init_ip_session_common line=6003 msg="allocate a new
session-00005efb, tun_id=0.0.0.0"
id=20085 trace_id=204 func=vf_ip_route_input_common line=2604 msg="find a route: flag
=04000000 gw-203.0.113.9 via port5"
id=20085 trace_id=204 func=fw_forward_handler line=874 msg="Allowed by Policy-1: SNAT"
!
BR1-FW1 (FG-traffic) # show firewall policy 1
set srcintf "LAN"
set dstintf "WAN"
set service "ALL_ICMP" "FTP" "DNS" "HTTP" "HTTPS"
```

11 Source NAT

In this section, we will cover the following objectives.

Objective

- Understand FortiGate SNAT Concepts.
- Configure NAT using Interface address overload, ip pool overload, one-to-one, Fixed Port and PBA NAT methods.
- verification of NAT translations.

In FortiGate, NAT can be configured in two different ways:

- **Policy-based NAT** is the simpler approach where NAT is enabled directly inside a firewall policy. When you set nat enable, the firewall automatically translates source addresses using the outgoing interface IP or an IP pool if defined. This is quick to configure and works well for straightforward deployments where NAT is tied to specific rules.
- **Central NAT** separates NAT from firewall policies and manages it in a dedicated NAT table. Policies reference these central rules, allowing more flexible source/destination translations, many-to-many mappings, and consistent NAT behavior across multiple policies. It's better suited for complex or large-scale environments.

In FortiGate, there are two main types of NAT (Network Address Translation):

- **SNAT (Source NAT):** Internal → External (outbound traffic).
- **DNAT (Destination NAT):** External → Internal (inbound traffic).

11.1 IP Pool

In FortiGate, SNAT (Source NAT) and IP Pools are closely related concepts – IP Pools are essentially a more flexible way of performing SNAT. An IP Pool is a defined range of public IP addresses that FortiGate can use for SNAT instead of just the interface IP. Use the following IP Pool for SNAT configuration.

Device	IP Pool Range
BR1-FW1	203.0.113.64/29

Table 11: IP Pool Ranges

Configuration Field Breakdown

- **Name:** Identifies the IP pool object (e.g. br1-fw1-ip-pool-snat). Used for reference in firewall policies and NAT configurations.
- **Comments:** Optional description for administrative clarity. Helps document the pool's purpose or associated interface.
- **Type:** Defines how source NAT (SNAT) is performed:
  - Overload: Many internal hosts share one or more public IPs using port translation (PAT).
  - One-to-One: Each internal IP maps directly to a unique external IP.
  - Fixed Port Range: Assigns specific port ranges per internal host.

  - Port Block Allocation: Allocates blocks of ports dynamically for scalability.

- **External IP Address/Range:** Specifies the public IPs used for translation. These addresses are assigned by the ISP and represent network externally.
- **NAT64:** Converts IPv6 traffic to IPv4 for compatibility with legacy networks. Disabled here since the configuration is purely IPv4.
- **ARP Reply:** When enabled, FortiGate responds to ARP requests for the IPs in the pool. Ensures proper Layer-2 resolution for devices on the same broadcast domain.

11.2 SNAT (Source NAT)

In FortiGate, SNAT (Source NAT) is the most common NAT method – it translates internal private IP addresses into public IPs when traffic flows outbound to the internet. This ensures multiple devices in the LAN can share a single public IP while still being uniquely identified by their source ports.

> ☞ **Note**
> After each SNAT Method demo we need to delete created IP Pool. If it is not deleted, we will receive the error "This external IP range overlaps with an existing IP pool range".

SNAT Methods in FortiGate

11.2.1 Overload (PAT – Port Address Translation)

- Many internal hosts share a single public IP.
- FortiGate differentiates sessions by assigning unique source ports.
- Default and most common method when public IPs are limited.
- Efficient for large LANs with only one WAN IP.
- Can be implemented using:
  - **Interface IP:** Outbound traffic is translated to the WAN interface IP directly.
    For BR1-FW1 we can create Interface Overload SNAT (PAT) as per below GUI screenshot or CLI Commands.

GUI PATH: **Policy & Object --> Firewall Policy --> br1-fw1-lan-to-internet --> edit**

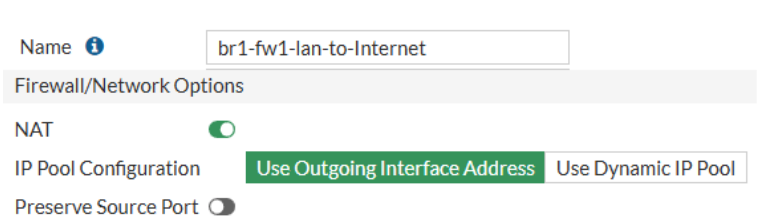


Figure: BR1-FW1 Configure Interface Overload SNAT

BR1-FW1 Configure Interface Overload SNAT

```
config vdom
 edit FG-traffic
  config firewall policy
```

```
    edit 1
     set nat enable
    next
end
```

The GUI screenshot below shows verification of NAT Overload using interface IP – all internal hosts are translated to the WAN interface address (203.0.113.10) for outbound traffic.

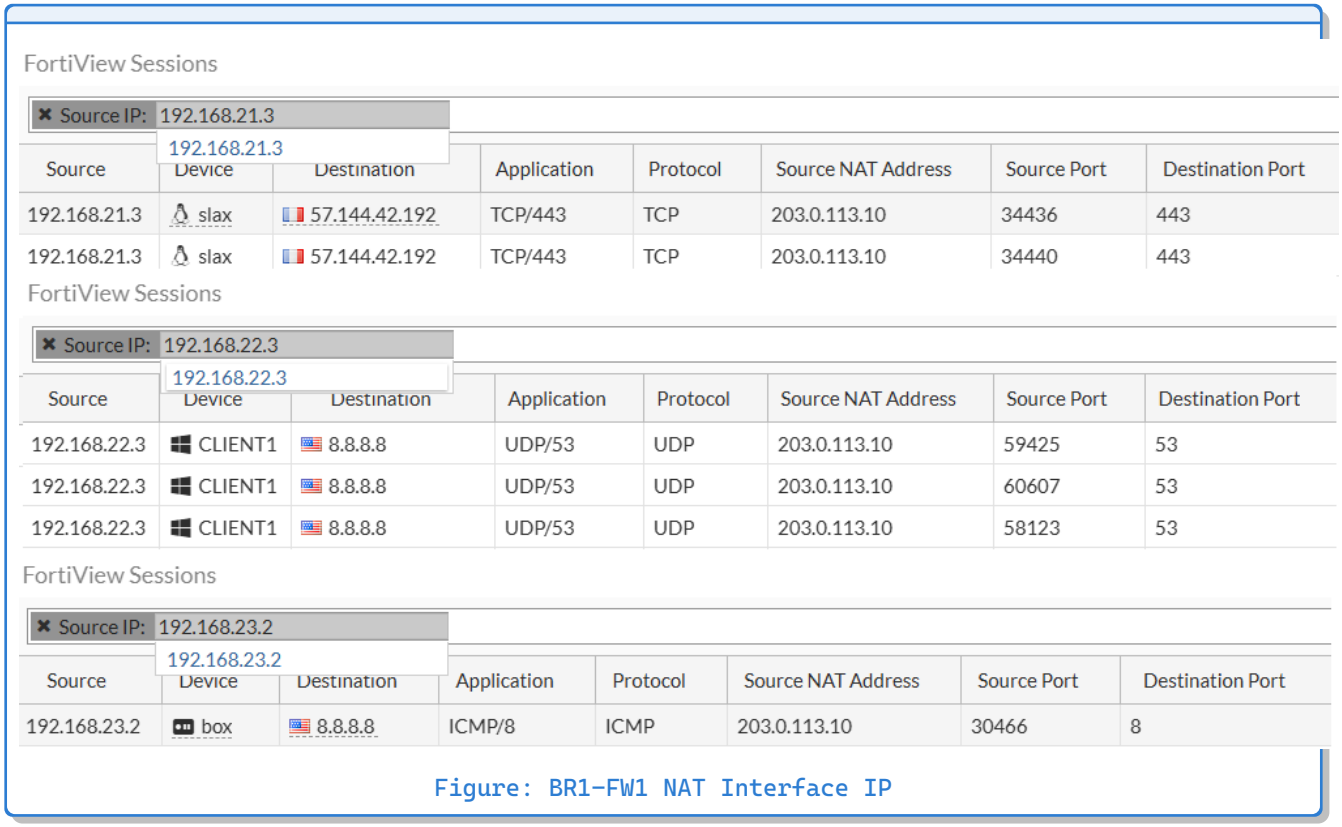


Figure: BR1-FW1 NAT Interface IP

- **IP Pool (Overload type):** A defined public IP or range is used for NAT, still applying port translation.

  For BR1-FW1 we can create IP Pool for Overload SNAT (PAT) as per below GUI screenshot or CLI Commands.

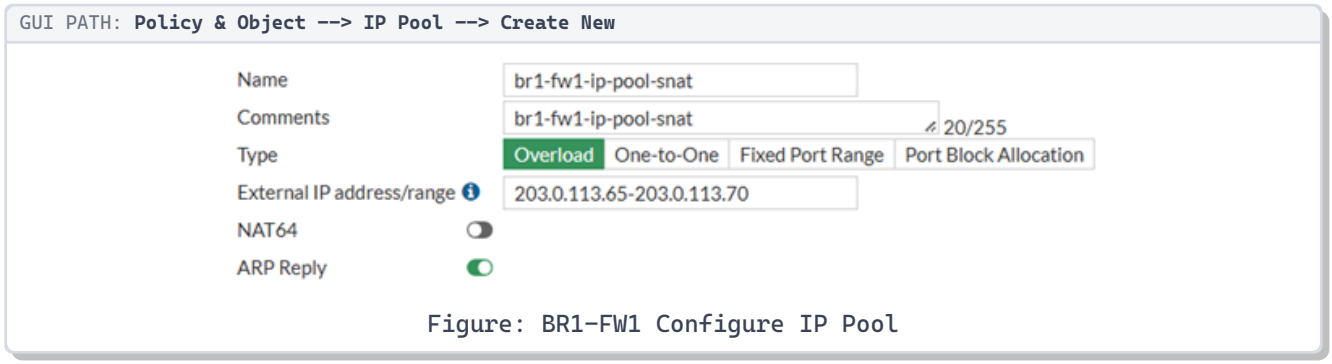


Figure: BR1-FW1 Configure IP Pool

BR1-FW1 Configure IP Pool Overload SNAT

```
config vdom
edit FG-traffic
 config firewall ippool
```

```
  edit br1-fw1-ip-pool-snat
   set startip 203.0.113.65
   set endip 203.0.113.70
   set comments br1-fw1-ip-pool-snat
  next
 end
```

Now we need to call this ip pool inside firewall policy, which can be done via both CLI and GUI as follows.

GUI PATH: **Firewall Policy --> br1-fw1-lan-to-internet --> edit**

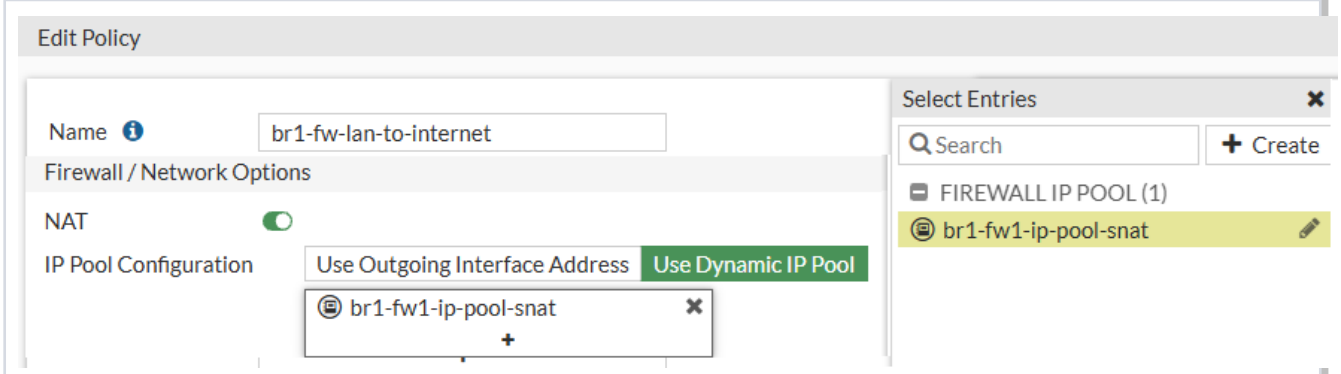
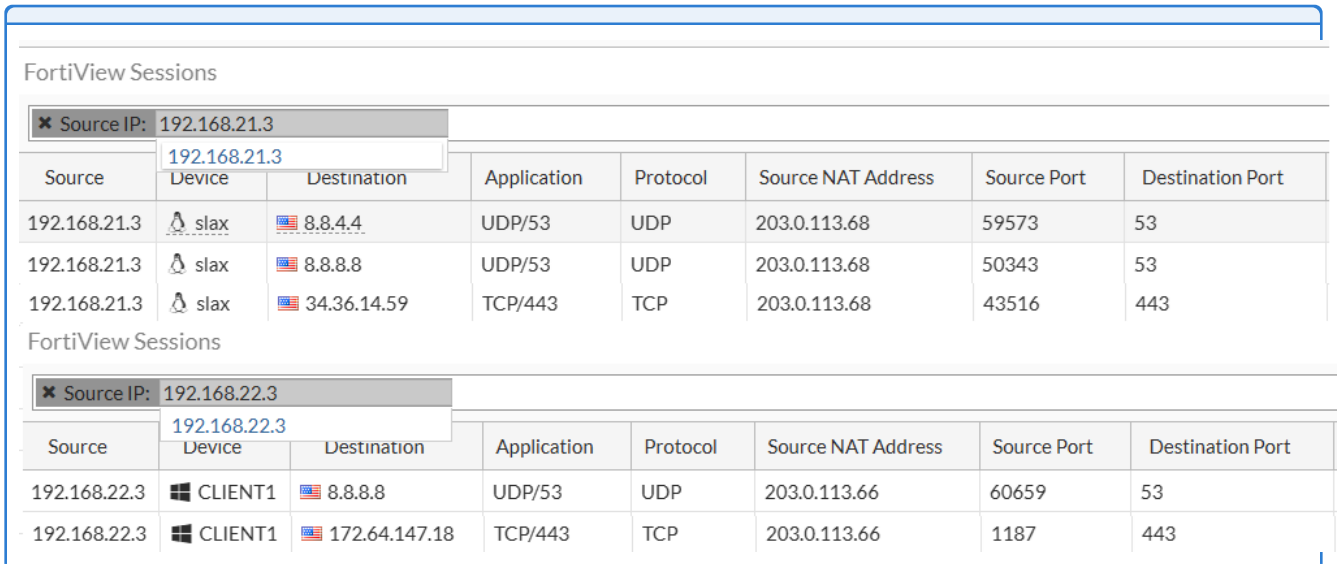


Figure: BR1-FW1 Firewall Policy NAT IP Pool

BR1-FW1 Firewall Policy IP Pool Overload SNAT

```
config vdom
edit FG-traffic
 config firewall policy
  edit 1
   set ippool enable
   set poolname br1-fw1-ip-pool-snat
  next
 end
```

The GUI screenshot below shows verification of NAT Overload using IP Pool – internal hosts are translated to multiple public IPs (203.0.113.65–70) from the configured pool, confirming dynamic SNAT distribution across departments.



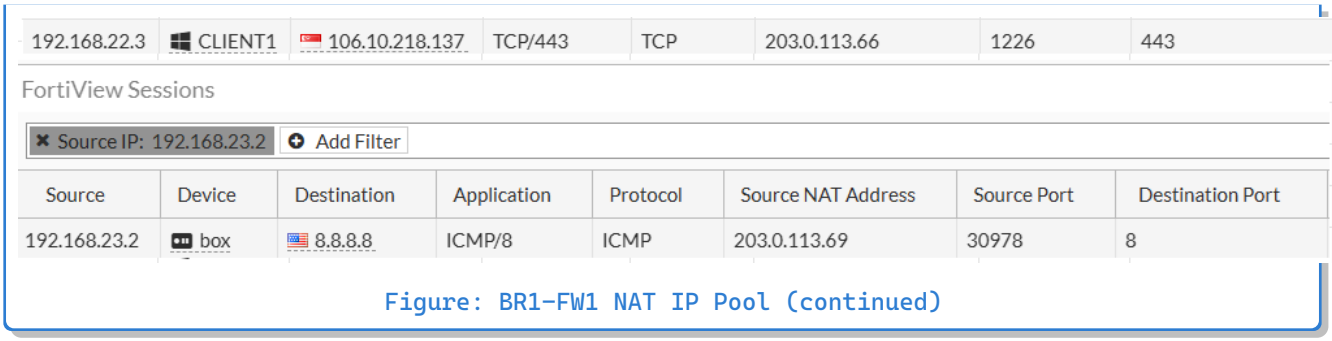


Figure: BR1-FW1 NAT IP Pool (continued)

11.2.2 One-to-One IP Pool

- A One-to-One SNAT (Static NAT) in FortiGate means that each internal private IP address is permanently mapped to a unique public IP address from the IP pool.
- Unlike overload or dynamic NAT, there's no port translation or sharing – it's a dedicated one-to-one mapping.

For BR1-FW1 we can create IP Pool for One-to-One SNAT as per below GUI screenshot or CLI Commands.

GUI PATH: **Policy & Object --> IP Pool --> Create New**

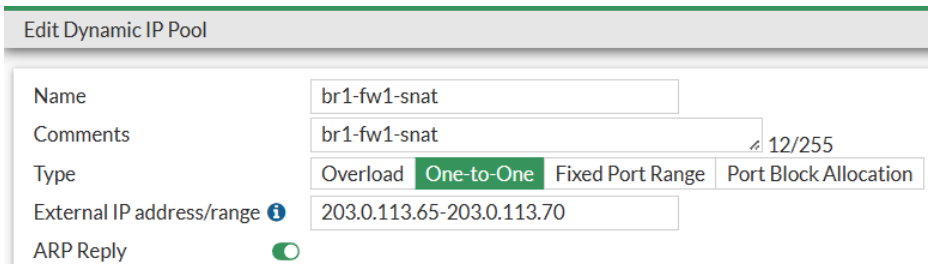


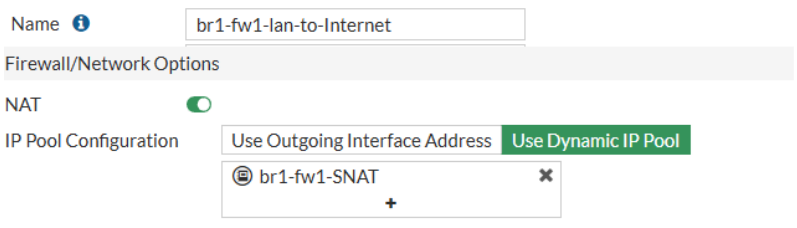
Figure: BR1-FW1 Configure IP Pool one-to-one SNAT

BR1-FW1 Configure IP Pool one-to-one SNAT

```
config vdom
edit FG-traffic
 config firewall ippool
  edit br1-fw1-SNAT
   set type one-to-one
   set startip 203.0.113.65
   set endip 203.0.113.70
   set comments br1-fw1-SNAT
  next
 end
```

Now we need to call this one-to-one ip pool inside firewall policy, which can be done via both CLI and GUI as follows.

GUI PATH: **Firewall Policy --> br1-fw1-lan-to-internet --> edit**



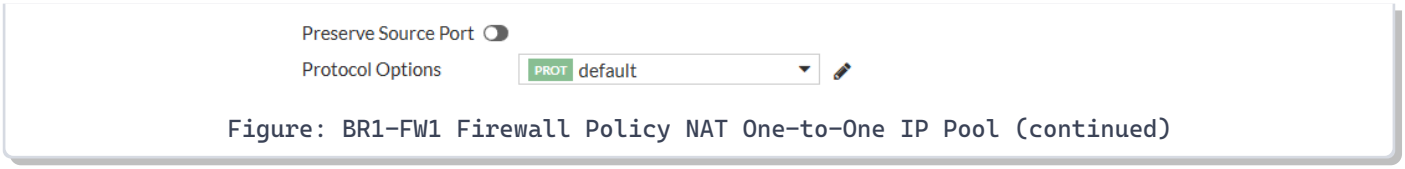


Figure: BR1-FW1 Firewall Policy NAT One-to-One IP Pool (continued)

BR1-FW1 Firewall Policy NAT One-to-One IP Pool

```
config vdom
 edit FG-traffic
  config firewall policy
   edit 1
    set nat enable
    set ippool enable
    set poolname br1-fw1-SNAT
   next
end
```

In the FortiView sessions screenshot below, the one-to-one NAT configuration is clearly visible through the distinct mapping between internal VLAN source IPs and unique public NAT addresses.

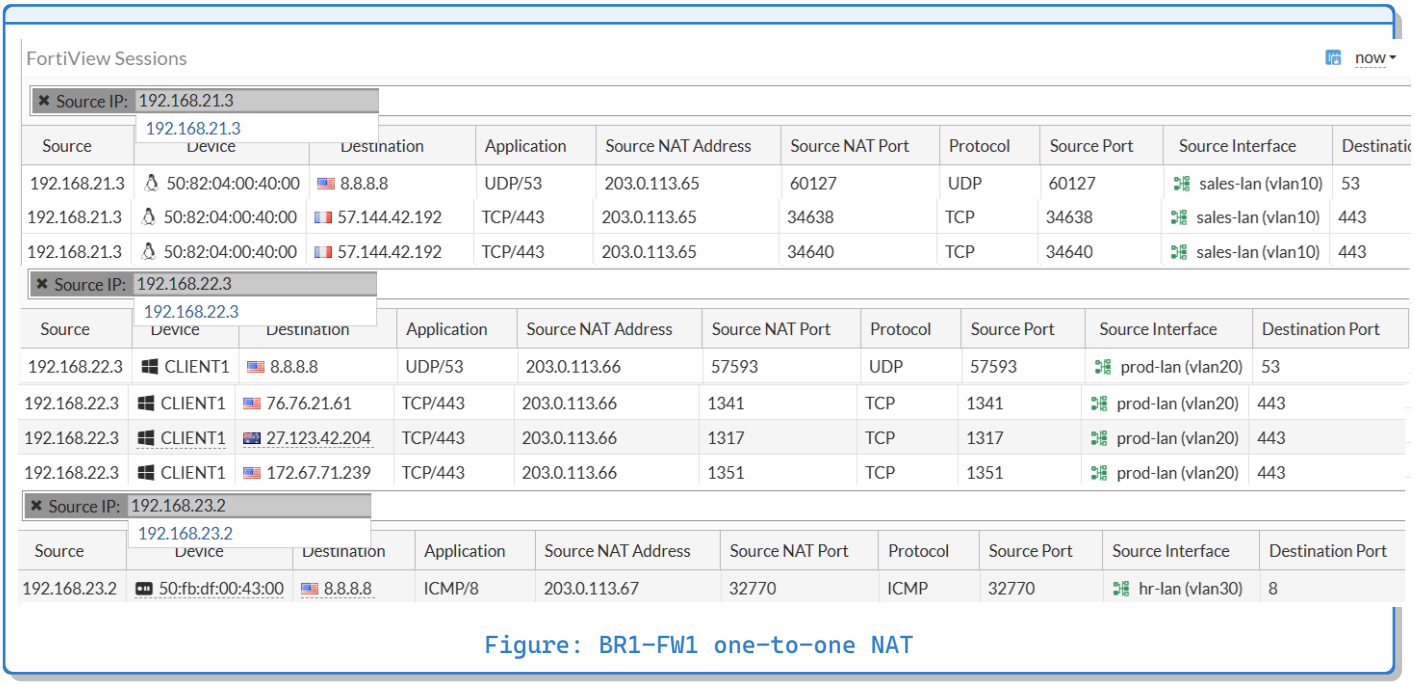


Figure: BR1-FW1 one-to-one NAT

11.2.3 Fixed Port Range

Each internal host is mapped to a fixed range of ports on a public IP. Ensures predictable port usage for specific hosts. Useful when applications require consistent port allocation. For BR1-FW1 we can create IP Pool for Fixed Port Range SNAT as per below GUI screenshot. GUI screenshot shown for vlan10, 20 and CLI for vlan 30.

BR1-FW1 Configure IP Pool Fixedport SNAT VLAN 30

```
config vdom
edit FG-traffic
 config firewall ippool
  edit br1-fw1-snat-fixedport-vlan30
   set type fixed-port-range
```

```
    set startip 203.0.113.67
    set endip 203.0.113.67
    set source-startip 192.168.23.2
    set source-endip 192.168.23.254
    set comments br1-fw1-snat-fixedport-vlan30
   next
  end
```

GUI PATH: **Policy & Object --> IP Pool --> Create New**

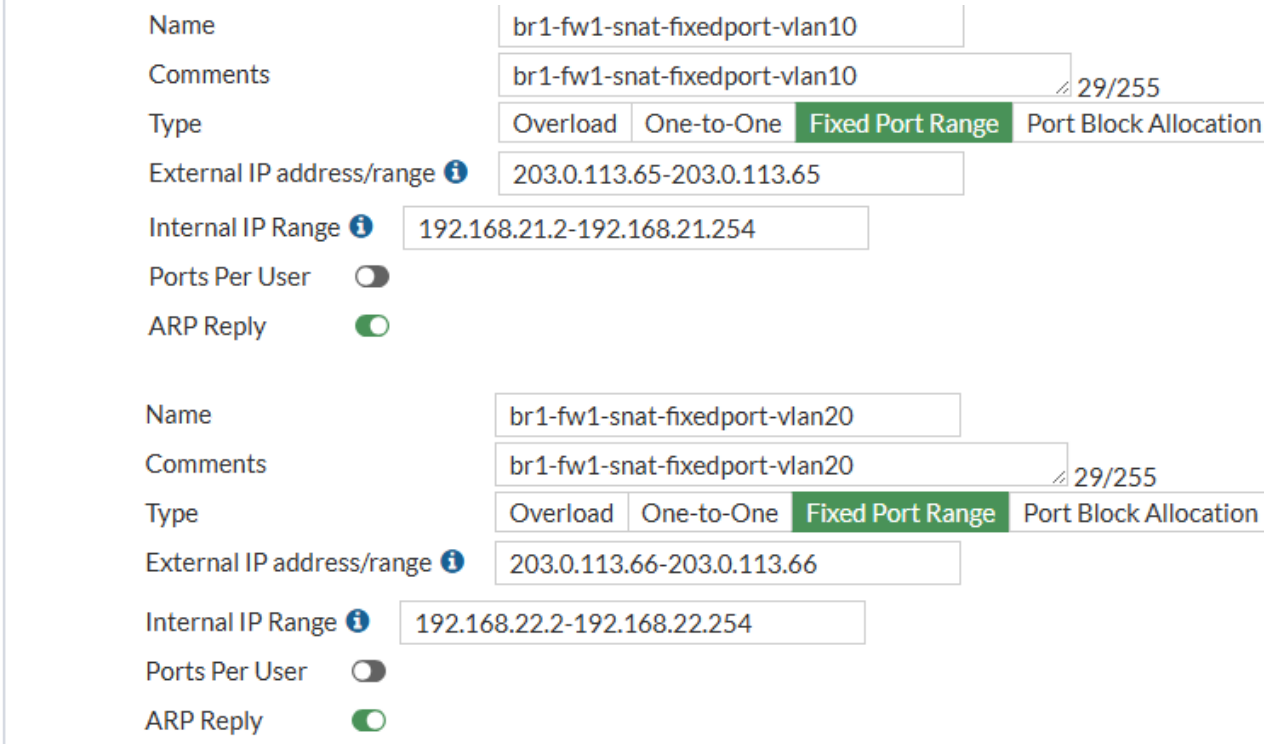


Figure: BR1-FW1 Configure IP Pool Fixedport VLAN 10,20

Apply these create IP pools inside the firewall policy as per following using CLI and GUI method.

BR1-FW1 Firewall Policy NAT Fixed Port

```
config vdom
 edit FG-traffic
  config firewall policy
   edit 1
   set nat enable
    set ippool enable
    set poolname br1-fw1-snat-fixedport-vlan10 br1-fw1-snat-fixedport-vlan20
    br1-fw1-snat-fixedport-vlan30
   next
end
```

GUI PATH: **Firewall Policy --> br1-fw1-lan-to-internet --> edit**

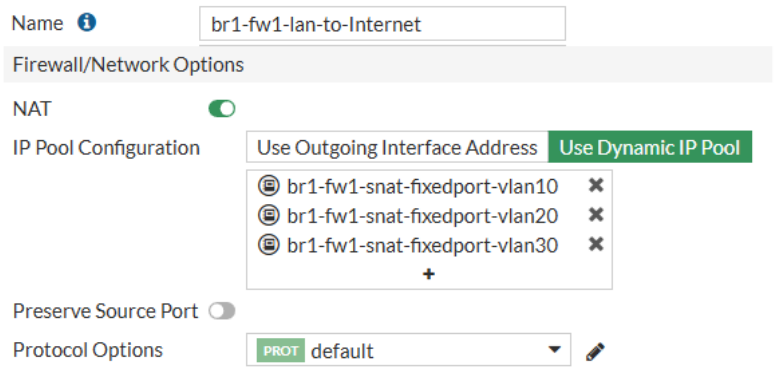


Figure: BR1-FW1 Firewall Policy NAT Fixed Port

11.2.4 Port Block Allocation (PBA)

Allocates a block of ports from a public IP to each internal host. Provides deterministic NAT mapping, reducing port conflicts. Useful in high-density environments where predictable port assignment is critical.

For BR1-FW1 we can create IP Pool for Port Block Allocation (PBA) as per below GUI screenshot.

GUI PATH: **Policy & Object --> IP Pool --> Create New**

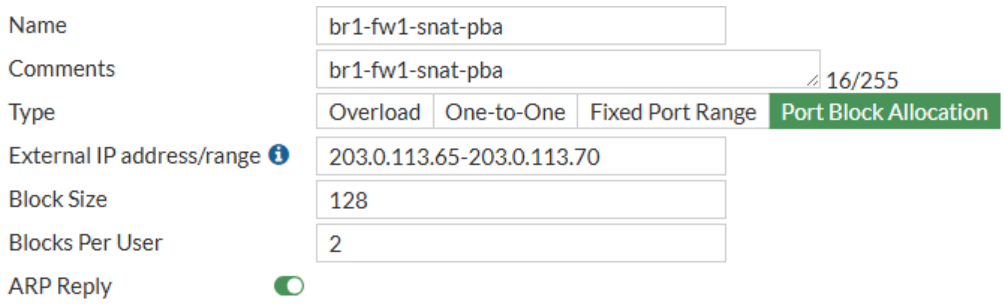


Figure: BR1-FW1 Configure IP Pool Port Block Allocation (PBA)

BR1-FW1 Configure IP Pool Port Block Allocation (PBA)

```
config vdom
edit FG-traffic
 config firewall ippool
  edit br1-fw1-snat-pba
   set type port-block-allocation
   set startip 203.0.113.65
   set endip 203.0.113.70
   set num-blocks-per-user 2
   set comments br1-fw1-snat-pba
  next
 end
```

Apply these create IP pools inside firewall policy as per following GUI or CLI Screenshot.

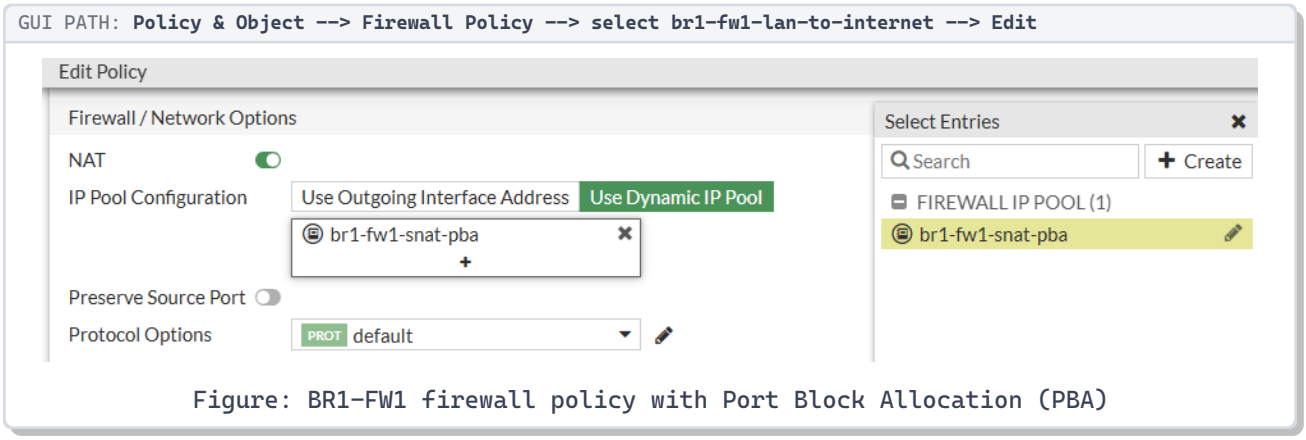


Figure: BR1-FW1 firewall policy with Port Block Allocation (PBA)

BR1-FW1 firewall policy with Port Block Allocation (PBA)

```
config vdom
 edit FG-traffic
  config firewall policy
   edit 1
    set nat enable
    set ippool enable
    set poolname br1-fw1-snat-pba
   next
end
```

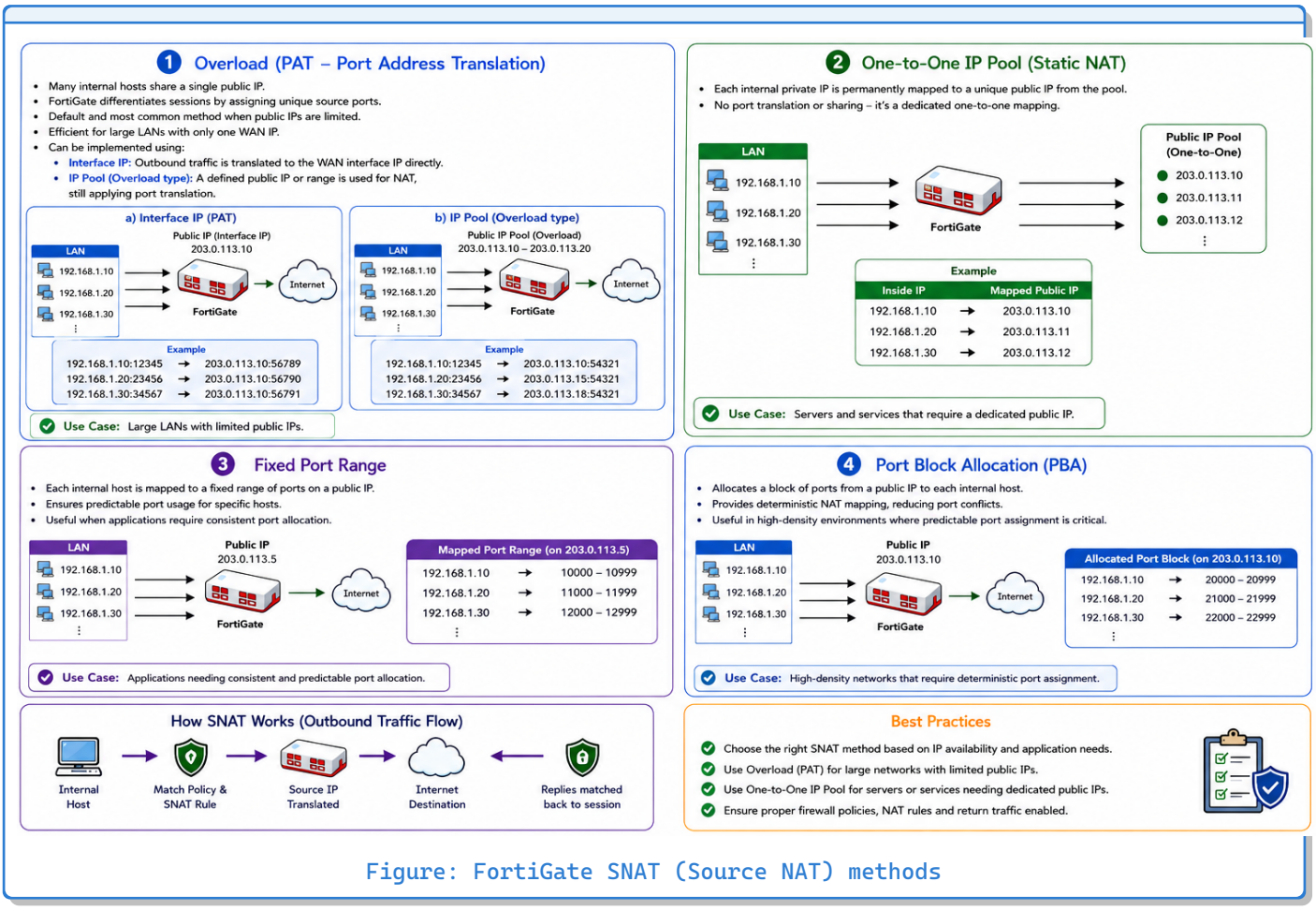


Figure: FortiGate SNAT (Source NAT) methods

The figure above provides an overview of FortiGate SNAT (Source NAT) methods – illustrating Overload (PAT), Dynamic IP Pool, Fixed Port Range, and Port Block Allocation. Each

method shows how internal private IPs are translated to public IPs for outbound traffic, with examples, use cases, and best practices for network design.

11.3 Central SNAT

In FortiGate, Central SNAT provides a centralized method for controlling Source NAT. Instead of configuring source NAT directly inside individual firewall policies, NAT translation rules are maintained separately in the Central SNAT table.

- **Separation of NAT and Security Policy:** Firewall policies determine whether traffic is permitted, while Central SNAT rules independently determine how the source address is translated.
- **Top-Down Rule Evaluation:** Central SNAT rules are evaluated from top to bottom, and the first matching rule is applied.
- **Granular Matching:** Rules can match incoming interface, outgoing interface, source address, destination address, protocol, and source port.
- **IP Pool Selection:** A matching Central SNAT rule can use the outgoing interface address or a configured IP pool for source address translation.
- **Scalability:** Central SNAT is useful in environments where NAT configuration must be managed independently from firewall security policies.

We can enable central NAT via both GUI and CLI. We use following CLI to enable it via CLI.

BR1-FW1 Central SNAT Enable

```
config vdom
 edit FG-traffic
  config system settings
   set central-nat enable
  end
```

This will enable Central SNAT GUI option inside Policy & Object as shown in following GUI screenshot.

GUI PATH: **Policy & Object --> Central SNAT**

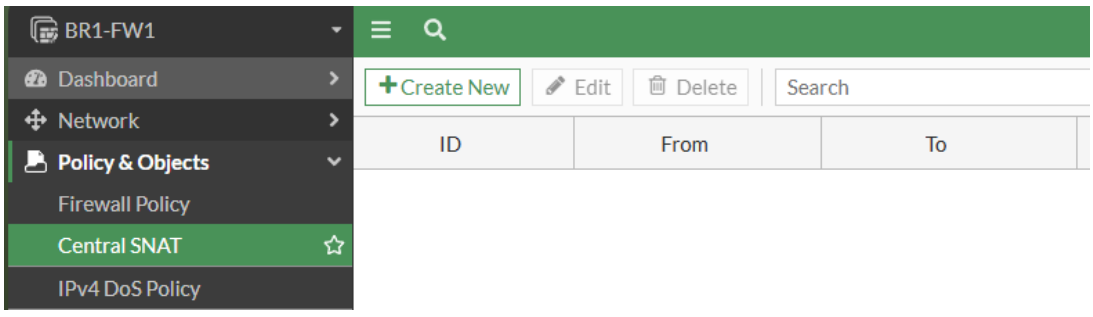


Figure: BR1-FW1 Central SNAT

Now we will create Central SNAT Policy as per following GUI SNAP. Configuration breakdown provide details about why each field configure.

Configuration Field Breakdown

- **Incoming Interface:** Defines where traffic originates (e.g. LAN). FortiGate checks packets entering this interface against the policy.
- **Outgoing Interface:** Specifies where traffic exits (e.g. WAN). Determines the egress path for NAT translation.
- **Source Address:** Identifies internal hosts or subnets initiating connections (e.g. br1-lan-address).
- **Destination Address:** Target network or host; "all" allows access to any external destination.
- **NAT:** Enables Source NAT, translating private source IPs to public ones for outbound traffic.
- **IP Pool Configuration:** Chooses the translation method:
  - **Use Outgoing Interface Address** – SNAT uses the WAN interface's IP.
  - **Use Dynamic IP Pool** – SNAT uses a defined IP pool for load balancing or multi-IP mapping.
- **Protocol:** Defines which protocols are affected (e.g. any, TCP, UDP, SCTP).
- **Explicit Port Mapping:** Allows manual source-to-translated port mapping for advanced NAT control.
- **Enable This Policy:** Activates the rule immediately so traffic matching criteria is processed.

GUI PATH: **Policy & Object --> Central SNAT --> Create New**

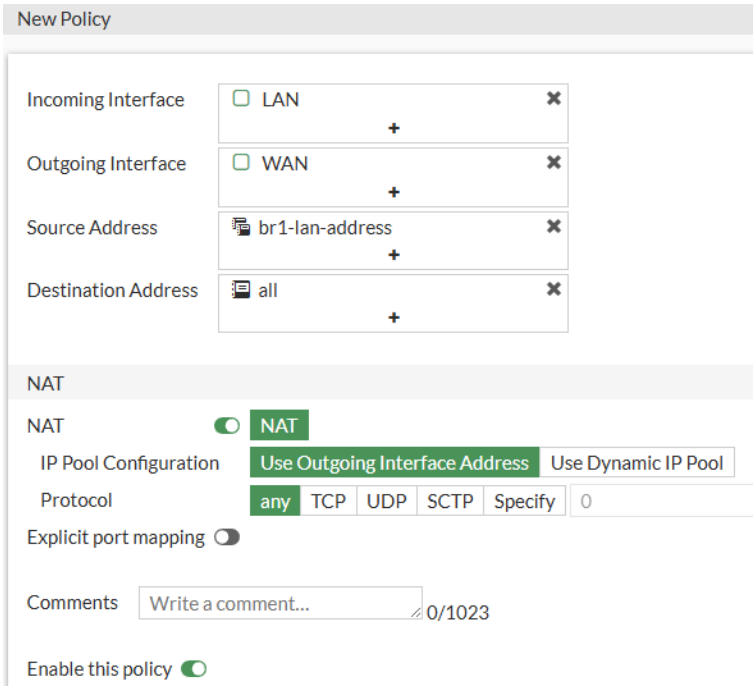


Figure: BR1-FW1 Central SNAT Policy

After this Central SNAT policy creation we will see **Central NAT is enabled so NAT**

settings from matching Central SNAT policies will be applied in the firewall policies. Following are few useful commands for central NAT verification.

★ **Verification Command**

- **Check Central NAT Status:** show system settings | grep central-nat
- **List Central NAT Rules:** show firewall central-snat-map
- **Active SNAT Sessions:** diagnose firewall snat list
- **Session Table Verification:** diagnose sys session list

BR1-FW1 Central SNAT Verification

```
BR1-FW1 (FG-traffic) # show system settings | grep central
    set central-nat enable
!
BR1-FW1 (FG-traffic) # show firewall central-snat-map
config firewall central-snat-map
    edit 1
        set srcintf "LAN"
        set dstintf "WAN"
        set orig-addr "br1-lan-address"
        set dst-addr "all"
    next
end
!
BR1-FW1 (FG-traffic) # diagnose debug enable
BR1-FW1 (FG-traffic) # diagnose debug flow filter addr 192.168.21.2
BR1-FW1 (FG-traffic) # diagnose debug flow trace start 10
BR1-FW1 (FG-traffic) # id=65308 trace_id=1 func=print_pkt_detail line=5895 msg="vd-FG-traffic
    :0 received a packet(proto=1, 192.168.21.2:512->8.8.8.8:2048) tun_id=0.0.0.0 from vlan10.
     type=8, code=0, id=512, seq=2048."
id=65308 trace_id=1 func=init_ip_session_common line=6076 msg="allocate a new
    session-00002309, tun_id=0.0.0.0"
id=65308 trace_id=1 func=vf_ip_route_input_common line=2605 msg="find a route: flag=04000000
    gw-203.0.113.9 via port5"
id=65308 trace_id=1 func=get_new_addr line=1228 msg="find DNAT: IP-203.0.113.10, port-60928"
id=65308 trace_id=1 func=fw_forward_handler line=903 msg="Allowed by Policy-1: SNAT"
id=65308 trace_id=1 func=__ip_session_run_tuple line=3502 msg="SNAT 192.168.21.2-
    >203.0.113.10:60928"
 8.8.8.8:60928->203.0.113.10:0) tun_id=0.0.0.0 from port5. type=0, code=0, id=60928, seq
    =2048."
id=65308 trace_id=2 func=resolve_ip_tuple_fast line=5983 msg="Find an existing session,
    id-00002309, reply direction"
id=65308 trace_id=2 func=__ip_session_run_tuple line=3515 msg="DNAT 203.0.113.10:0-
    >192.168.21.2:512"
id=65308 trace_id=2 func=vf_ip_route_input_common line=2605 msg="find a route: flag=00000000
    gw-192.168.21.2 via vlan10"
id=65308 trace_id=2 func=npu_handle_session44 line=1173 msg="Trying to offloading session
    from port5 to vlan10,
id=65308 trace_id=2 func=fw_forward_dirty_handler line=414 msg="state=00000204, state2
    =00000001, npu_state=00000100"
```

12 Initial Configuration Data-Center

In this section, we will cover the following objectives.

> ➲ **Objective**
>
> - Topology and IP address for Data Center firewall.
> - **Build this yourself** exercise.

12.1 Data Center Topology and IP Addressing Table

We will use the following topology in this lab demonstration.

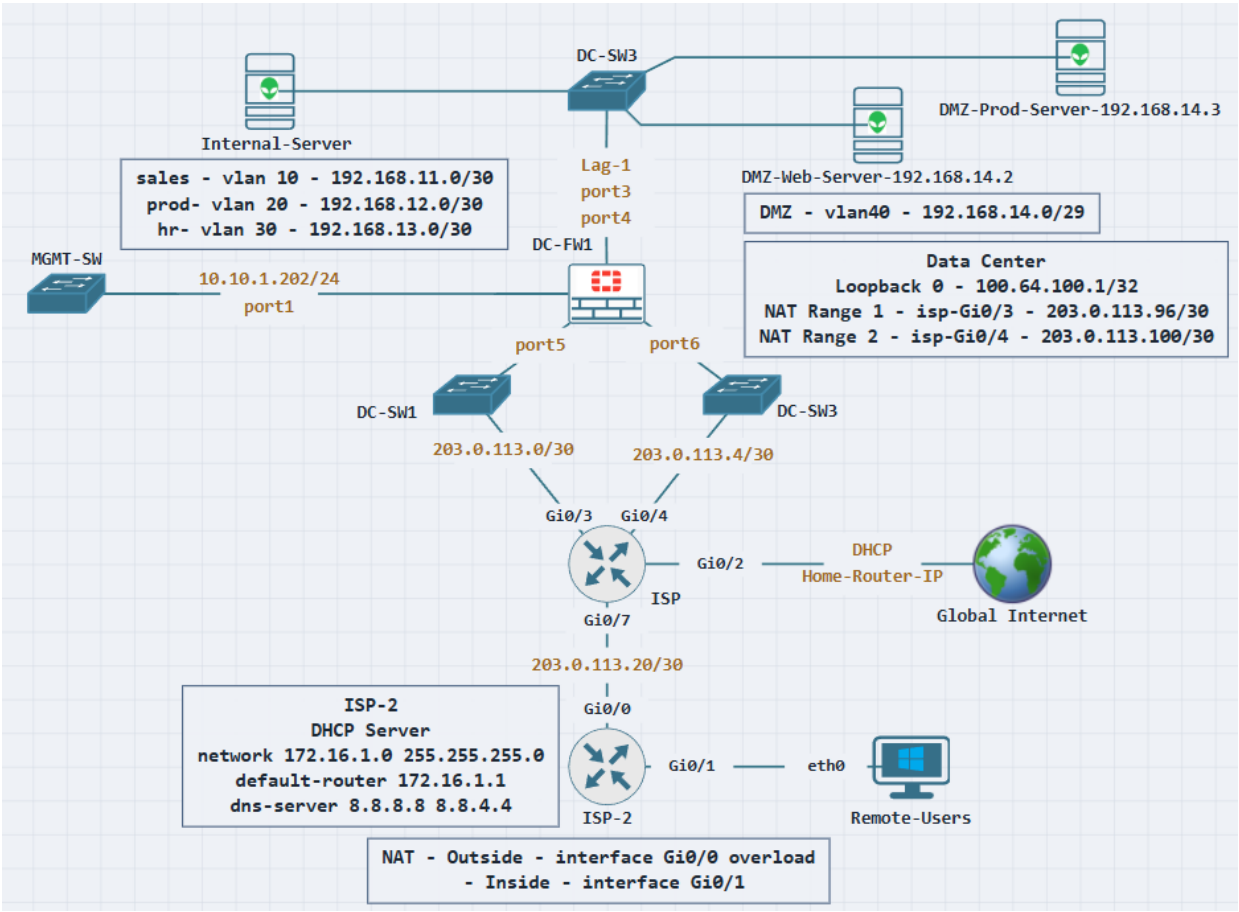


Figure: Data Center Topology

> ☞ **Note**
>
> Configure the Data Center firewall as per following sequence. We can take reference of BR1-FW1 for DC-FW1 Configuration.
>
> - **VDOM Configuration** – Create and manage Virtual Domains (VDOMs) for multi-tenant or lab segmentation.
> - **Zone Configuration** – Define logical zones (WAN, LAN, DMZ) and group interfaces for simplified policy referencing.
> - **Interface Configuration** – Physical Interfaces: Assign IPs, link status, and role.

– VLAN Interfaces: Create sub-interfaces for segmentation. – Aggregate Interfaces: Configure LACP bundles for redundancy and throughput.

- **Address Objects and Groups** – Define subnet address objects. – Group related addresses for simplified policy use.
- **Firewall Policy with Interface IP Overload SNAT** – Configure LAN → WAN policy. – Apply source NAT using interface IP overload (PAT).

We will use the following IP address in this lab demonstration.

Device	Interface	IP Address / Subnet	Description
DC-FW1	port1	10.10.1.202/24	Management network
Sales server		192.168.11.2/30	Sales server
Prod server		192.168.12.2/30	Prod server
HR server		192.168.13.2/30	HR server
DC-FW1	vlan10	192.168.11.1/30	Sales
DC-FW1	vlan20	192.168.12.1/30	Prod
DC-FW1	vlan30	192.168.13.1/30	HR
DC-FW1	vlan40	192.168.14.1/29	DC-FW1 DMZ
DMZ Web server	vlan40	192.168.14.2/29	DC-FW1 DMZ Web server
DMZ Prod server	vlan40	192.168.14.3/29	DC-FW1 DMZ Prod server
DC-FW1	Loopback0	100.64.100.1/32	Data center loopback
DC-FW1	port5	203.0.113.2/30	WAN Link to ISP-Gi0/3
DC-FW1	port6	203.0.113.6/30	WAN Link to ISP-Gi0/4
DC-FW1	–	203.0.113.96/30	Public NAT pool 1
DC-FW1	–	203.0.113.100/30	Public NAT pool 2
DC-FW1	port3-4 (Lag-1)	–	Po1 to DC-SW3

Table 12: Data Center IP Addressing Summary

We can do the verification for DC-FW1 configuration as per following.

DC-FW1 Verification commands

```
DC-FW1 # config vdom
DC-FW1 (vdom) # edit
<vdom>    Virtual Domain Name
FG-traffic
root
!
DC-FW1 (FG-traffic) # show system zone
config system zone
    edit "LAN"
        set interface "vlan10" "vlan20" "vlan30"
    next
    edit "WAN"
        set interface "port5" "port6"
    next
    edit "DMZ"
        set interface "vlan40"
    next
end
!
DC-FW1 (FG-traffic) # show system interface
```

```
name     Name.
Lag-1    static    0.0.0.0 0.0.0.0   0.0.0.0 0.0.0.0   up    disable    aggregate   enable
Loopback0    static    0.0.0.0 0.0.0.0   100.64.100.1 255.255.255.255   up    disable    loopback
port1    static    0.0.0.0 0.0.0.0   10.10.1.202 255.255.255.0   up    disable    physical
port5    static    0.0.0.0 0.0.0.0   203.0.113.2 255.255.255.252   up    disable    physical
port6    static    0.0.0.0 0.0.0.0   203.0.113.6 255.255.255.252   up    disable    physical
vlan10    static    0.0.0.0 0.0.0.0   192.168.11.1 255.255.255.252   up    disable    vlan   enable
vlan20    static    0.0.0.0 0.0.0.0   192.168.12.1 255.255.255.252   up    disable    vlan   enable
vlan30    static    0.0.0.0 0.0.0.0   192.168.13.1 255.255.255.252   up    disable    vlan   enable
vlan40    static    0.0.0.0 0.0.0.0   192.168.14.1 255.255.255.248   up    disable    vlan   enable
!
DC-FW1 (FG-traffic) # sh firewall address dc-vlan10-sales
config firewall address
    edit "dc-vlan10-sales"
        set comment "dc-vlan10-sales"
        set subnet 192.168.11.0 255.255.255.252
DC-FW1 (FG-traffic) # sh firewall address dc-vlan20-prod
config firewall address
    edit "dc-vlan20-prod"
        set comment "dc-vlan20-prod"
        set subnet 192.168.12.0 255.255.255.252
DC-FW1 (FG-traffic) # sh firewall address dc-vlan30-hr
config firewall address
    edit "dc-vlan30-hr"
        set comment "dc-vlan30-hr"
        set subnet 192.168.13.0 255.255.255.252
DC-FW1 (FG-traffic) # sh firewall address dc-vlan40-dmz
config firewall address
    edit "dc-vlan40-dmz"
        set comment "dc-vlan40-dmz"
        set subnet 192.168.14.0 255.255.255.248
!
DC-FW1 (FG-traffic) # sh firewall addrgrp dc-lan-address
config firewall addrgrp
    edit "dc-lan-address"
        set member "dc-vlan10-sales" "dc-vlan20-prod" "dc-vlan30-hr"
DC-FW1 (FG-traffic) # sh firewall addrgrp dc-dmz-address
config firewall addrgrp
    edit "dc-dmz-address"
        set member "dc-vlan40-dmz"
!
DC-FW1 (FG-traffic) # sh firewall policy 1
config firewall policy
    edit 1
        set name "dc-lan-to-wan"
        set srcintf "LAN"
        set dstintf "WAN"
        set action accept
        set srcaddr "dc-lan-address"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
        set logtraffic all
        set nat enable
        set comments "dc-lan-to-wan"
    next
end
```

12.2 Data Center WAN Connectivity – Static Route

Now configure static routes on DC-FW1 as per the following table. In the DNAT section we will demonstrate access DMZ servers via ISP as primary and secondary role for the link. Two equal-cost default routes provide redundant WAN connectivity through ISP. The DMZ server uses the FortiGate as its default gateway so return traffic is sent back through the firewall. FortiGate then selects and maintains the appropriate WAN path based on session state, routing, ECMP, and NAT/VIP configuration. The server's default gateway ensures return traffic reaches the FortiGate, but does not by itself guarantee selection of the same ISP path.

Device	Destination Network	Gateway	Interface	AD Value
DC-FW1	0.0.0.0/0	203.0.113.1	port5	10
	0.0.0.0/0	203.0.113.5	port6	10

Table 13: DC-FW1 Static Route Configuration

Configuration Field Breakdown

- **Destination:** Defines the target network or subnet for the route. In this case, 0.0.0.0/0 represents the default route – meaning all traffic not matching other routes will be forwarded to the next hop (Internet).
- **Gateway Address:** Specifies the next-hop IP address (e.g. 203.0.113.1). This is the ISP router's IP that the FortiGate uses to reach external networks.
- **Interface:** Indicates the outgoing interface used to reach the gateway (e.g., wan-to-isp-Gi0/3 (port5)).
- **Administrative Distance:** Determines route preference when multiple routes exist to the same destination. Lower values have higher priority; 10 is typical for static routes.
- **Comments:** Optional field for documentation (e.g., Default route to ISP). Helps administrators quickly identify the route's purpose.
- **Status:** Enables or disables the route. When set to Enabled, the route becomes active in the routing table.
- **Advanced Options:** Provides additional parameters such as route tagging, priority, or policy-based routing.

GUI PATH: **Network --> Static Routes --> Create New**

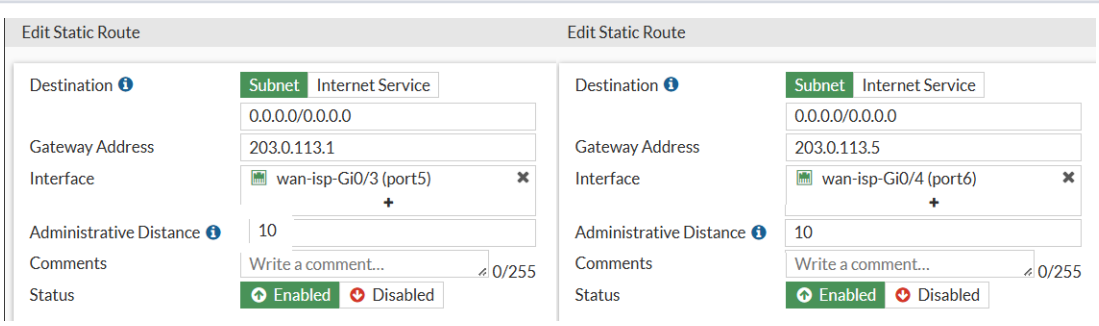


Figure: FortiGate Static Route Configuration

DC-FW1 Static Route to ISP

```
config vdom
 edit FG-traffic
  config router static
   edit 1
    set gateway 203.0.113.1
    set distance 10
    set device port5
   next
   edit 2
    set gateway 203.0.113.5
    set distance 10
    set device port6
   next
  end
 next
end
```

We can verify the static route on the firewall with the CLI command below.

DC-FW1 Static Route Verification

```
DC-FW1 (FG-traffic) # get router info routing-table static
Routing table for VRF=0
S*      0.0.0.0/0 [10/0] via 203.0.113.1, port5, [1/0]
          0.0.0.0/0 [10/0] via 203.0.113.5, port6, [1/0]
```

DC-FW1 Routing Table

```
DC-FW1 (FG-traffic) # get router info routing-table all vrf 0
Routing table for VRF=0
S*      0.0.0.0/0 [10/0] via 203.0.113.1, port5, [1/0]
        0.0.0.0/0 [10/0] via 203.0.113.5, port6, [1/0]
C       100.64.100.1/32 is directly connected, Loopback0
C       192.168.11.0/30 is directly connected, vlan10
C       192.168.12.0/30 is directly connected, vlan20
C       192.168.13.0/30 is directly connected, vlan30
C       192.168.14.0/29 is directly connected, vlan40
C       203.0.113.0/30 is directly connected, port5
C       203.0.113.4/30 is directly connected, port6
```

We can do the verification of DC-FW1 to BR1-FW1 connectivity as per following.

DC-FW1 ping to BR1-FW

```
DC-FW1 (FG-traffic) # execute ping 203.0.113.10
PING 203.0.113.10 (203.0.113.10): 56 data bytes
64 bytes from 203.0.113.10: icmp_seq=0 ttl=254 time=1.3 ms
!
64 bytes from 203.0.113.10: icmp_seq=4 ttl=254 time=1.1 ms
5 packets transmitted, 5 packets received, 0\% packet loss
round-trip min/avg/max = 1.1/1.2/1.8 ms
```

13 DNAT (Destination NAT)

In this section, we will cover the following objectives.

> **Objective**
>
> - Configure Virtual IPs.
> - Configure static and Port Forwarding based DNAT.
> - Troubleshoot DNAT connectivity issue.
> - Configure and validate WAN to DMZ firewall policy.

In FortiGate, Destination NAT (DNAT) is used to allow external users to access internal servers by translating the destination IP address of incoming packets (public → private). It's the reverse of SNAT – instead of hiding internal IPs, DNAT exposes internal services securely through controlled mappings.

DNAT Methods in FortiGate

13.1 Virtual IP (Static DNAT)

- Maps a single public IP to a single internal private IP.
- Commonly used for hosting internal servers (e.g. web, mail, FTP).

Static DNAT maps all inbound traffic destined for a public/external IP directly to an internal host. Before starting the configuration note the following points regard to external interface.

- **Purpose:** The extintf any option allows a Virtual IP (VIP) to match traffic arriving on *any interface*, rather than binding it to a single WAN port.
- **Use Case:**
  - Dual WAN or SD-WAN environments where the same public IP block is reachable via multiple interfaces.
  - High Availability (HA) clusters where failover between WAN links should not break DNAT rules.
  - Simplifies configuration by avoiding duplicate VIP entries per interface.
- **Caution:**
  - Avoid using any if different ISPs provide distinct public IPs – this can cause ambiguity.
  - Ensure inbound firewall policies are properly restricted to prevent exposing services unintentionally on all interfaces.
- In short: We can use extintf for redundancy or shared public IPs, but not required if each WAN has unique external IPs or separate ISPs.

In this lab document we will use 2 separate VIP for primary and secondary ISP as per following.

Below screenshot shows the GUI method for DNAT configuration for web server use the parameter as per table.

ISP Role	Interface	External IP (VIP)	Server	Internal IP
Primary	port5	203.0.113.97	web server	192.168.14.2
Secondary	port6	203.0.113.101		
Primary	port6	203.0.113.102	prod server	192.168.14.3
Secondary	port5	203.0.113.98		

Table 14: VIP allocation for DNAT across Primary and Secondary ISP

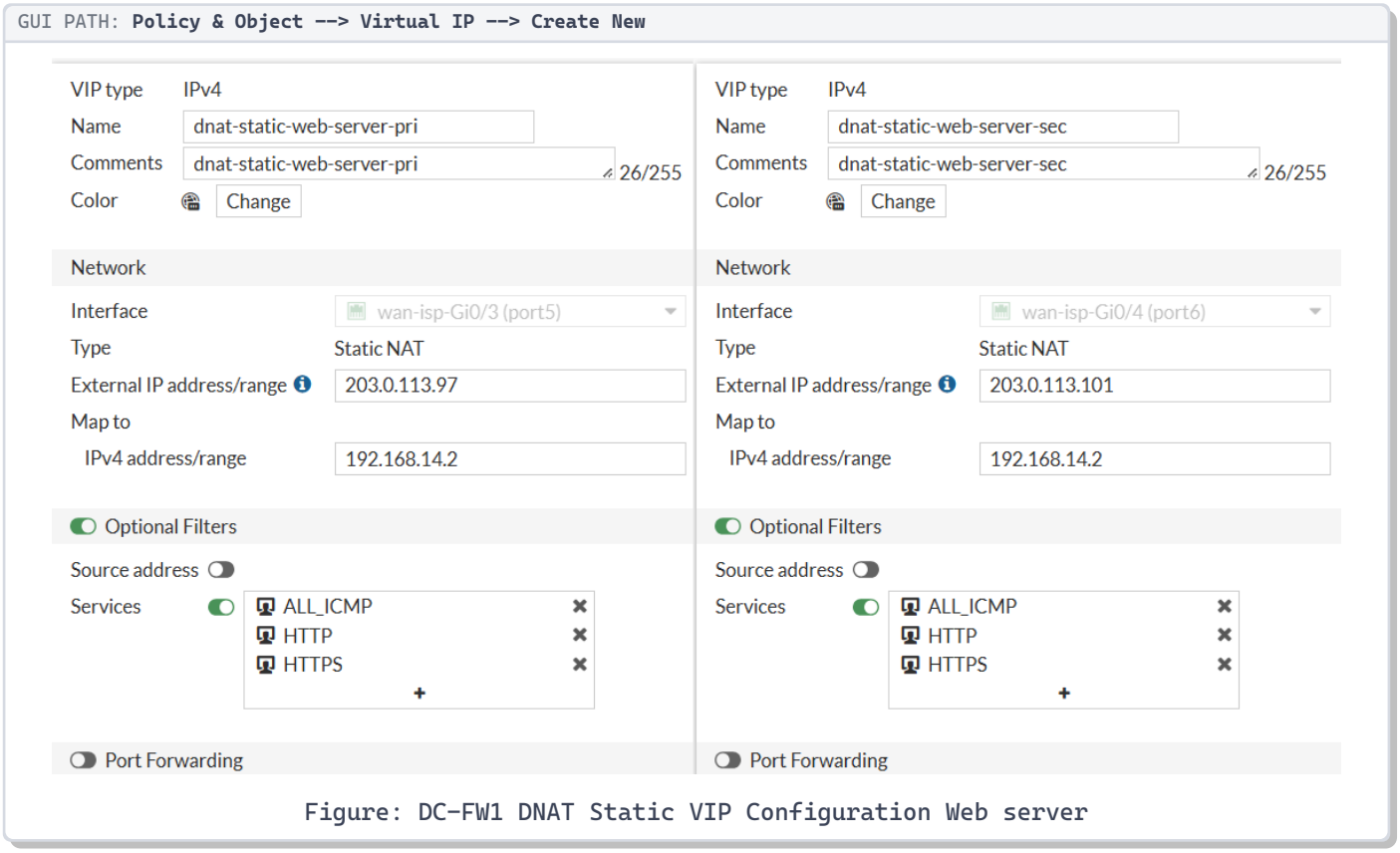


Figure: DC-FW1 DNAT Static VIP Configuration Web server

Now configure static DNAT for prod server as per below CLI Configuration.

DC-FW1 DNAT Static VIP Configuration prod server

```
config vdom
edit FG-traffic
 config firewall vip
  edit dnat-static-prod-server-pri
  set comment dnat-static-prod-server-pri
  set service ALL_ICMP HTTP HTTPS
  set extip 203.0.113.102
  set mappedip 192.168.14.3
  set extintf port6
 next
 edit dnat-static-prod-server-sec
  set comment dnat-static-prod-server-sec
  set service ALL_ICMP HTTP HTTPS
  set extip 203.0.113.98
  set mappedip 192.168.14.3
  set extintf port5
 next
end
```

Now create firewall policy to forward traffic from WAN TO DMZ as per below GUI Screenshot.

GUI PATH: **Policy & Object --> Firewall Policy --> Create New**

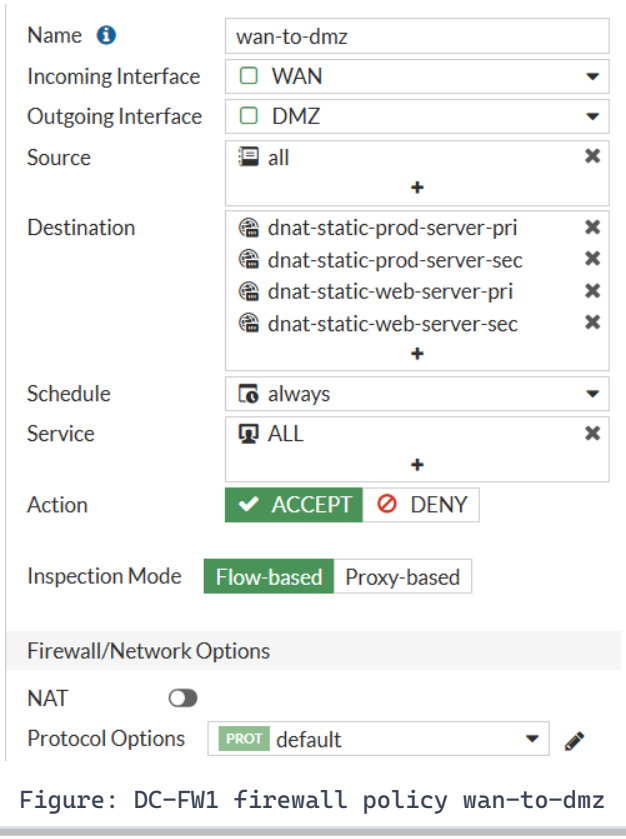


Figure: DC-FW1 firewall policy wan-to-dmz

Also we can configure DNAT using CLI method as per following.

DC-FW1 DNAT Configuration

```
config vdom
 edit FG-traffic
  config firewall policy
   edit 8
    set name wan-to-dmz
    set srcintf WAN
    set dstintf DMZ
    set action accept
    set srcaddr all
    set dstaddr dnat-static-web-server-pri dnat-static-web-server-sec
    dnat-static-prod-server-pri dnat-static-prod-server-sec
    set schedule always
    set service ALL
    set logtraffic all
   next
end
```

In the FortiView sessions screenshot below, Verified active DNAT sessions from source IP 203.0.113.22 to VIPs 203.0.113.97, 203.0.113.98, 203.0.113.101, and 203.0.113.102 over TCP/80.

FortiView Sessions

Source IP: 203.0.113.22 | Add Filter

Source	Device	Destination	Application	Protocol	Source Port	Destination Port	SPU	Bytes	Packets
203.0.113.22		203.0.113.97	TCP/80	TCP	52478	80		12.09 kB	21
203.0.113.22		203.0.113.102	TCP/80	TCP	52474	80		5.80 kB	18
203.0.113.22		203.0.113.101	TCP/80	TCP	52476	80		13.33 kB	23
203.0.113.22		203.0.113.98	TCP/80	TCP	52480	80		5.80 kB	18
203.0.113.22		203.0.113.97	TCP/80	TCP	52479	80		3.34 kB	15
203.0.113.22		203.0.113.101	TCP/80	TCP	52477	80		3.34 kB	15
203.0.113.22		203.0.113.98	TCP/80	TCP	52481	80		237 B	5

Figure: DC-FW1 DNAT Sessions

Below is the DNAT session list CLI output.

DC-FW1 DNAT Session List

```
DC-FW1 (FG-traffic) # diagnose sys session list | grep 192.168.14.2
orgin->sink: org pre->post, reply pre->post dev=7->24/24->7 gwy=192.168.14.2/203.0.113.1
hook=pre dir=org act=dnat 203.0.113.22:52050->203.0.113.97:80(192.168.14.2:80)
hook=post dir=reply act=snat 192.168.14.2:80->203.0.113.22:52050(203.0.113.97:80)
orgin->sink: org pre->post, reply pre->post dev=8->24/24->8 gwy=192.168.14.2/203.0.113.5
hook=pre dir=org act=dnat 203.0.113.22:52051->203.0.113.101:80(192.168.14.2:80)
hook=post dir=reply act=snat 192.168.14.2:80->203.0.113.22:52051(203.0.113.101:80)
!
DC-FW1 (FG-traffic) # diagnose sys session list | grep 192.168.14.3
orgin->sink: org pre->post, reply pre->post dev=8->24/24->8 gwy=192.168.14.3/203.0.113.5
hook=pre dir=org act=dnat 203.0.113.22:52052->203.0.113.102:80(192.168.14.3:80)
hook=post dir=reply act=snat 192.168.14.3:80->203.0.113.22:52052(203.0.113.102:80)
orgin->sink: org pre->post, reply pre->post dev=7->24/24->7 gwy=192.168.14.3/203.0.113.1
hook=pre dir=org act=dnat 203.0.113.22:52048->203.0.113.98:80(192.168.14.3:80)
hook=post dir=reply act=snat 192.168.14.3:80->203.0.113.22:52048(203.0.113.98:80)
!
DC-FW1 (FG-traffic) # diagnose sys session list | grep 203.0.113.97
hook=pre dir=org act=dnat 203.0.113.22:52050->203.0.113.97:80(192.168.14.2:80)
hook=post dir=reply act=snat 192.168.14.2:80->203.0.113.22:52050(203.0.113.97:80)
!
DC-FW1 (FG-traffic) # diagnose sys session list | grep 203.0.113.98
hook=pre dir=org act=dnat 203.0.113.22:52048->203.0.113.98:80(192.168.14.3:80)
hook=post dir=reply act=snat 192.168.14.3:80->203.0.113.22:52048(203.0.113.98:80)
!
DC-FW1 (FG-traffic) # diagnose sys session list | grep 203.0.113.101
hook=pre dir=org act=dnat 203.0.113.22:52051->203.0.113.101:80(192.168.14.2:80)
hook=post dir=reply act=snat 192.168.14.2:80->203.0.113.22:52051(203.0.113.101:80)
!
DC-FW1 (FG-traffic) # diagnose sys session list | grep 203.0.113.102
hook=pre dir=org act=dnat 203.0.113.22:52052->203.0.113.102:80(192.168.14.3:80)
hook=post dir=reply act=snat 192.168.14.3:80->203.0.113.22:52052(203.0.113.102:80)
```

13.2 Virtual IP DNAT with Port Forwarding

> ☞ **Note**
> In this section we will reuse the same external IP addresses (203.0.113.97 and 203.0.113.101 from the NAT pools) for Port Forwarding DNAT. Before configuring the port-forwarding VIPs, the Static VIPs created in the previous section must be removed.

Destination NAT (DNAT) in FortiGate is used to translate the **destination IP address** of

incoming traffic so that external clients can reach internal servers. When configured with **port forwarding**, DNAT allows mapping of specific external ports to internal server ports, enabling multiple services or servers to share the same public IP. Following are the some **Advantages of Port Forwarding DNAT**

- **Efficient IP Usage:** Multiple services (HTTP, HTTPS, SSH, RDP) can be hosted on one public IP by forwarding different ports.
- **Security Hardening:** Only required ports are exposed, reducing the attack surface.
- **Multi-Server Hosting:** Different servers can be accessed via unique port mappings on the same external IP.
- **ISP Redundancy:** Port forwarding can be configured per WAN interface (e.g. Primary ISP vs Secondary ISP).

Thus, DNAT with port forwarding in FortiGate enables precise service publishing, efficient use of public IP addresses, and enhanced security control. We will configure VIP with port forwarding as per the following table.

ISP Role	Interface	External IP (VIP)	External→Internal Port	Server	Internal IP
Primary	port5	203.0.113.97	TCP/5555 → TCP/80	Web Server	192.168.14.2
Secondary	port6	203.0.113.101	TCP/5555 → TCP/80	Web Server	192.168.14.2
Primary	port6	203.0.113.101	TCP/5556 → TCP/80	Prod Server	192.168.14.3
Secondary	port5	203.0.113.97	TCP/5556 → TCP/80	Prod Server	192.168.14.3

Table 15: Port-forwarding DNAT allocation across Primary and Secondary ISP

Now configure DNAT VIP port forwarding for prod server as per below CLI Configuration.

DC-FW1 DNAT VIP port forwarding Configuration prod server

```
config firewall vip
 edit dnat-port-fw-prod-server-pri
  set comment dnat-port-fw-prod-server-pri
  set extip 203.0.113.101
  set mappedip 192.168.14.3
  set extintf port6
  set portforward enable
  set extport 5556
  set mappedport 80
 next
 edit dnat-port-fw-prod-server-sec
  set comment dnat-port-fw-prod-server-sec
  set extip 203.0.113.97
  set mappedip 192.168.14.3
  set extintf port5
  set portforward enable
  set extport 5556
  set mappedport 80
 next
end
```

Below GUI screenshot below shows the GUI method for DNAT configuration for web server use the parameter as per above table.

GUI PATH: **Policy & Object --> Virtual IP --> Create New**

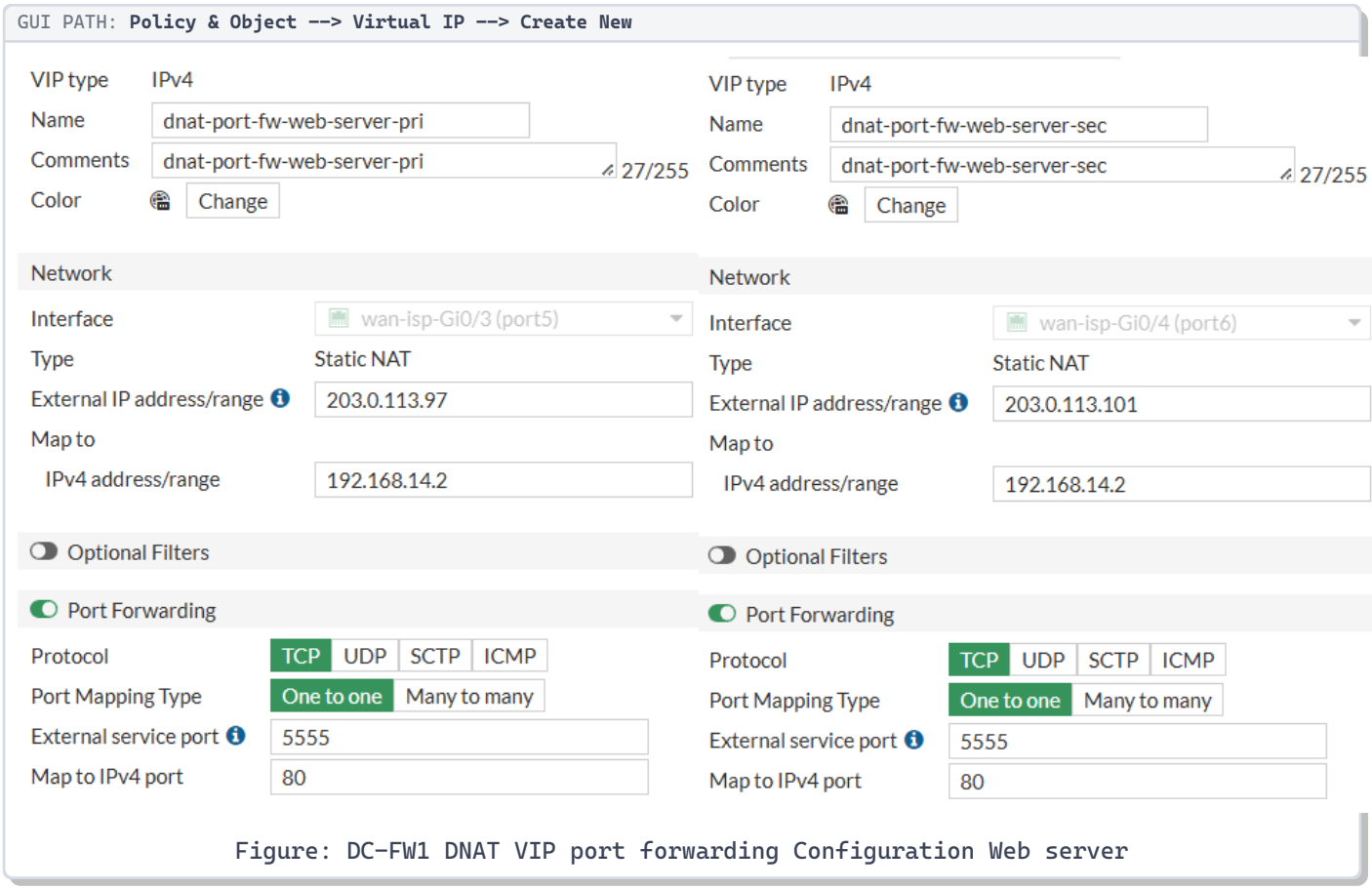


Figure: DC-FW1 DNAT VIP port forwarding Configuration Web server

We can perform verification for the DNAT port forwarding as per below CLI output.

DC-FW1 DNAT port forwarding packet sniffer

```
DC-FW1 (FG-traffic) # diagnose sniffer packet any 'host 192.168.14.2 and port 80' 4
filters=[host 192.168.14.2 and port 80]
32.413296 vlan40 out 203.0.113.22.54258 -> 192.168.14.2.80: syn 1765673712
32.415325 vlan40 in 192.168.14.2.80 -> 203.0.113.22.54258: syn 1002898368 ack 1765673713
32.419512 vlan40 out 203.0.113.22.54258 -> 192.168.14.2.80: ack 1002898369
!
32.439624 vlan40 out 203.0.113.22.54259 -> 192.168.14.2.80: syn 4099634670
32.441395 vlan40 in 192.168.14.2.80 -> 203.0.113.22.54259: syn 3035128838 ack 4099634671
32.445078 vlan40 out 203.0.113.22.54259 -> 192.168.14.2.80: ack 3035128839
!
DC-FW1 (FG-traffic) # diagnose sniffer packet any 'host 192.168.14.3 and port 80' 4
filters=[host 192.168.14.3 and port 80]
10.4294928055 vlan40 out 203.0.113.22.54264 -> 192.168.14.3.80: syn 2242226524
10.4294931919 vlan40 in 192.168.14.3.80 -> 203.0.113.22.54264: syn 1993138525 ack 2242226525
10.4294935442 vlan40 out 203.0.113.22.54264 -> 192.168.14.3.80: ack 1993138526
!
10.4294956503 vlan40 out 203.0.113.22.54265 -> 192.168.14.3.80: syn 2311032459
10.4294958516 vlan40 in 192.168.14.3.80 -> 203.0.113.22.54265: syn 863964209 ack 2311032460
10.4294961974 vlan40 out 203.0.113.22.54265 -> 192.168.14.3.80: ack 863964210
```

Below Image shows the Overview of FortiGate DNAT (Destination NAT) methods.

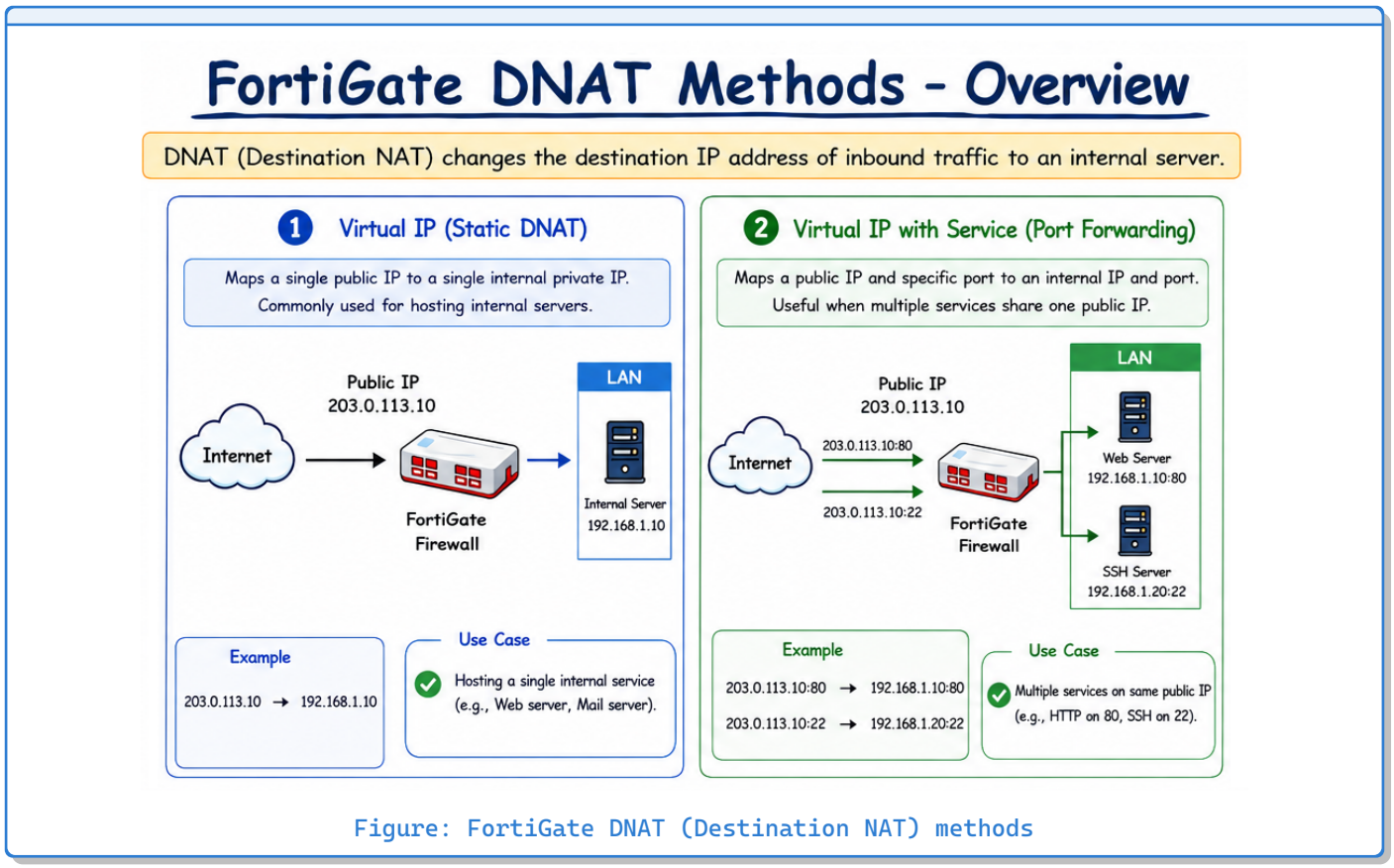


Figure: FortiGate DNAT (Destination NAT) methods

13.3 DNAT Troubleshooting Scenarios

13.3.1 DNAT – External Users Cannot Reach DMZ Server

- **Symptoms:** Public users cannot access the DMZ server via VIP. Connection times out or is dropped.
- **Troubleshooting Focus:**
  - Verify VIP object configuration (extip, mappedip, portforward).
  - Check firewall policy referencing the VIP.
  - Confirm routing symmetry (return traffic flows back through FortiGate).

> ★ **Verification Command**
>
> - show firewall vip – confirm VIP mapping.
>
> - show firewall policy– ensure policy references VIP.
>
> - diagnose sniffer packet any 'host <dmz-ip>' 4 – capture traffic.
>
> - diagnose debug flow filter addr <dmz-ip> – trace flow.

➲ Observation

```
DC-FW1 (FG-traffic) # diagnose sniffer packet any 'host 192.168.14.2' 4
Using Original Sniffing Mode
interfaces=[any]
filters=[host 192.168.14.2]
```

```
13.919392 vlan40 in arp who-has 192.168.14.1 tell 192.168.14.2
13.919410 vlan40 out arp reply 192.168.14.1 is-at 00:09:0f:09:00:02
13.919412 Lag-1 out arp reply 192.168.14.1 is-at 00:09:0f:09:00:02
13.919415 port5 out arp reply 192.168.14.1 is-at 00:09:0f:09:00:02
!
DC-FW1 (FG-traffic) # diagnose debug flow filter addr 192.168.14.2
DC-FW1 (FG-traffic) # diagnose debug enable
.
no logs
.
!
DC-FW1 (FG-traffic) # sh firewall policy 8
config firewall policy
    edit 8
        set name "wan-to-dmz"
        set action accept
        set srcaddr "all"
        set dstaddr "all"
        set service "ALL_ICMP" "HTTP" "HTTPS"
DC-FW1 (FG-traffic) # show firewall vip
config firewall vip
    edit "dnat-static-web-server-pri"
        set service "ALL_ICMP" "HTTP" "HTTPS"
        set extip 203.0.113.97
        set mappedip "192.168.14.2"
        set extintf "port5"
```

- **Root Cause:** The firewall policy is not referencing the VIP object. Because of that, traffic destined for 203.0.113.97 is not matched to the DNAT VIP, so FortiGate doesn't create a session → no debug flow logs.
- **Fix Note:** Update the firewall policy to use the VIP object as destination.

Observation after Fix

```
config firewall policy
    edit 8
        set name "wan-to-dmz"
        set action accept
        set srcaddr "all"
        set dstaddr "dnat-static-web-server-pri"
        set service "ALL_ICMP" "HTTP" "HTTPS"
    next
end
!
DC-FW1 (FG-traffic) # diagnose sniffer packet any 'host 192.168.14.2' 4
filters=[host 192.168.14.2]
4.415231 vlan40 out 203.0.113.22.4214 -> 192.168.14.2.80: syn 3497204221
4.415237 Lag-1 out 203.0.113.22.4214 -> 192.168.14.2.80: syn 3497204221
4.415239 port3 out 203.0.113.22.4214 -> 192.168.14.2.80: syn 3497204221
```

13.3.2 DNAT – Asymmetric Routing / Missing Default Gateway

- **Symptoms:** Public users cannot access the DMZ server via VIP. Connection times out or is dropped.
- **Troubleshooting Focus:**
  - Verify VIP object configuration (extip, mappedip, portforward).
  - Check firewall policy referencing the VIP.
  - Confirm routing symmetry (return traffic flows back through FortiGate).

★ **Verification Command**

- show firewall vip – confirm VIP mapping
- show firewall policy – ensure policy references VIP
- diagnose sniffer packet any 'host <Client_Public_IP>' 4 – capture traffic
- diagnose debug flow filter addr <Client_Public_IP> – trace flow
- diagnose debug flow trace start 50
- diagnose debug enable

Observation before Fix

```
DC-FW1 (FG-traffic) # diagnose debug flow filter addr 203.0.113.101
DC-FW1 (FG-traffic) # diagnose debug flow trace start 50
DC-FW1 (FG-traffic) # diagnose debug enable
DC-FW1 (FG-traffic) # id=20085 trace_id=94 func=print_pkt_detail line=5824 msg="
    vd-FG-traffic:0 received a packet(proto=6, 203.0.113.22:4282->203.0.113.101:80) tun_id
    =0.0.0.0 from port6. flag [S], seq 1943730220, ack 0, win 65535"
id=20085 trace_id=94 func=get_new_addr line=1227 msg="find DNAT: IP-192.168.14.3, port-0(
    fixed port)" msg="VIP-192.168.14.3:80, outdev-port6"
id=20085 trace_id=94 func=__ip_session_run_tuple line=3502 msg="DNAT 203.0.113.101:80-
    >192.168.14.3:80"
!
DC-FW1 (FG-traffic) # diagnose sniffer packet any 'host 203.0.113.101' 4
filters=[host 203.0.113.101]
5.929905 port6 in 203.0.113.22.4288 -> 203.0.113.101.80: syn 1253805675
6.171221 port6 in 203.0.113.22.4289 -> 203.0.113.101.80: syn 4241364157
9.122561 port6 in 203.0.113.22.4288 -> 203.0.113.101.80: syn 1253805675
!
DC-FW1 (FG-traffic) # show firewall vip
config firewall vip
    edit "dnat-port-fw-prod-server-pri"
        set service "ALL_ICMP" "HTTP" "HTTPS"
        set extip 203.0.113.101
        set mappedip "192.168.14.3"
        set extintf "port6"
DC-FW1 (FG-traffic) # show firewall policy 8
config firewall policy
        set name "wan-to-dmz"
        set dstaddr "dnat-port-fw-web-server-pri" "dnat-port-fw-prod-server-pri"
        set schedule "always"
        set service "ALL_ICMP" "HTTP" "HTTPS"
```

Observation before Fix

```
internet-cloud#sh ip route | i 203.0.113.100
S        203.0.113.100/30 [5/0] via 203.0.113.6
internet-cloud#
!
DC-FW1 (FG-traffic) # get router info routing-table all
Routing table for VRF=0
S*      0.0.0.0/0 [10/0] via 203.0.113.1, port5, [1/0]
!
DC-FW1 (FG-traffic) # sh router static
config router static
    edit 1
        set gateway 203.0.113.1
        set device "port5"
    edit 2
        set gateway 203.0.113.5
        set distance 15
        set device "port6"
```

- **Root Cause:**
  - **Server Default Gateway mis-configuration:** Inbound DNAT and Firewall Policy match successfully, but if host 192.168.14.3 does not use the FortiGate DMZ interface (192.168.14.1) as its default gateway, return traffic bypasses FortiGate entirely and fails to reach the client.
  - **WAN Route Asymmetry / Inactive Egress Route:** Inbound traffic arrives on port6, but FortiGate only has an active default route via port5 (Gateway 203.0.113.1, Distance 10). The static route via port6 (Gateway 203.0.113.5) has a higher distance (15) and is inactive in the routing table. As a result, return traffic attempts to exit out port5, breaking stateful connection symmetry or failing RPF checks.
- **Fix Note:** Lowering the port6 route distance to 10 makes both default routes active. Lower the distance of static route 2 (port6) from 15 to 10 so it enters the routing table.

Observation

```
DC-FW1 (FG-traffic) # get router info routing-table all
Routing table for VRF=0
S*      0.0.0.0/0 [10/0] via 203.0.113.1, port5, [1/0]
                  [10/0] via 203.0.113.5, port6, [1/0]
!
DC-FW1 (FG-traffic) #  diagnose sniffer packet any 'host 203.0.113.101' 4
Using Original Sniffing Mode
interfaces=[any]
filters=[host 203.0.113.101]
2.420329 port6 in 203.0.113.22.4315 -> 203.0.113.101.80: syn 150979000
2.422906 port6 out 203.0.113.101.80 -> 203.0.113.22.4315: syn 2207811497 ack 150979001
2.426728 port6 in 203.0.113.22.4315 -> 203.0.113.101.80: ack 2207811498
2.427838 port6 in 203.0.113.22.4315 -> 203.0.113.101.80: psh 150979001 ack 2207811498
```

14 FortiGate Local User Authentication

In this section, we will cover the following objectives.

> ➲ **Objective**
>
> • Create and verification of local user account and User Groups.

Local user authentication on a FortiGate firewall allows administrators to create user accounts directly on the FortiGate device, which can then be used for VPN access, captive portal login, or firewall policy authentication.

- Local authentication is best suited for small deployments or testing labs.
- For production, integrate with LDAP, RADIUS, or SAML for scalability.
- Passwords are stored locally on the FortiGate, so enforce strong password policies.

FortiGate Local User configuration blocks are shown in below section.

- **Local User Objects (config user local):** Defines individual accounts consisting of a unique username, password hash, status (enabled/disabled), optional contact info (email/SMS for OTP), and optional 2FA settings (e.g., FortiToken).

We will create local user as per below table on DC-FW1. Following section shows both GUI (sales1 user created with GUI) and CLI configuration for User Definition creation.

Table 16: Local Users by Department

Department (Group)	User 1	User 2
prod	prod1	prod2
sales	sales1	sales2
hr	hr1	hr2

> ☞ **Note**
> Two Factor authentication disabled in lab demo as evaluation version does not support Forti cloud or Forti Token.

GUI PATH: **User and Authentication --> User Definition --> Create New --> Local User**

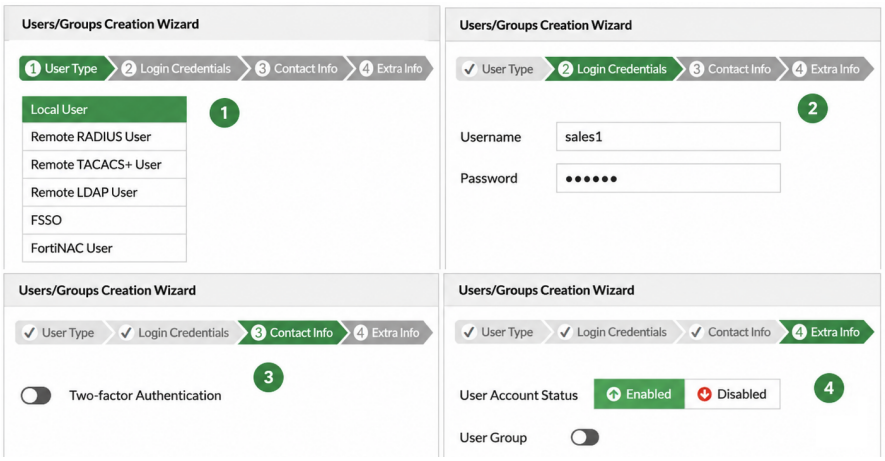


Figure: DC-FW1 User Definition

With following command we can created User using CLI.

Local User Creation

```
config vdom
 edit FG-traffic
  config user local
   edit sales2
    set type password
    set passwd ***
   next
   edit hr1
    set type password
    set passwd ***
   next
   edit hr2
    set type password
    set passwd ***
   next
   edit prod1
    set type password
    set passwd ***
   next
   edit prod2
    set type password
    set passwd ***
   next
end
```

User Definition Verification

```
DC-FW1 (FG-traffic) # get user local
== [ sales1 ]
name: sales1   id: 16777217
.
output omitted
.
== [ prod2 ]
name: prod2   id: 16777222
!
DC-FW1 (FG-traffic) # show user local
config user local
    edit "sales1"
        set type password
        set passwd ***
    next
.
output omitted
.
    edit "sales2"
        set type password
```

- **User Groups (config user group):** Acts as the logical container grouping local users together. FortiGate policies evaluate group memberships rather than mapping directly to individual users. Following section shows both GUI (sales1 user created with GUI) and CLI configuration for User Groups creation.

GUI PATH: **User and Authentication --> User Groups --> Create New**

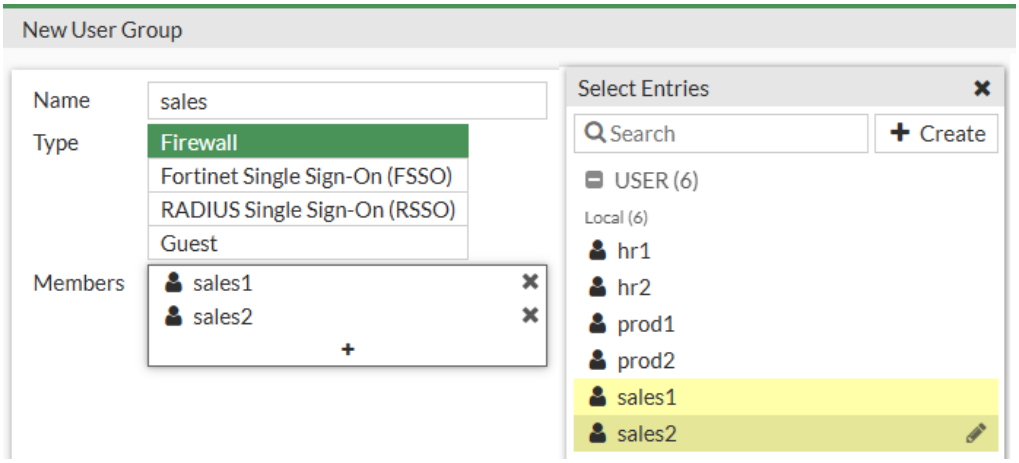


Figure: DC-FW1 User Group Creation

Local User Group Creation

```
config vdom
 edit FG-traffic
  config user group
   edit hr
    set member hr1 hr2
   next
   edit prod
    set member prod1 prod2
   next
  end
```

User Group Verification

```
DC-FW1 (FG-traffic) # sh user group
config user group
    edit "sales"
        set member "sales1" "sales2"
..output omitted..
    edit "prod"
        set member "prod1" "prod2"
    next
end
!
DC-FW1 (FG-traffic) # get user group
== [ SSO_Guest_Users ]
name: SSO_Guest_Users   id: 16777215
== [ sales ]
name: sales   id: 1
..output omitted..
== [ prod ]
name: prod   id: 3
```

15 VPN (Virtual Private Network)

In this section, we will cover the following objectives.

> **Objective**
>
> - Understand IPsec and SSL VPN architectures.
> - Configure Site-to-Site IPsec VPNs, Remote access VPN using SSL and IPsec.
> - Understand IKEv1 operations.
> - Troubleshoot VPN Phase 1, Phase 2 and traffic flow issues.

A VPN (Virtual Private Network) creates a secure, encrypted tunnel across an untrusted network (like the internet). Within VPNs, two common deployment models are Site-to-Site VPNs and Remote Access VPNs – both often built on IPsec for encryption, authentication, and data integrity.

- **Site-to-Site VPN:** Connects entire networks (e.g. headquarters ⇔ branch office) using firewalls or routers at each site. The tunnel is persistent and transparent to internal hosts, requiring no VPN client software. Best suited for branch connectivity, data center integration, and B2B partner links.
- **Remote Access VPN:** Connects individual devices (laptops, phones) to the corporate LAN via VPN client software. Requires user authentication (passwords, MFA, certificates). Best suited for employees working remotely, contractors, and mobile workforce on untrusted networks.

15.1 Cryptography

Cryptography is the science of securing information by transforming it into a form that is unreadable to unauthorized parties. It ensures confidentiality, integrity, authentication, and non-repudiation in communication and data storage. Modern cryptography relies on mathematical algorithms to protect sensitive information from cyber threats.

- **Symmetric Cryptography:** Symmetric cryptography uses a single secret key for both encryption and decryption. It is fast and efficient, making it suitable for encrypting large amounts of data. However, the challenge lies in securely sharing the secret key between parties. Common symmetric encryption algorithms include AES, DES, and Blowfish. AES is widely used in modern security deployments, while DES is considered obsolete because of its short key length and should not be used for production security.
- **Asymmetric Cryptography:** Asymmetric cryptography, also known as public-key cryptography, uses a mathematically related key pair consisting of a public key and a private key. Depending on the cryptographic mechanism, the public key can be used to encrypt data or verify a digital signature, while the private key can be used to decrypt data or create a digital signature. Asymmetric cryptography is commonly used for authentication, digital signatures, certificate-based security, and secure key establishment. RSA and Elliptic Curve Cryptography (ECC) are commonly used public-key technologies. Diffie-Hellman (DH) and Elliptic Curve Diffie-Hellman (ECDH) are key-agreement mechanisms used to establish a shared secret; they do not directly encrypt user data.

15.2 IPSec Tunnel

An IPsec tunnel is a secure communication channel that protects traffic across untrusted networks. It provides multiple security services to ensure confidentiality, integrity, and authenticity of data. Let's break down the key options:

- **Authentication:** Verifies the identity of VPN peers using pre-shared keys (PSK) or digital certificates (PKI). Ensures only trusted endpoints can establish the tunnel.
- **Data Integrity:** Protects against packet tampering by using HMAC algorithms (e.g., SHA-256). Generates an Integrity Check Value (ICV) to confirm data has not been altered in transit.
- **Confidentiality:** Encrypts the payload so unauthorized parties cannot read it. Common algorithms include AES-256, AES-128, and 3DES (legacy). Provides privacy for sensitive traffic.
- **Anti-Replay Protection:** Prevents attackers from re-sending captured packets. Sequence numbers in the IPsec header detect and reject duplicates.

15.2.1 IPsec Security Building Blocks

- **Hashing (Integrity & Authentication):** Hashing in IPsec is all about data integrity and authentication. It ensures that packets haven't been altered during transmission and that they truly come from the claimed sender.
  - HMAC-MD5 (128-bit) → legacy, weak today.
  - HMAC-SHA1 (160-bit, truncated to 96 bits) → widely used, moderate security.
  - HMAC-SHA256 / SHA384 / SHA512 → modern, strong integrity protection.
- **Encryption Algorithms (Confidentiality):** An encryption algorithm is a mathematical procedure used to transform readable data (plaintext) into an unreadable format (ciphertext) to ensure confidentiality. Only someone with the correct key can reverse the process (decrypt) and recover the original information.
  - AES-128 / AES-256 → industry standard, strong security.
  - 3DES → legacy, slower, less secure.
  - ChaCha20-Poly1305 → modern alternative, efficient on low-power devices.
- **Diffie-Hellman (DH) Groups (Key Exchange Strength):** Diffie-Hellman (DH) Groups are sets of predefined mathematical parameters used in the Diffie-Hellman key exchange process. In IPsec, they determine the strength of the cryptographic keys generated during IKE negotiation.
  - Group 2 (1024-bit) → legacy, weak today.
  - Group 5 (1536-bit) → stronger but outdated.
  - Group 14 (2048-bit) → widely used, secure baseline.
  - Group 19/20 (Elliptic Curve, 256/384-bit) → modern, efficient, strong.
  - Group 24 (2048-bit prime) → advanced security.
- **IKE (Internet Key Exchange):** Protocol that negotiates keys, algorithms, and Security Associations (SAs) to establish and manage IPsec tunnels securely.
- **IPsec Process Steps**
  - **Initiation:** VPN peers detect interesting traffic (matching crypto ACL/policy) and begin negotiation. ISAKMP/IKE is triggered to establish a secure tunnel.

  - **IKE Phase I:** Establishes a secure, authenticated channel between peers. Negotiates encryption, hashing, and Diffie-Hellman parameters. Creates the IKE Security Association (SA).
  - **IKE Phase II:** Within the secure channel, peers negotiate IPsec parameters (encryption, integrity, anti-replay). Builds IPsec SAs for actual data traffic.
  - **Data Transfer:** Protected traffic flows through the IPsec tunnel. Confidentiality (AES), integrity (HMAC-SHA), authentication, and anti-replay mechanisms ensure secure communication.
  - **Termination:** Tunnel is torn down when idle timeout, manual shutdown, or rekey failure occurs. Security Associations are deleted, and traffic is matching the VPN policy will fail.
- **ISAKMP (Internet Security Association and Key Management Protocol):** Framework that defines procedures for establishing, negotiating, modifying, and deleting Security Associations (SAs) in IPsec tunnels.

15.2.2 Ports and IP Protocols

- **UDP 500 (ISAKMP/IKE):**
  - Used for IKE Phase 1 and Phase 2 negotiations.
  - Default port for initiating and negotiating Security Associations.
- **UDP 4500 (NAT-T / NAT-Traversal):**
  - Used when one or both peers are behind NAT devices.
  - Encapsulates ESP packets inside UDP to traverse NAT.
- **Protocol 50 (ESP – Encapsulating Security Payload):**
  - Used for actual encrypted data traffic once the tunnel is established.
  - Provides confidentiality, integrity, and authentication.
- **Protocol 51 (AH – Authentication Header):**
  - Optional, used for integrity/authentication without encryption.
  - Less common today compared to ESP.

15.2.3 IPsec Tunnel Lifecycle with IKEv1

- **HEGAL Parameters:** Set of attributes negotiated during IPsec tunnel setup.
  - H → Hashing (integrity/authentication)
  - E → Encryption (confidentiality)
  - G → Diffie-Hellman Group (key exchange strength)
  - A → Authentication method (PSK or certificates)
  - L → Lifetime (validity of the SA before rekeying)
- **Security Association (SA):** Logical contract between two peers that defines how traffic will be protected.
  - Contains the negotiated H-E-G-A-L parameters (algorithms, keys, lifetimes, modes).
  - Unidirectional – one SA for inbound traffic, one SA for outbound traffic.

Phase 1 – Main Mode (Messages 1–6): Purpose is to establish a secure, authenticated channel (IKE SA).

- **Message 1:** Initiator → Proposes encryption, hashing, DH group, authentication method.
- **Message 2:** Responder → Accepts or modifies proposals.
- **Message 3:** Initiator → Sends Diffie–Hellman public value + nonce.
- **Message 4:** Responder → Sends Diffie–Hellman public value + nonce.
- **Message 5:** Initiator → Sends identity + authentication (PSK or certificate).
- **Message 6:** Responder → Sends identity + authentication (PSK or certificate).

Result: The IKE SA is established, providing a secure authenticated channel that is used to protect subsequent IPsec negotiation.

Phase 2 – Quick Mode (Messages 7–9): Purpose is to negotiate IPsec parameters (IPsec SAs).

- **Message 7:** Initiator → Proposes IPsec parameters such as encryption, integrity, lifetime, mode, and traffic selectors; also sends a nonce.
- **Message 8:** Responder → Selects an acceptable IPsec proposal and returns its nonce. If PFS is enabled, it also sends its Diffie–Hellman public value.
- **Message 9:** Initiator → Sends the final authenticated confirmation that completes the Quick Mode exchange.

Result: IPsec SAs established – data traffic can now flow securely.

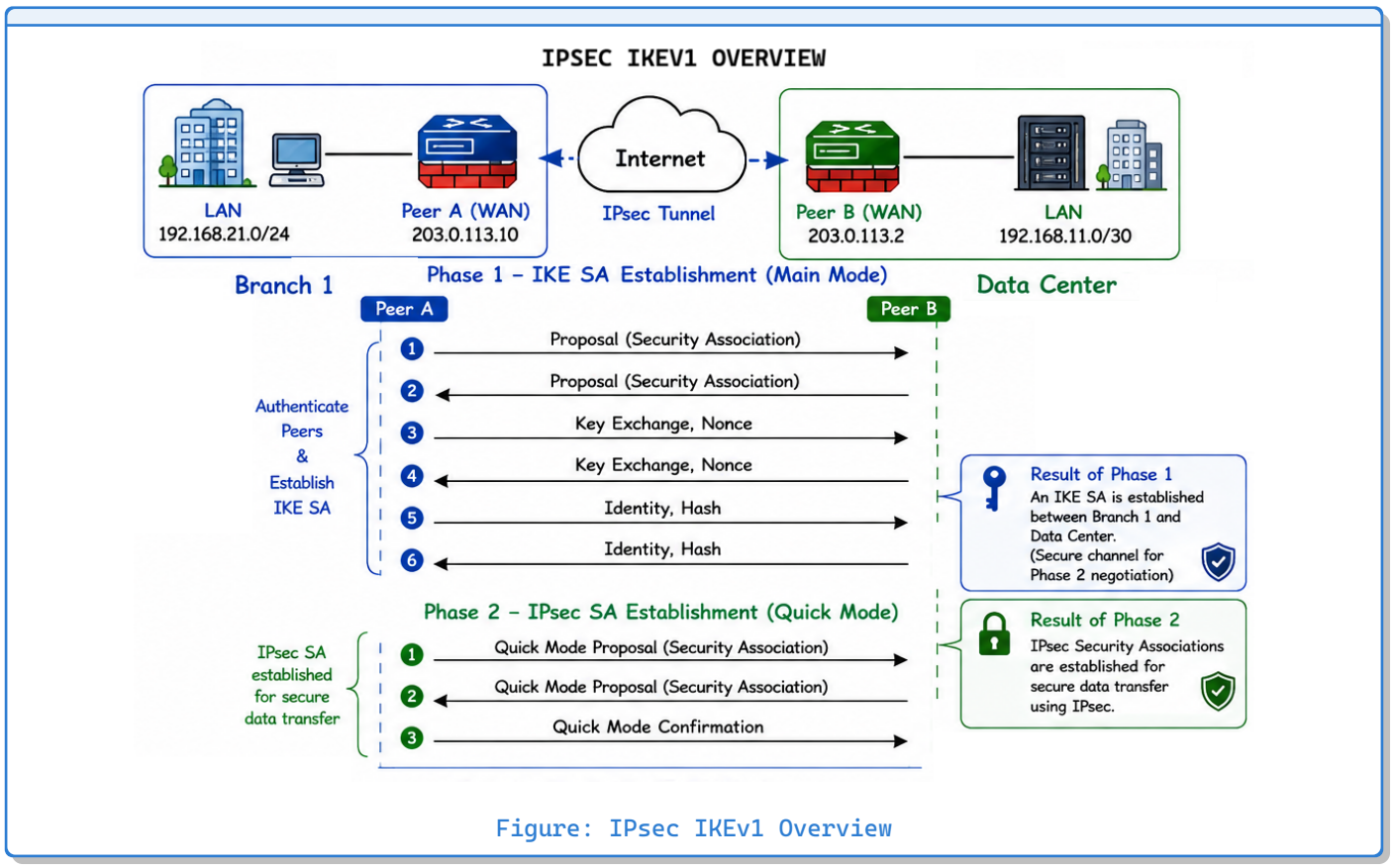


Figure: IPsec IKEv1 Overview

Figure above illustrates the complete process of secure tunnel establishment between Branch and Data Center using IKEv1. It shows Phase 1 (Main Mode) and Phase 2 (Quick Mode)

message exchanges, negotiated parameters, and resulting Security Associations (IKE SA and IPsec SA) for end-to-end encrypted communication

15.2.4 Black Hole Route

A black hole route (also called a Null route or Discard route) is a network routing table entry that directs packets destined for specific IP addresses to a virtual discard interface (such as Null0 on Cisco, discard/reject on Juniper, or blackhole on Linux/FortiGate).

In FortiOS, a Black Hole Route instructs the FortiGate to silently drop any packet matching that destination prefix without forwarding it out a physical/logical interface and without generating an ICMP unreachable message. This Route is active when IPSEC tunnel is down. This help to make the network more secure because traffic which supported to enter the encrypted tunnel is now not pushed to default gateway. AD value 254 for a blackhole route on an IPsec VPN is standard practice for FortiGate.

15.2.5 Dead Peer Detection

Dead Peer Detection (DPD) in FortiGate IPsec VPN verifies the availability of a remote IKE peer. It sends keepalive probe packets and listens for acknowledgments. If the remote peer stops responding to DPD probes, FortiGate tears down the Phase 1 and Phase 2 Security Associations (SAs), flushes associated routes, and allows dynamic routing or backup paths to take over immediately.

FortiGate supports three operational modes configured under Phase 1 for DPD.

- **On-Demand (on-demand):**
  - **Trigger Condition:** Triggered only when there is active outbound traffic traversing the tunnel, but no corresponding inbound return traffic is received from the remote peer.
  - **Behavior:** FortiGate sends DPD probe messages (ISAKMP R-U-THERE) solely when it actively expects a response. If the link is entirely idle, no DPD traffic is generated.
  - **Primary Use Case:** Standard production deployments and metered/bandwidth-constrained links to minimize probe overhead and conserve bandwidth.
- **On-Idle (on-idle):**
  - **Trigger Condition:** Triggered continuously whenever the tunnel is completely idle (no active inbound or outbound user data traffic).
  - **Behavior:** FortiGate periodically transmits DPD keepalive probes at the configured interval regardless of traffic flow.
  - **Primary Use Case:** Environments requiring NAT traversal (maintaining intermediate firewall/carrier NAT session state tables) and redundant/failover tunnels where rapid, proactive dead-peer detection is required before user traffic is dispatched.
- **Disable (disable):**
  - **Trigger Condition:** Never triggered.
  - **Behavior:** Completely disables DPD probing mechanisms. The Security Associations (Phase 1 and Phase 2 SAs) remain installed and active in the security database until their configured lifetime counters expire, regardless of whether the physical link or remote peer fails.
  - **Primary Use Case:** Interoperability with legacy third-party VPN appliances or non-standard IPsec endpoints that do not support DPD negotiations.

15.3 Site-To-Site VPN

On FortiGate firewall we have options to configure IPsec Tunnel using Template Based, custom and policy-based VPN. In industry practice, we do not create a separate IPsec tunnel per department. The standard approach is to configure a single site-to-site IPsec tunnel between the Data Center and each Branch, and then define multiple Phase 2 selectors (subnet pairs) inside that tunnel to cover each department's LAN. Below Table shows the IPSec Local and Remote LAN segment details.

Table 17: FortiGate VPN Phase 1 and 2 selector (Department-wise) for Tunnel Between DC and BR1

Site	Tunnel-Name	Department	DC-LAN	Branch-LAN
Branch 1	dc-br1-tunnel	Sales	192.168.11.0/30	192.168.21.0/24
		Prod	192.168.12.0/30	192.168.22.0/24
		HR	192.168.13.0/30	192.168.23.0/24

Also we are going to use loopback interface as tunnel source so we need to create new policy to allow ping and isakmp protocol as per below GUI screenshot or CLI also.

GUI PATH: **Policy & Objects --> Firewall Policy --> Create New**

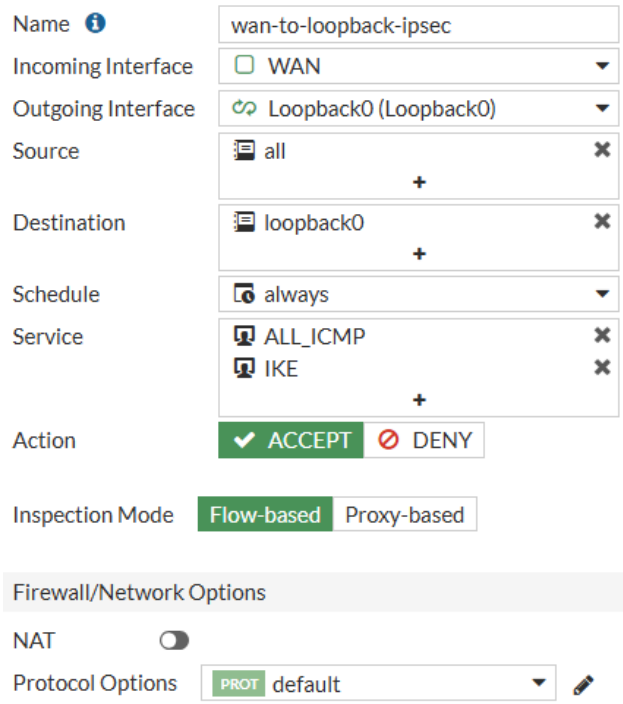


Figure: Policy for IKE Service

Policy for IKE Service

```
config vdom
 edit FG-traffic
  config firewall policy
   edit (Policy-id)
    set name wan-to-loopback-ipsec
    set srcintf WAN
    set dstintf Loopback0
    set action accept
    set srcaddr all
```

```
    set dstaddr loopback0
    set schedule always
    set service ALL_ICMP IKE
    set logtraffic all
    set comments wan-to-loopback-ipsec
   next
end
```

We can verify loopback interface connectivity between DC-FW1 and BR1-FW1.

Loopback connectivity verification

```
BR1-FW1 (FG-traffic) # execute ping 100.64.100.1
PING 100.64.100.1 (100.64.100.1): 56 data bytes
64 bytes from 100.64.100.1: icmp_seq=0 ttl=254 time=1.5 ms
..
64 bytes from 100.64.100.1: icmp_seq=4 ttl=254 time=0.9 ms
--- 100.64.100.1 ping statistics ---
5 packets transmitted, 5 packets received, 0\% packet loss
round-trip min/avg/max = 0.8/1.2/1.8 ms
!
DC-FW1 (FG-traffic) # execute ping 100.64.100.2
PING 100.64.100.2 (100.64.100.2): 56 data bytes
64 bytes from 100.64.100.2: icmp_seq=0 ttl=254 time=1.0 ms
.
64 bytes from 100.64.100.2: icmp_seq=4 ttl=254 time=1.4 ms
--- 100.64.100.2 ping statistics ---
5 packets transmitted, 5 packets received, 0\% packet loss
round-trip min/avg/max = 0.8/1.0/1.4 ms
```

15.3.1 Site-to-Site VPN using Template / Wizard

Before start VPN configuration please note below points regarding Phase-1 and 2 Parameters.

Note

- DES MD5 and DH Group are included in this lab for demonstration and troubleshooting purpose only.
- Production deployment should use AES-256, SHA-256 OR SHA-384, DH Group 14 or higher.

On FortiGate firewall Site-to-Site vpn template, the wizard automatically creates one Phase1 interface (the tunnel itself) and One Phase2 selector pair (local and remote subnets) as per following.

Configuration Field Breakdown

- **VPN Setup:**
  - **Name** – Identifies the tunnel object for configuration and policy reference.
  - **Template Type** – Defines VPN mode (Site-to-Site, Hub-and-Spoke, etc.).
  - **NAT Configuration** – Specifies whether NAT is applied between sites.
  - **Remote Device Type** – Optimizes parameters for peer device (FortiGate or Cisco).
- **Authentication:**

  - **Remote Device / IP Address** – Defines peer identification method.
  - **Outgoing Interface** – Interface used for IKE negotiation.
  - **Authentication Method** – Chooses Pre-shared Key or Certificate.
  - **Pre-shared Key** – Shared secret used for Phase 1 authentication.
- **Policy & Routing:**
  - **Local Interface** – Interface representing internal LAN.
  - **Local Subnets** – Networks behind this FortiGate.
  - **Remote Subnets** – Networks behind the peer FortiGate.
  - **Internet Access** – Defines whether local or remote internet is used.
- **Review Settings:**
  - **Object Summary** – Lists auto-created objects (interfaces, address groups, policies).
  - Confirms all components are ready before committing the VPN configuration.

Below GUI Screenshot show the configuration for dc-br1-tunnel.

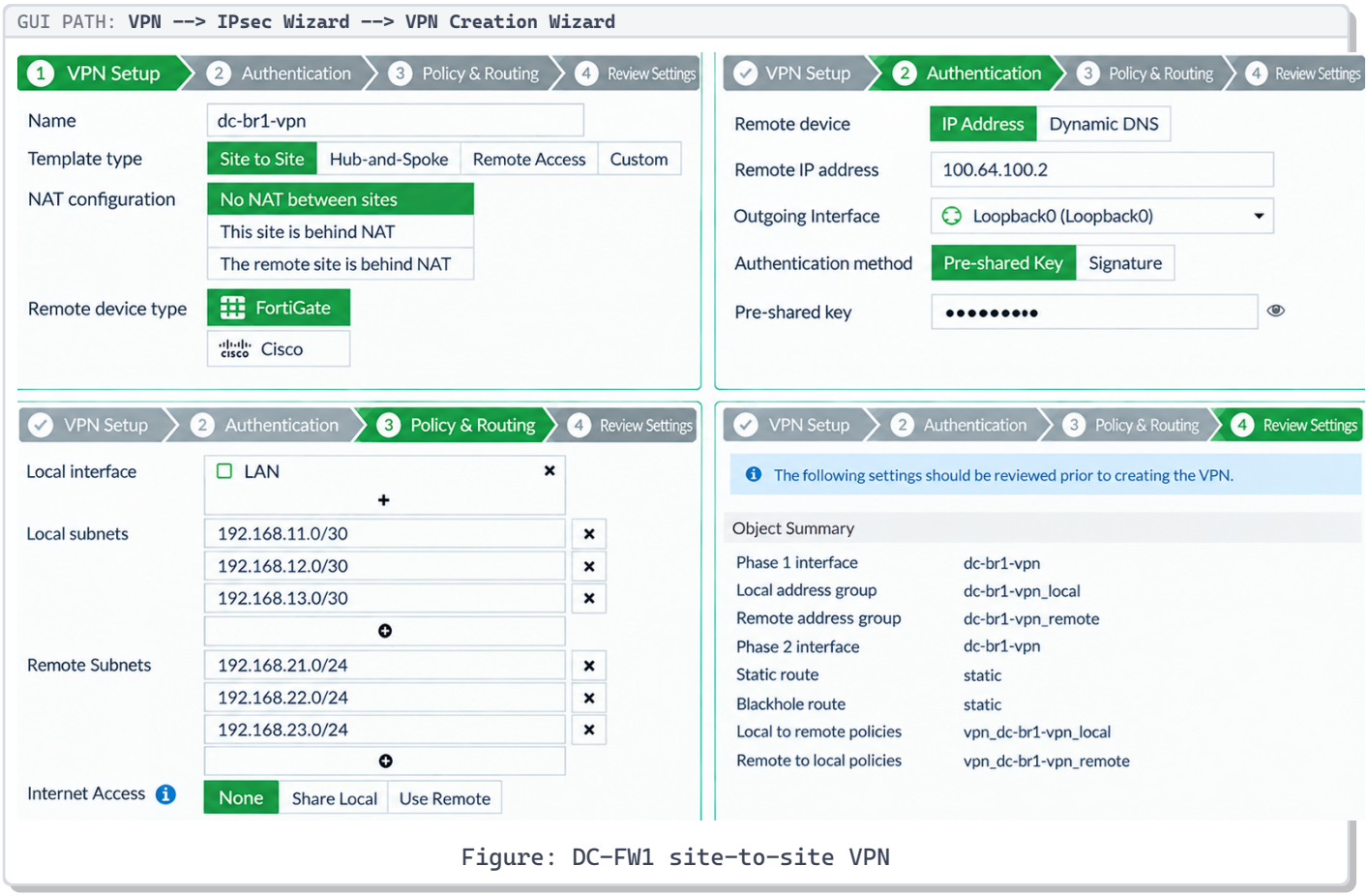


Figure: DC-FW1 site-to-site VPN

Below GUI Screenshot show the configuration for br1-dc-tunnel.

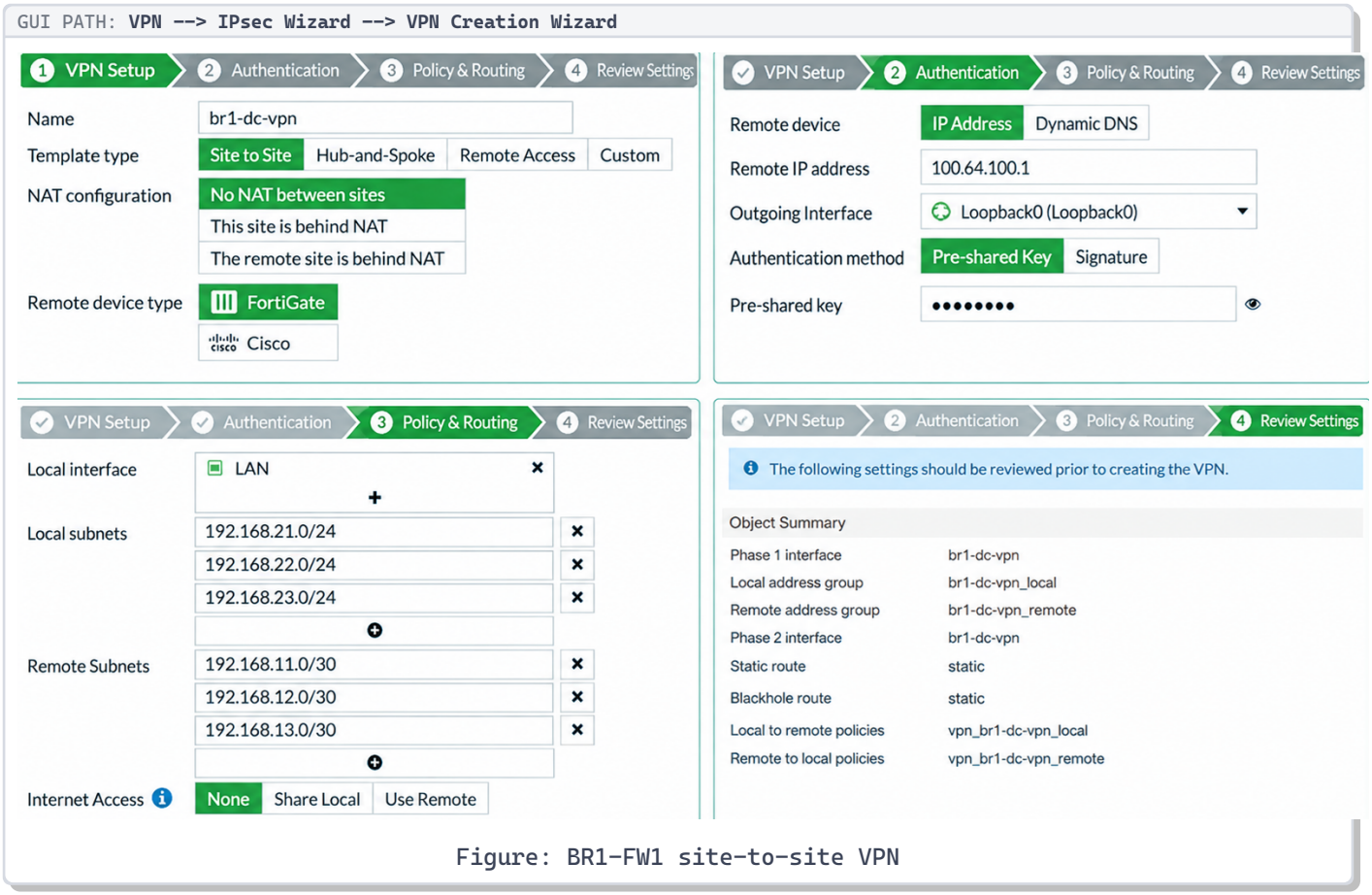


Figure: BR1-FW1 site-to-site VPN

The GUI screenshot below shows ipsec tunnel status and VPN events (only shown for BR1-FW1).

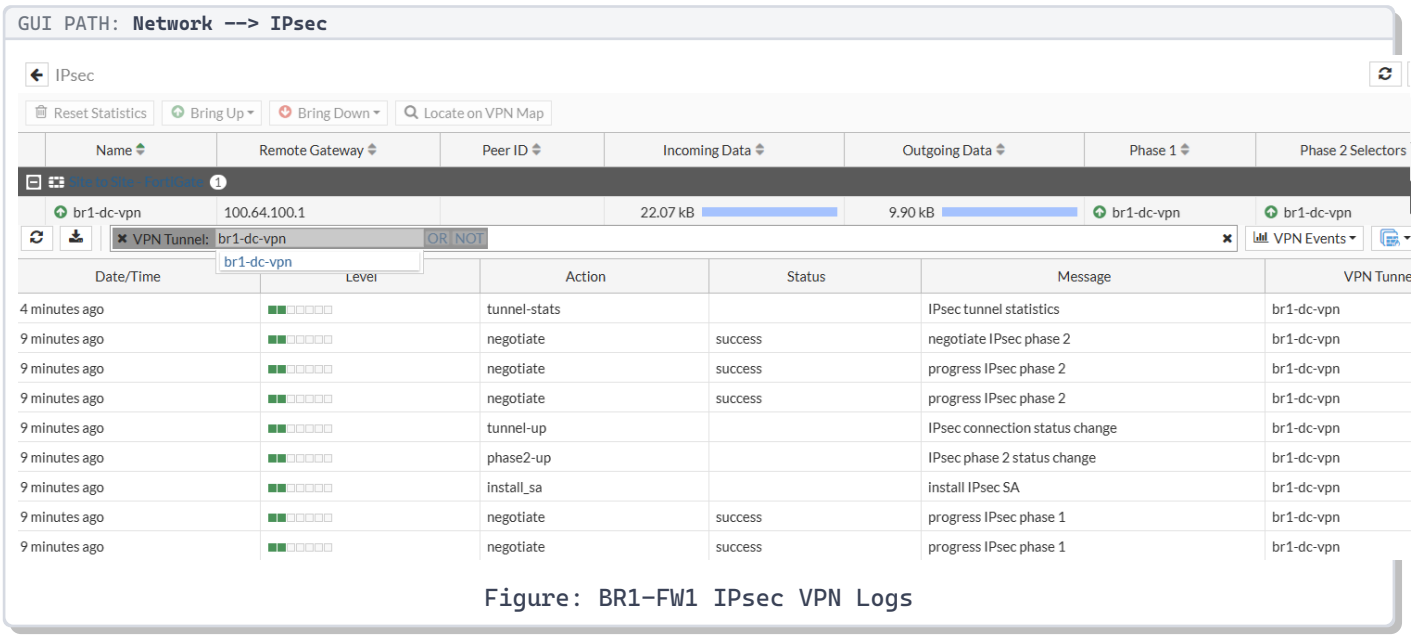


Figure: BR1-FW1 IPsec VPN Logs

In FortiGate, template-based IPsec tunnels simplify VPN deployment by auto-creating Phase 1 and 2 settings. Also with tunnel setup, address objects for remote subnets and group them for easier policy management. Add static routes pointing traffic to the tunnel interface and finally, configure a firewall policy allowing LAN-to-VPN communication using the address group. All this shown in below screenshots.

- Addresses and Address Group

GUI PATH: **Policy & Objects --> Addresses/Address Group**

Name	Details	Interface	Type	Ref.
IP Range/Subnet 15				
br1-dc-vpn_local_subnet_1	192.168.21.0/24		Address	1
br1-dc-vpn_local_subnet_2	192.168.22.0/24		Address	1
br1-dc-vpn_local_subnet_3	192.168.23.0/24		Address	1
br1-dc-vpn_remote_subnet_1	192.168.11.0/30		Address	1
br1-dc-vpn_remote_subnet_2	192.168.12.0/30		Address	1
br1-dc-vpn_remote_subnet_3	192.168.13.0/30		Address	1
Address Group 5				
br1-dc-vpn_local	br1-dc-vpn_local_subnet_1 br1-dc-vpn_local_subnet_2 br1-dc-vpn_local_subnet_3		Address Group	3
br1-dc-vpn_remote	br1-dc-vpn_remote_subnet_1 br1-dc-vpn_remote_subnet_2 br1-dc-vpn_remote_subnet_3		Address Group	5

Figure: BR1-FW1 addresses and Address Object

- Static Route – VPN wizard automatically created both the forward route and a blackhole route for the template-based IPsec tunnel.

GUI PATH: **Network --> Static Routes**

Destination	Gateway IP	Interface	Status	Comments
br1-dc-vpn_remote	100.64.100.1	br1-dc-vpn	Enabled	VPN: br1-dc-vpn (Created by VPN wizard)
br1-dc-vpn_remote		Blackhole	Enabled	VPN: br1-dc-vpn (Created by VPN wizard)

Figure: BR1-FW1 static route

- Firewall Policy – Auto-generated firewall policies for dc-br1-tunnel – LAN-to-VPN and VPN-to-LAN communication enabled, permitting all services for seamless bidirectional traffic flow.

GUI PATH: **Policy & Objects --> Firewall Policy**

Name	From	To	Source	Destination	Schedule	Service	Action	Hit Count	Bytes
vpn_br1-dc-vpn_local_0	LAN	br1-dc-vpn	br1-dc-vpn_local	br1-dc-vpn_remote	always	ALL	✔ ACCEPT	25	33.32 kB
vpn_br1-dc-vpn_remote_0	br1-dc-vpn	LAN	br1-dc-vpn_remote	br1-dc-vpn_local	always	ALL	✔ ACCEPT	0	0 B

Figure: BR1-FW1 firewall policy

The following CLI commands can be used to validate the IP sec tunnel status.

➟ **diagnose vpn ike gateway list**

The command **diagnose vpn ike gateway list** is used on a FortiGate firewall to quickly display IKE Gateway phase 1 details.

- Displays all configured IKE gateways (Phase 1 interfaces).
- Shows their status (up/down), peer IP, local interface, and authentication method.
- Useful for quickly checking whether IPsec tunnels are established or stuck in negotiation.

➲ diagnose vpn ike gateway list

```
BR1-FW1 (FG-traffic) # diagnose vpn ike gateway list
name: br1-dc-tunnel
version: 1
interface:  0
addr: 100.64.100.2:500 -> 100.64.100.1:500
tun_id: 100.64.100.1/::100.64.100.1
remote_location: 0.0.0.0
created: 2290s ago
IKE SA: created 2/2  established 2/2  time 0/5/10 ms
IPsec SA: created 1/1  established 1/1  time 0/0/0 ms
  id/spi: 1 6476cfe056efaf8d/e212e75dc2cbb438
  direction: responder
  status: established 2285-2285s ago = 0ms
  proposal: des-md5
  key: b7d86960072d3e13
  lifetime/rekey: 86400/83844
  id/spi: 0 e6905af884c27457/0da3833ad2230d13
  direction: initiator
  status: established 2290-2290s ago = 10ms
  proposal: des-md5
  key: 0a3b439e817b6b52
  lifetime/rekey: 86400/83809
```

➟ diagnose vpn tunnel list

The command **diagnose vpn tunnel list** is used on a FortiGate firewall to check below details.

- Phase 1 status → whether the IKE SA is up.
- Phase 2 selectors → local/remote subnet pairs negotiated.
- Encryption/authentication algorithms in use.
- Uptime and byte counters for each tunnel.
- Peer IP and interface used for the tunnel.

➲ diagnose vpn tunnel list

```
BR1-FW1 (FG-traffic) # diagnose vpn tunnel list
name=br1-dc-tunnel ver=1 serial=7 100.64.100.2:0->100.64.100.1:0 tun_id=100.64.100.1 tun_id6
    =::100.64.100.1 dst_mtu=1500 dpd-link=on
  src: 0:192.168.21.0/255.255.255.0:0 0:192.168.22.0/255.255.255.0:0
    0:192.168.23.0/255.255.255.0:0
  dst: 0:192.168.11.0/255.255.255.252:0 0:192.168.12.0/255.255.255.252:0
    0:192.168.13.0/255.255.255.252:0
  dec: spi=4db7790d esp=des key=8 bbe3e5437c5bf5d5
       ah=md5 key=16 22fa81952bea684715aae55907985da9
  enc: spi=bca01885 esp=des key=8 a90d2f91d6891b66
       ah=md5 key=16 27a57f466aad8d9136a5225c3258d9fe
  dec:pkts/bytes=1934/168902, enc:pkts/bytes=1979/270000
run_tally=0
```

➟ execute ping

The command **execute ping (remote-subnet)** is used on a FortiGate firewall to confirms that:

- Our Phase 2 selectors for each subnet are correctly negotiated in the IPsec tunnel.
- Firewall policies are allowing ICMP traffic from each source subnet to its destination.

execute ping

```
BR1-FW1 (FG-traffic) # execute ping-options source 192.168.21.1
BR1-FW1 (FG-traffic) # execute ping 192.168.11.2
PING 192.168.11.2 (192.168.11.2): 56 data bytes
64 bytes from 192.168.11.2: icmp_seq=0 ttl=63 time=3.1 ms
.
64 bytes from 192.168.11.2: icmp_seq=4 ttl=63 time=3.9 ms
--- 192.168.11.2 ping statistics ---
5 packets transmitted, 5 packets received, 0\% packet loss
round-trip min/avg/max = 3.1/3.8/5.2 ms
!
BR1-FW1 (FG-traffic) # execute ping-options source 192.168.22.1
BR1-FW1 (FG-traffic) # execute ping 192.168.12.2
PING 192.168.12.2 (192.168.12.2): 56 data bytes
64 bytes from 192.168.12.2: icmp_seq=0 ttl=63 time=3.7 ms
.
64 bytes from 192.168.12.2: icmp_seq=4 ttl=63 time=3.0 ms
--- 192.168.12.2 ping statistics ---
5 packets transmitted, 5 packets received, 0\% packet loss
round-trip min/avg/max = 2.8/3.4/4.1 ms
!
BR1-FW1 (FG-traffic) # execute ping-options source 192.168.23.1
BR1-FW1 (FG-traffic) # execute ping 192.168.13.2
PING 192.168.13.2 (192.168.13.2): 56 data bytes
64 bytes from 192.168.13.2: icmp_seq=0 ttl=63 time=5.3 ms
.
64 bytes from 192.168.13.2: icmp_seq=4 ttl=63 time=2.7 ms

--- 192.168.13.2 ping statistics ---
5 packets transmitted, 5 packets received, 0\% packet loss
round-trip min/avg/max = 2.7/4.3/5.3 ms
```

- Common Template based IPSEC tunnel Limitations
  - **Single Address Group Limitation**
    The wizard groups all remote subnets into one address group. This makes it harder to apply department-specific firewall policies (e.g., HR restricted, Sales open).
  - **Policy Granularity**
    Auto-generated policies are broad (LAN ↔ VPN). They don't differentiate between HR, Prod, and Sales traffic. If you need per-department rules, you must manually split policies.
  - **Routing Conflicts**
    Static routes created by the wizard point all traffic through the same tunnel interface. Overlapping or multiple subnets can cause ambiguity unless you add separate Phase 2 selectors and routes.

- **Troubleshooting Complexity**
  Since everything is bundled under one tunnel, isolating issues for a specific department (say HR subnet not passing traffic) becomes harder compared to custom VPNs where each department can have its own tunnel.
  To overcome this issue we will create custom site-to-site vpn in next section.

15.3.2 Site-to-Site VPN using custom method

A Site-to-Site VPN using the custom method on FortiGate means we're not relying on the wizard's pre-built templates – instead, you manually configure Phase 1 and 2, firewall policies, and routes. This gives you granular control over parameters and is often used when:

- We need **non-standard encryption or authentication algorithms** that differ from FortiGate defaults.
- We want to **fine-tune Phase 1 and Phase 2 parameters**, such as lifetimes, DH groups, or proposals.
- We must **integrate with third-party devices** (Cisco, Juniper, Palo Alto, etc.) that don't match FortiGate's wizard templates.
- We want to **segment traffic per department** by defining multiple Phase 2 selectors and firewall policies.

To start the custom method for IPsec VPN from Data Center to Branch-1 and vice versa. We first create address objects on DC-FW and BR1-FW1 for there Remote and Local subnets.

Configuration Field Breakdown

- **set allow-routing enable** – Allows this address object to be used in routing (e.g., static routes), not just firewall policies.

Configure addresses object and address group on both DC-FW1 and BR1-FW1 Device as per following.

➲ Address Object for DC-FW1 LAN Segment on both Firewall

```
config vdom
edit FG-traffic
config firewall address
 edit dc-tunnel_hr
  set comment dc-tunnel_hr
  set subnet 192.168.13.0 255.255.255.252
  set allow-routing enable
 next
 edit dc-tunnel_sales
  set comment dc-tunnel_sales
  set subnet 192.168.11.0 255.255.255.252
  set allow-routing enable
 next
 edit dc-tunnel_prod
  set comment dc-tunnel_prod
  set subnet 192.168.12.0 255.255.255.252
  set allow-routing enable
 next
end
```

➲ Address Object for BR1-FW1 LAN Segment on both Firewall

```
config vdom
edit FG-traffic
config firewall address
 edit br1-tunnel_hr
  set comment br1-tunnel_hr
  set subnet 192.168.23.0 255.255.255.0
  set allow-routing enable
 next
 edit br1-tunnel_sales
  set comment br1-tunnel_sales
  set subnet 192.168.21.0 255.255.255.0
  set allow-routing enable
 next
 edit br1-tunnel_prod
  set comment br1-tunnel_prod
  set subnet 192.168.22.0 255.255.255.0
  set allow-routing enable
 next
end
```

➲ Address Group for DC-FW1 on both Firewall

```
config vdom
edit FG-traffic
 config firewall addrgrp
  edit dc-tunnel
   set member dc-tunnel_hr dc-tunnel_prod
    dc-tunnel_sales
   set allow-routing enable
  next
end
```

➲ Address Group for BR1-FW1 on both Firewall

```
config vdom
edit FG-traffic
 config firewall addrgrp
  edit br1-tunnel
   set member br1-tunnel_hr br1-tunnel_prod
     br1-tunnel_sales
   set allow-routing enable
  next
end
```

Now we will configure custom IPsec VPN on DC-FW1 as per below GUI screenshot.

Phase-1 Parameter:-

GUI PATH: **VPN --> IPsec Wizard --> Custom**

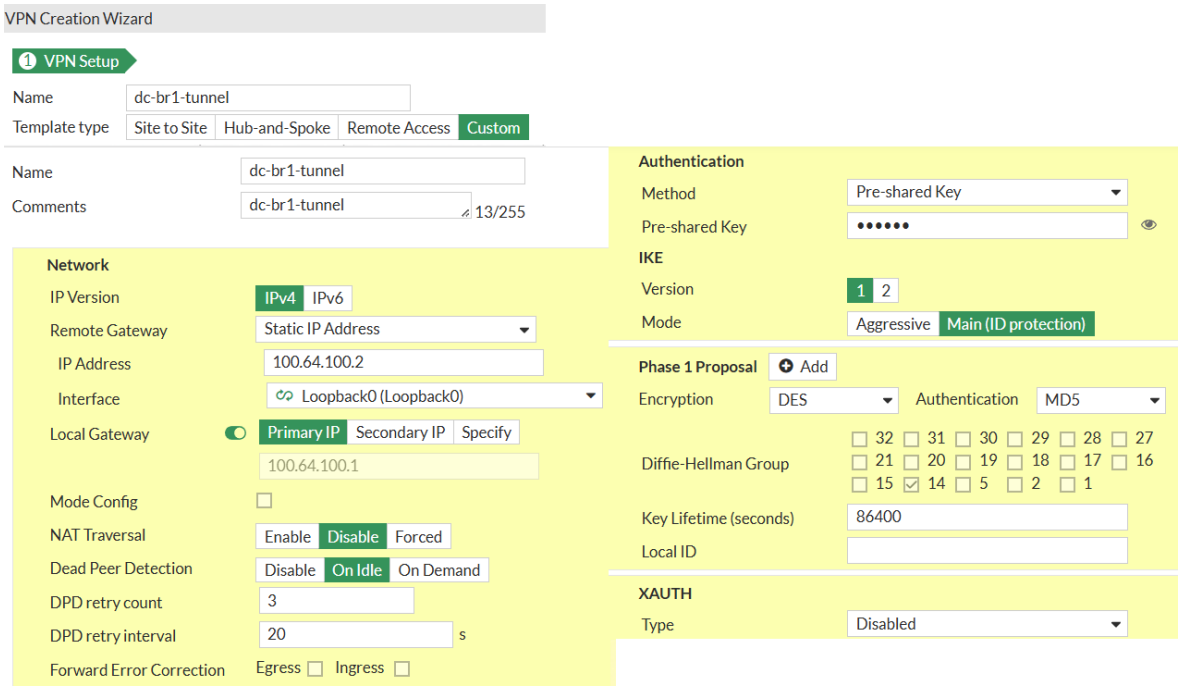


Figure: DC-FW1 Phase 1 Parameter

Phase-2 Selector:- The following configuration defines the **Phase 2 selectors** for the IPsec tunnel. It specifies local and remote address pairs with encryption and authentication parameters for secure departmental communication.

GUI PATH: **VPN --> IPsec Wizard --> Custom**

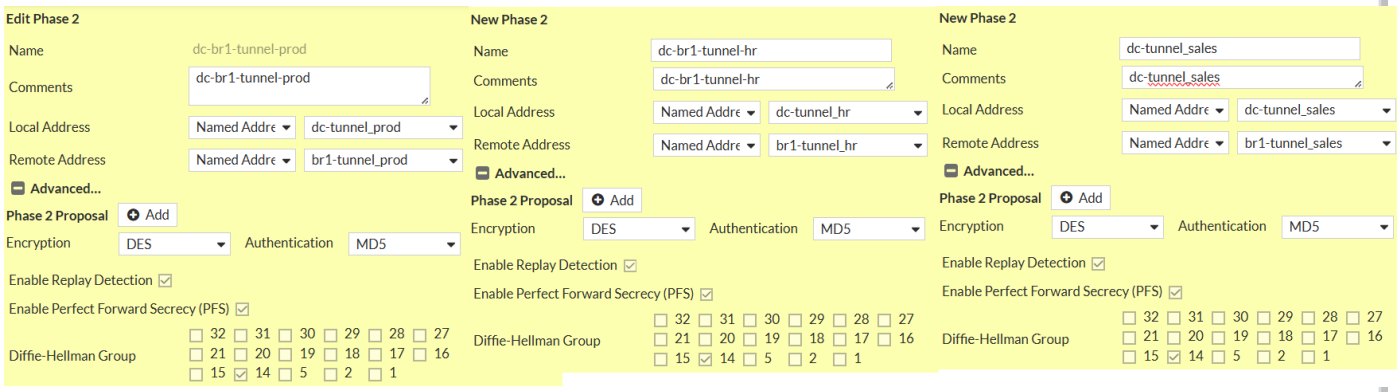


Figure: DC-FW1 Phase 2 Parameter

Add Static Routes:- Purpose of Each Static Route as per follow.

- **Primary Route**
  - Interface: dc-br1-tunnel
  - Administrative Distance: 10
  - Function: Main active route through the IPsec tunnel to BR1-FW1. Used when the tunnel is up and reachable.
- **Backup Route**
  - Interface: Blackhole
  - Administrative Distance: 254
  - Function: Failsafe route that discards traffic when the tunnel is down, preventing route flapping or unwanted fallback to default routes.

GUI PATH: **Network --> Static Routes**

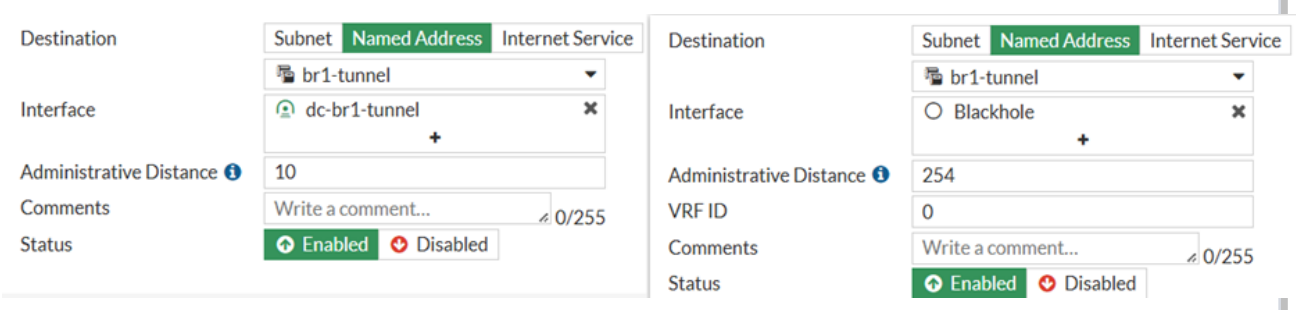
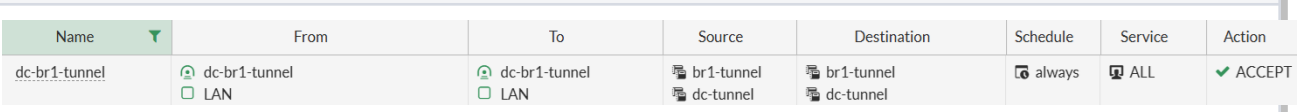


Figure: DC-FW1 Static Routes

Create Firewall Policies:-

GUI PATH: **Policy & Objects --> Firewall Policy**



Name	From	To	Source	Destination	Schedule	Service	Action
dc-br1-tunnel	dc-br1-tunnel LAN	dc-br1-tunnel LAN	br1-tunnel dc-tunnel	br1-tunnel dc-tunnel	always	ALL	ACCEPT

Figure: DC-FW1 Custom IPsec VPN Firewall Policy

We can configure custom IPSEC tunnel on BR1-FW1 firewall using CLI Commands as per following.

BR1-FW1 IPsec VPN Configuration

```
config vdom
edit FG-traffic
config vpn ipsec phase1-interface
 edit br1-dc-tunnel
  set interface Loopback0
  set local-gw 100.64.100.2
  set proposal des-md5
  set dhgrp 14
  set dpd on-idle
  set comments br1-dc-tunnel
  set nattraversal disable
  set remote-gw 100.64.100.1
  set psksecret *****
 next
end
```

BR1-FW1 firewall policy Configuration

```
config vdom
edit FG-traffic
config firewall policy
 edit 3
  set name br1-dc-lan
  set srcintf LAN br1-dc-tunnel
  set dstintf br1-dc-tunnel LAN
  set action accept
  set srcaddr br1-tunnel dc-tunnel
  set dstaddr dc-tunnel br1-tunnel
  set schedule always
  set service ALL
  set logtraffic all
 next
end
```

BR1-FW1 static route Configuration

```
config vdom
edit FG-traffic
config router static
 edit 3
  set device br1-dc-tunnel
  set dstaddr dc-tunnel
 next
 edit 4
  set distance 254
  set blackhole enable
  set dstaddr dc-tunnel
 next
end
```

BR1-FW1 IPsec phase-2 selector

```
config vpn ipsec phase2-interface
 edit dc-br1-tunnel-hr
  set phase1name br1-dc-tunnel
  set proposal des-md5
  set dhgrp 14
  set comments dc-br1-tunnel-hr
  set src-addr-type name
  set dst-addr-type name
  set src-name br1-tunnel_hr
  set dst-name dc-tunnel_hr
 next
 edit dc-br1-tunnel-prod
  set phase1name br1-dc-tunnel
  set proposal des-md5
  set dhgrp 14
  set comments dc-br1-tunnel-prod
  set src-addr-type name
  set dst-addr-type name
```

```
    set src-name br1-tunnel_prod
    set dst-name dc-tunnel_prod
  next
  edit dc-br1-tunnel-sales
    set phase1name br1-dc-tunnel
    set proposal des-md5
    set dhgrp 14
    set comments dc-br1-tunnel-sales
    set src-addr-type name
    set dst-addr-type name
    set src-name br1-tunnel_sales
    set dst-name dc-tunnel_sales
  next
 end
```

We can do the validation of created ipsec tunnels as per below CLI outputs. In Template based method already shown details about each command.

In the following CLI output of static routing table confirms that the custom IPsec tunnel (br1-dc-tunnel) is correctly advertising the three departmental subnets through the VPN interface.

diagnose vpn tunnel list

```
BR1-FW1 (FG-traffic) # get router info routing-table static
Routing table for VRF=0
S*      0.0.0.0/0 [10/0] via 203.0.113.9, port5, [1/0]
S       192.168.11.0/30 [10/0] via br1-dc-tunnel tunnel 100.64.100.1, [1/0]
S       192.168.12.0/30 [10/0] via br1-dc-tunnel tunnel 100.64.100.1, [1/0]
S       192.168.13.0/30 [10/0] via br1-dc-tunnel tunnel 100.64.100.1, [1/0]
```

Following CLI output shows a custom IPsec tunnel (dc-br1-tunnel-c) with three Phase□2 selectors – one each for HR, Prod, and Sales. That's exactly how custom tunnels differ from template-based ones: we manually define multiple subnets instead of the wizard grouping them together.

diagnose vpn tunnel list

```
DC-FW1 (FG-traffic) # diagnose vpn tunnel list
name=dc-br1-tunnel ver=1 serial=5 100.64.100.1:0->100.64.100.2:0
dpd: mode=on-idle on=1 idle=20000ms retry=3 count=0 seqno=165
natt: mode=none draft=0 interval=0 remote_port=0
proxyid=dc-br1-tunnel-hr proto=0 sa=1 ref=3 serial=3
  src: 0:192.168.13.0/255.255.255.252:0
  dst: 0:192.168.23.0/255.255.255.0:0
  dec:pkts/bytes=754/63336, enc:pkts/bytes=754/102544
proxyid=dc-br1-tunnel-prod proto=0 sa=1 ref=2 serial=2
  src: 0:192.168.12.0/255.255.255.252:0
  dst: 0:192.168.22.0/255.255.255.0:0
  dec:pkts/bytes=4/240, enc:pkts/bytes=4/448
proxyid=dc-br1-tunnel-sales proto=0 sa=1 ref=2 serial=4
  src: 0:192.168.11.0/255.255.255.252:0
  dst: 0:192.168.21.0/255.255.255.0:0
  dec:pkts/bytes=17/1675, enc:pkts/bytes=14/3896
run_tally=0
!
BR1-FW1 (FG-traffic) # diagnose vpn tunnel list
```

```
name=br1-dc-tunnel ver=1 serial=1 100.64.100.2:0->100.64.100.1:0 tun_id=100.64.100.1
dpd: mode=on-idle on=1 idle=20000ms retry=3 count=0 seqno=2
natt: mode=none draft=0 interval=0 remote_port=0
proxyid=dc-br1-tunnel-hr proto=0 sa=1 ref=2 serial=3
  src: 0:192.168.23.0/255.255.255.0:0
  dst: 0:192.168.13.0/255.255.255.252:0
  dec:pkts/bytes=0/0, enc:pkts/bytes=0/0
proxyid=dc-br1-tunnel-prod proto=0 sa=1 ref=2 serial=2
  src: 0:192.168.22.0/255.255.255.0:0
  dst: 0:192.168.12.0/255.255.255.252:0
  dec:pkts/bytes=0/0, enc:pkts/bytes=0/0
proxyid=dc-br1-tunnel-sales proto=0 sa=1 ref=2 serial=4
  src: 0:192.168.21.0/255.255.255.0:0
  dst: 0:192.168.11.0/255.255.255.252:0
  dec:pkts/bytes=502/518424, enc:pkts/bytes=502/531476
run_tally=0
```

diagnose vpn tunnel list

```
BR1-FW1 (FG-traffic) # get vpn ike gateway
vd: FG-traffic/1
name: br1-dc-tunnel
addr: 100.64.100.2:500 -> 100.64.100.1:500
IKE SA  created: 1/1  established: 1/1  time: 0/0/0 ms
IPsec SA  created: 3/3  established: 3/3  time: 0/6/10 ms
  direction: responder
  proposal: des-md5
!
DC-FW1 (FG-traffic) # get vpn ike gateway
vd: FG-traffic/1
name: dc-br1-tunnel
addr: 100.64.100.1:500 -> 100.64.100.2:500
IKE SA  created: 1/1  established: 1/1  time: 21000/21000/21000 ms
IPsec SA  created: 3/3  established: 3/3  time: 0/0/0 ms
  direction: initiator
  proposal: des-md5
!
DC-FW1 (FG-traffic) # get vpn ipsec tunnel summary
'dc-br1-tunnel' 100.64.100.2:0  selectors(total,up): 3/3  rx(pkt,err): 251/0  tx(pkt,err):
    251/1
!
BR1-FW1 (FG-traffic) # get vpn ipsec tunnel summary
'br1-dc-tunnel' 100.64.100.1:0  selectors(total,up): 3/3  rx(pkt,err): 251/0  tx(pkt,err):
    251/1
```

Following CLI output shows Successful pings when source matches Phase 2 selector and Failed pings when source doesn't match Phase 2 selector.

BR1-FW1 execute ping to diagnose vpn tunnel between BR1-FW1 to DC-FW1

```
BR1-FW1 (FG-traffic) # execute ping-options source 192.168.21.1
BR1-FW1 (FG-traffic) # execute ping 192.168.11.1
PING 192.168.11.1 (192.168.11.1): 56 data bytes
64 bytes from 192.168.11.1: icmp_seq=0 ttl=255 time=1.7 ms
!
--- 192.168.11.1 ping statistics ---
3 packets transmitted, 3 packets received, 0\% packet loss
```

```
round-trip min/avg/max = 1.7/1.7/1.9 ms
!
BR1-FW1 (FG-traffic) # execute ping-options source 192.168.22.1
BR1-FW1 (FG-traffic) # execute ping 192.168.12.1
PING 192.168.12.1 (192.168.12.1): 56 data bytes
64 bytes from 192.168.12.1: icmp_seq=0 ttl=255 time=1.9 ms
!
--- 192.168.12.1 ping statistics ---
2 packets transmitted, 2 packets received, 0\% packet loss
round-trip min/avg/max = 1.0/1.4/1.9 ms
!
BR1-FW1 (FG-traffic) # execute ping-options source 192.168.23.1
BR1-FW1 (FG-traffic) # execute ping 192.168.13.1
PING 192.168.13.1 (192.168.13.1): 56 data bytes
64 bytes from 192.168.13.1: icmp_seq=0 ttl=255 time=1.6 ms
!
--- 192.168.13.1 ping statistics ---
3 packets transmitted, 3 packets received, 0\% packet loss
round-trip min/avg/max = 1.3/1.6/2.1 ms
!
BR1-FW1 (FG-traffic) # execute ping-options source 192.168.23.1
BR1-FW1 (FG-traffic) # execute ping 192.168.12.1
PING 192.168.12.1 (192.168.12.1): 56 data bytes
--- 192.168.12.1 ping statistics ---
3 packets transmitted, 0 packets received, 100\% packet loss
!
BR1-FW1 (FG-traffic) # execute ping-options source 192.168.23.1
BR1-FW1 (FG-traffic) # execute ping 192.168.11.1
PING 192.168.11.1 (192.168.11.1): 56 data bytes
--- 192.168.11.1 ping statistics ---
5 packets transmitted, 0 packets received, 100\% packet loss
```

Observation from above output

- **Phase 1 (IKE SA):**
  - Both DC-FW1 and BR1-FW1 show IKE SA created: 1/1, established: 1/1.
  - Proposal negotiated: DES-MD5.
  - Direction: DC-FW1 acts as *initiator*, BR1-FW1 as *responder*.
- **Phase 2 (IPsec SA):**
  - Three selectors (proxy-IDs) defined: HR, Prod, Sales.
  - Each selector shows created: 3/3, established: 3/3.
  - Traffic counters confirm encryption/decryption activity per subnet.
- **Routing Table Verification:**
  - Static routes point to tunnel interface br1-dc-tunnel for HR, Prod, Sales subnets.
- **Ping Tests:**
  - Successful pings when source IP matches Phase 2 selector:
    - 192.168.21.1 → 192.168.11.1 (Sales).
    - 192.168.22.1 → 192.168.12.1 (Prod).

    - 192.168.23.1 → 192.168.13.1 (HR).
  - Failed pings when source does not match selector:
    - 192.168.23.1 → 192.168.12.1.
    - 192.168.23.1 → 192.168.11.1.

- **Observation:** Custom tunnel (dc-br1-tunnel) successfully enforces subnet-based Phase 2 selectors. Traffic passes only when source/destination match defined proxy-IDs, demonstrating correct IPsec policy enforcement.

15.4 Remote Access VPN

Remote access VPN in FortiGate connects remote workers securely to the internal corporate network. FortiGate natively supports two primary types: SSL-VPN and IPsec Remote Access (Dial-up IPsec).

15.4.1 Split and Full Tunnel mode

- **Split Tunneling:**
  - **Routing:** Encapsulates only specified corporate subnets; general internet traffic exits via the local ISP.
  - **Endpoint Table:** Retains local default gateway (0.0.0.0/0) and injects static routes for corporate networks.
  - **Bandwidth & Load:** Minimal impact on corporate WAN bandwidth and FortiGate CPU.
  - **Security:** Remote internet traffic bypasses corporate firewall UTM inspection.
  - **Use Case:** BYOD, remote workforces, and bandwidth-sensitive collaboration tools (e.g., Teams, Zoom).
- **Full Tunneling:**
  - **Routing:** Forces all endpoint traffic (corporate and public internet) through the VPN tunnel.
  - **Endpoint Table:** Overrides local default route (0.0.0.0/0) to point to the VPN interface.
  - **Bandwidth & Load:** High corporate WAN utilization and elevated FortiGate processing overhead.
  - **Security:** Complete visibility; enables full UTM inspection, web filtering, and DLP on all traffic.
  - **Use Case:** Regulated industries (finance, healthcare), managed corporate assets, and strict zero-trust policies.

15.4.2 SSL VPN Portals

- **Definition & Role:** An SSL-VPN Portal in FortiOS defines the specific user interface, access permissions, and session parameters presented to an authenticated remote user.
- **Built-in Default Portals:**
  - **full-access:** Supports both Web Mode (browser-based bookmarks) and Tunnel Mode (FortiClient Layer 3 access).
  - **tunnel-access:** Dedicated exclusively to Tunnel Mode; automatically initiates the FortiClient tunnel upon login.

  - **web-access:** Dedicated exclusively to clientless Web Mode; restricts access to browser-rendered bookmarks and quick connections.

- **Security & Endpoint Features:** Controls host check restrictions (OS version, AV/firewall posture), auto-login, idle timeouts, and parallel session limits.
- **Portal Mapping:** Mapped to specific users or user groups under VPN > SSL-VPN Settings, allowing role-based access (e.g. mapping contractors to web-access and employees to tunnel-access).

> ☞ **Note**
> FortiGate evaluation version only support one SSL VPN portal. So for the lab environments, we will use one portal (full-access) and toggle between web-mode and tunnel-mode as needed.

15.4.3 SSL VPN Web Mode (Client-less)

SSL VPN Web Mode (Clientless) acts as an application-layer reverse proxy, allowing users to securely access internal resources (like HTTP/HTTPS, RDP, and SSH) directly through a standard web browser without installing any client software. Because it does not assign a virtual IP address or join the endpoint to the corporate network, it is ideal for unmanaged devices, BYOD, and third-party contractors.

- **Web Mode Controls:** Configures bookmarks (HTTP/HTTPS, RDP, SSH, VNC, SMB), customizable UI themes, and browser-based login banners.

Due to evaluation version limitation we will modify full-access ssl vpn portal to be used as web-access portal only. Below GUI screenshot shown how to configure web access and bookmark for ssl vpn portal.

GUI PATH: **VPN --> SSL-VPN Portals --> Edit**

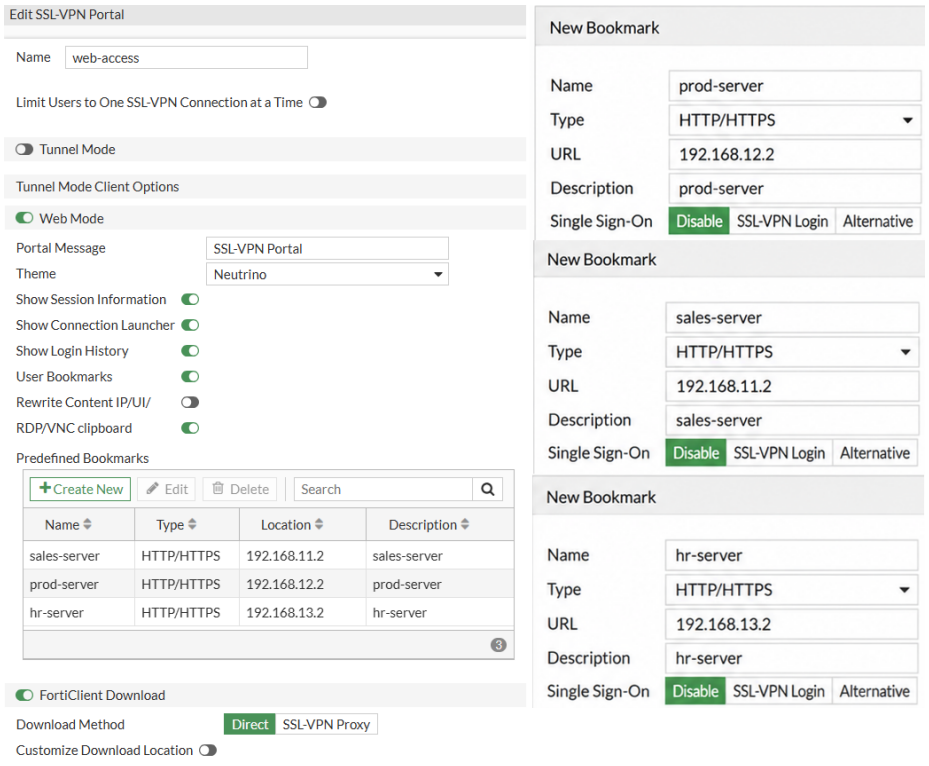


Figure: DC-FW1 SSL VPN Web Portal and Bookmark configuration

The GUI Screenshot below shows the SSL VPN and Firewall policy Configuration.

GUI PATH: **VPN --> SSL-VPN Settings --> Edit**

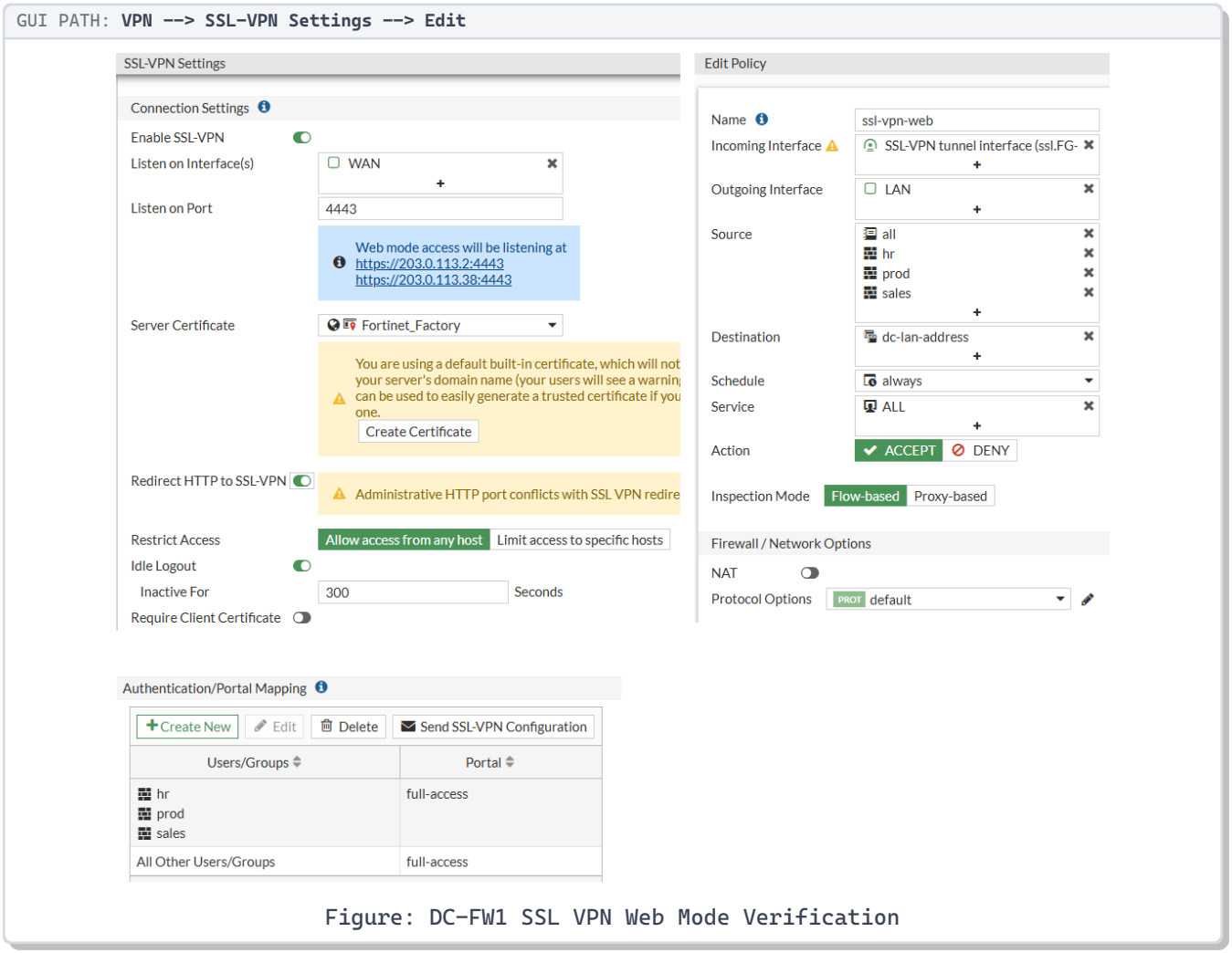


Figure: DC-FW1 SSL VPN Web Mode Verification

We can do the verification of ssl vpn web mode as per below GUI screenshot and cli commands.

SSL Verification

```
DC-FW1 (FG-traffic) # diagnose vpn ssl statistics
SSLVPN statistics (FG-traffic):
------------------
Memory unit:                1
SSLVPN state:               normal
Max number of users:        2
Max number of tunnels:      1
Max number of connections: 12
Current number of users:        2
Current number of tunnels:      0
Current number of connections: 10
!
DC-FW1 (FG-traffic) # diagnose vpn ssl list
[8599:FG-traffic]sconn=0x7f452a1f1000, from(203.0.113.22) task=ap_read_request, fd=33(1:1),-1
    (0:0),-1(0:0),-1(0:0),-1(0:0), pending=0, asic_pending=0.
!output omitted....
[8599:FG-traffic]sconn=0x7f452a1f4c00, from(203.0.113.22) task=ap_read_request, fd=38(1:1),-1
    (0:0),-1(0:0),-1(0:0),-1(0:0), pending=0, asic_pending=0.
```

GUI PATH: **Dashboard --> Network --> SSL-VPN**

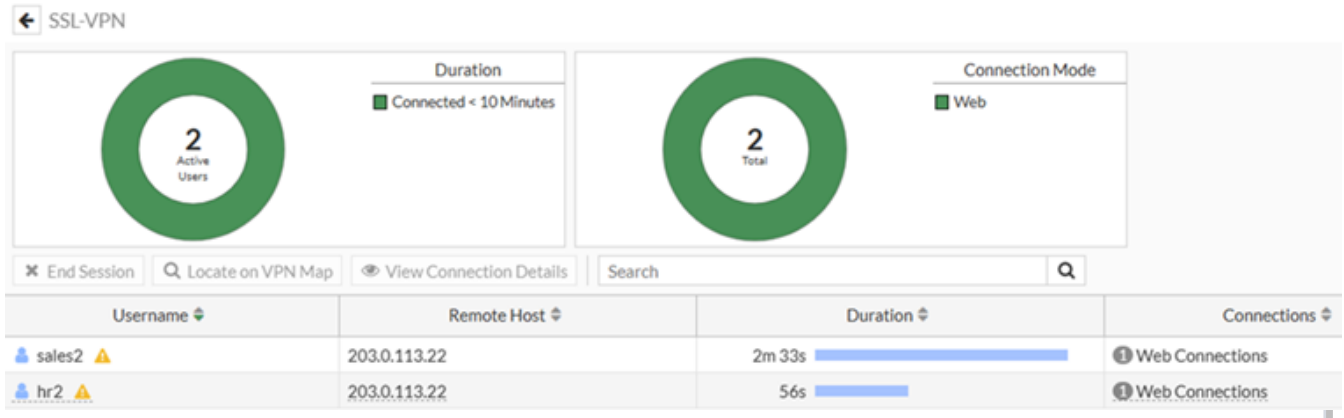


Figure: DC-FW1 SSL VPN configuration

In the following Screenshot we can see the ssl web vpn portal.

GUI PATH: **VPN --> SSL-VPN Portals --> Edit**

Figure: DC-FW1 SSL VPN Tunnel Mode

15.4.4 SSL VPN Tunnel mode (Client-based)

SSL VPN Tunnel Mode (Client-based) requires installing software like FortiClient to establish an encrypted Layer 3 tunnel to the firewall, assigning the endpoint a virtual IP address. This grants full network connectivity for all IP protocols and desktop applications, supporting both split-tunnel and full-tunnel routing for regular remote employees.

- **Tunnel Mode Controls:** Assigns virtual IP pools, DNS servers, client routing (split tunneling vs. full tunneling), and FortiClient auto-connect settings.

Below GUI screenshot shown how to configure ssl vpn tunnel mode.

GUI PATH: **VPN --> SSL-VPN Portals --> Edit**

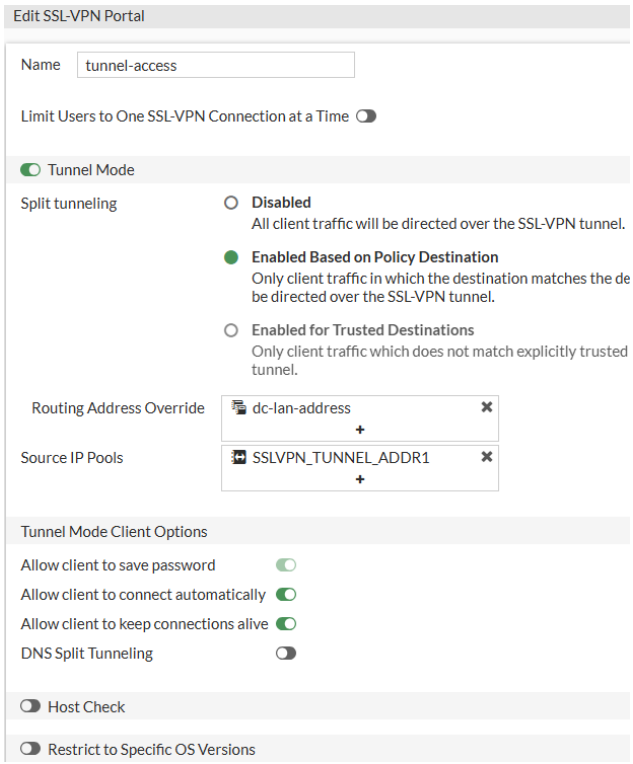


Figure: DC-FW1 SSL VPN Tunnel Mode

We will use firewall policy same as created for web-mode. Now we will see the FortiClient configuration and vpn status on client device as shown in the screenshot below.

GUI PATH:

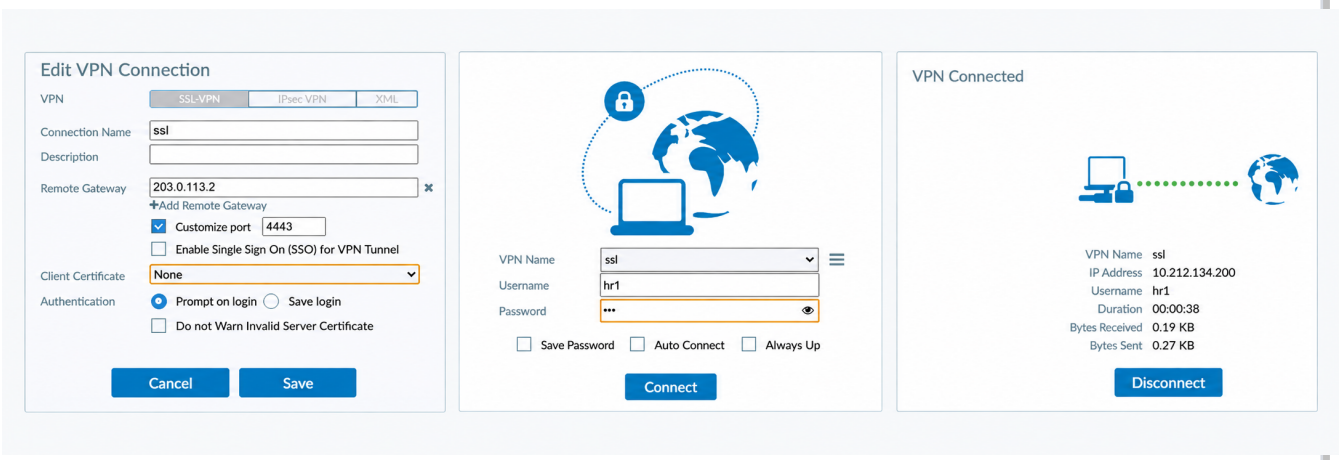


Figure: DC-FW1 SSL VPN FortiClient

We can do the verification of ssl vpn tunnel mode as per below GUI screenshot and cli commands.

DC-FW1 ssl_remote_vpn verification

```
DC-FW1 (FG-traffic) # get vpn ssl monitor
SSL-VPN Login Users:
 Index    User     Group    Auth Type       Timeout          Auth-Timeout     From      HTTP in/out
       HTTPS in/out     Two-factor Auth
```

```
 0       hr1     hr     1(1)              262     28760    203.0.113.22   0/0     0/0     0

SSL-VPN sessions:
 Index   User    Group   Source IP      Duration        I/O Bytes       Tunnel/Dest IP
 0       hr1     hr     203.0.113.22     38      208/190         10.212.134.200
!
DC-FW1 (FG-traffic) # get vpn ssl settings
status              : enable
reqclientcert       : disable
ciphersuite         : TLS-AES-128-GCM-SHA256 TLS-AES-256-GCM-SHA384
    TLS-CHACHA20-POLY1305-SHA256
ssl-insert-empty-fragment: enable
https-redirect      : enable
x-content-type-options: enable
ssl-client-renegotiation: disable
force-two-factor-auth: disable
servercert          : Fortinet_Factory
idle-timeout        : 300
auth-timeout        : 28800
login-attempt-limit : 2
login-block-time    : 60
login-timeout       : 30
tunnel-ip-pools     : "SSLVPN_TUNNEL_ADDR1"
tunnel-ipv6-pools   : "SSLVPN_TUNNEL_IPv6_ADDR1"
port                : 4443
port-precedence     : enable
auto-tunnel-static-route: enable
header-x-forwarded-for: add
source-interface    : "WAN"
source-address      : "all"
source-address-negate: disable
source-address6     : "all"
source-address6-negate: disable
default-portal      : tunnel-access
```

GUI PATH: **Dashboard --> Network --> SSL-VPN**

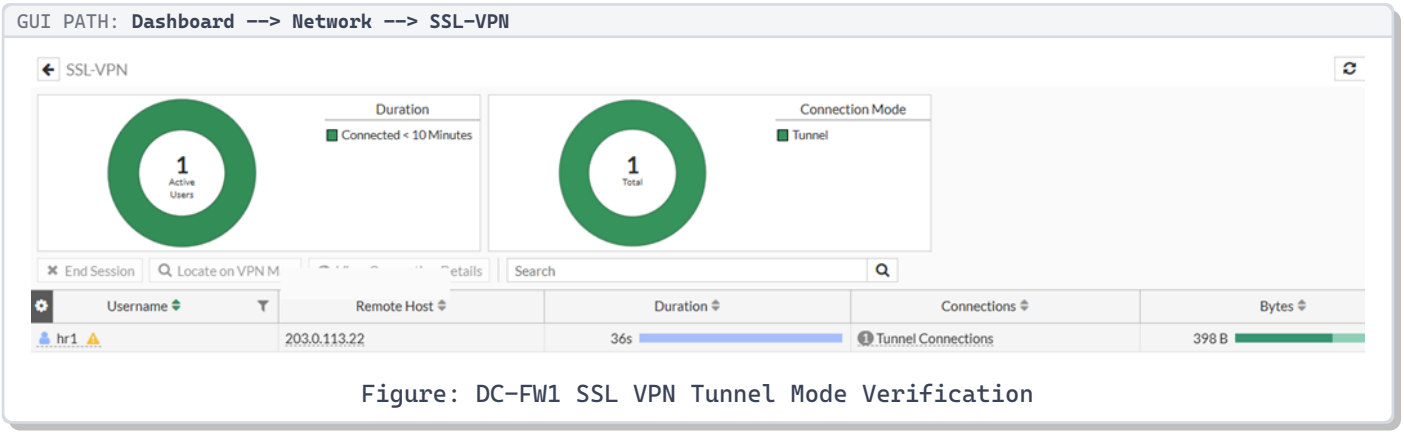


Figure: DC-FW1 SSL VPN Tunnel Mode Verification

15.4.5 Remote Access IPsec VPN

Remote Access IPsec VPN (also known as Dial-up IPsec VPN or Client-to-Gateway IPsec) enables remote endpoints (laptops, mobile devices) to securely connect to a central firewall/gateway over the public internet using standard IPsec protocols.

In the following screenshot we shown the ipsec remote vpn configuration steps.

Configuration Field Breakdown

- **Network:**
  - **IP Version:** IPv4 – Defines the protocol used for VPN communication.
  - **Remote Gateway:** Dialup User – Allows dynamic remote clients to initiate the VPN.
  - **Interface:** Loopback0 (Local Gateway 100.64.100.1) – Specifies the logical interface and local endpoint for tunnel termination.
  - **Mode Config:** Enabled – Permits FortiGate to assign IP addresses to remote users.
  - **Accessible Networks:** dc-lan-address – Determines which internal networks are reachable through the VPN.
- **IPv4 Mode Config:**
  - *General Purpose:* Defines how IP addresses and DNS settings are distributed to remote users, ensuring proper routing and name resolution inside the VPN.
  - **Client Address Range:** 10.20.20.1–10.20.20.254 – IP pool assigned to remote VPN clients.
  - **Subnet Mask:** 255.255.255.255 – Each client receives a unique host address.
  - **DNS Server:** 8.8.8.8 – Provides name resolution for connected clients.
- **Authentication (Phase 1):**
  - *General Purpose:* Establishes the secure channel parameters for initial tunnel negotiation, ensuring confidentiality and integrity during setup.
  - **Method:** Pre-shared Key – Simple shared secret for authentication.
  - **IKE Version:** 1 or 2 – Supports both legacy and modern negotiation protocols.
  - **IKEV1 Mode:** Main or Aggressive – Negotiation suitable for dynamic peers.
  - **Encryption:** DES, **Authentication:** MD5, **DH Group:** – 14 – Defines cryptographic strength and integrity. **Key Lifetime:** 86400 seconds – Tunnel rekey interval (24 hours).
- **NAT Traversal:**
  - *General Purpose:* Maintains tunnel reliability and connectivity even when NAT devices modify packet headers, ensuring automatic recovery from peer inactivity.
  - **Status:** Enabled – Allows VPN traffic to pass through NAT devices.
  - **Dead Peer Detection:** On Idle, **Retry Count:** 3, **Interval:** 60s – Monitors peer availability and re-establishes tunnels automatically.
  - **Advanced Options:** Route addition, interface IP exchange, and device creation all Enabled.
- **Phase 2 Selector:**
  - *General Purpose:* Defines which traffic is encrypted and how data integrity is maintained, ensuring secure communication between remote clients and internal networks.
  - **Local Address:** 0.0.0.0/0, **Remote Address:** 0.0.0.0/0 – Allows full access between endpoints.
  - **Encryption:** DES, **Authentication:** MD5, **DH Group:** – 14 – Defines cryptographic strength and integrity.

- **Replay Detection:** Enabled – Prevents packet replay attacks.
- **Perfect Forward Secrecy (PFS):** Enabled – Generates new keys per session. **Key Lifetime:** 43200 seconds – Rekey interval (12 hours).

GUI PATH: **VPN --> IPsec Wizard --> custom**

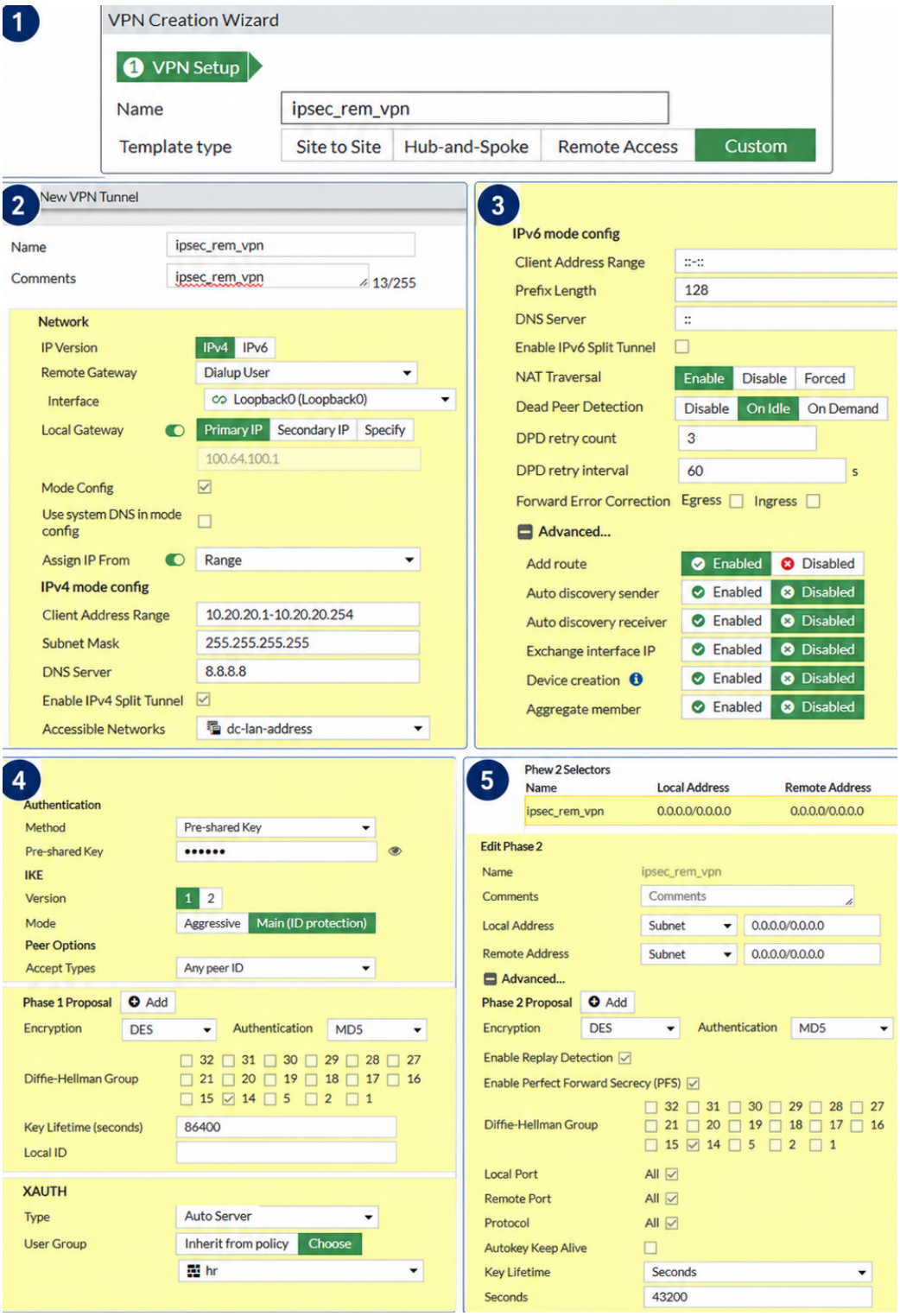


Figure: DC-FW1 IPSec Remote VPN

Now create address object for ipv4 address range from which remote user will get ip address i.e. Client Address Range.

GUI PATH: **Policy & Objects --> Addresses --> Create New --> Address**

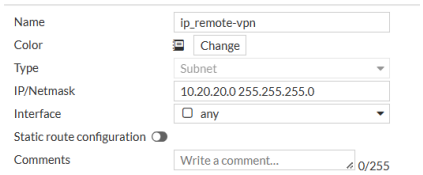


Figure: Address Object for Client IP Address Range

Now create Firewall Policy to allow traffic flow from ipsec remote vpn interface to LAN zone as per below image.

GUI PATH: **Policy & Objects --> Firewall Policy --> Create New**

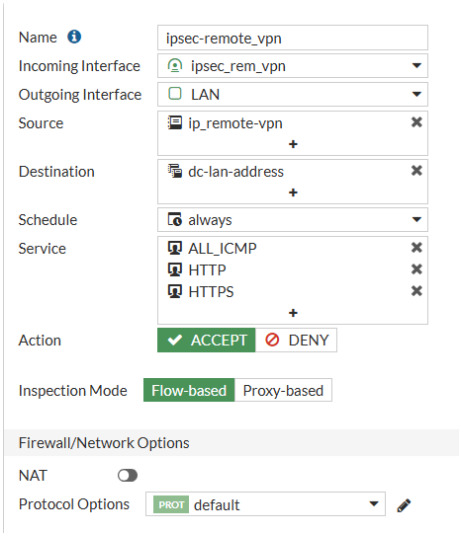


Figure: Firewall Policy for IPsec Remote VPN

On client machine we will do the forticlient configuration as per the following screenshot.

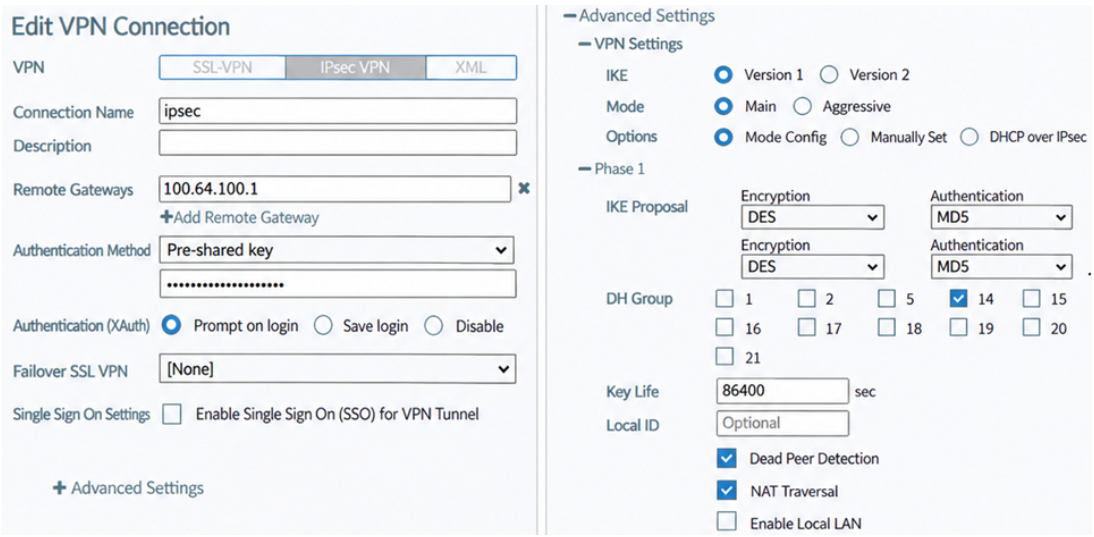


Figure: FortiClient Configuration

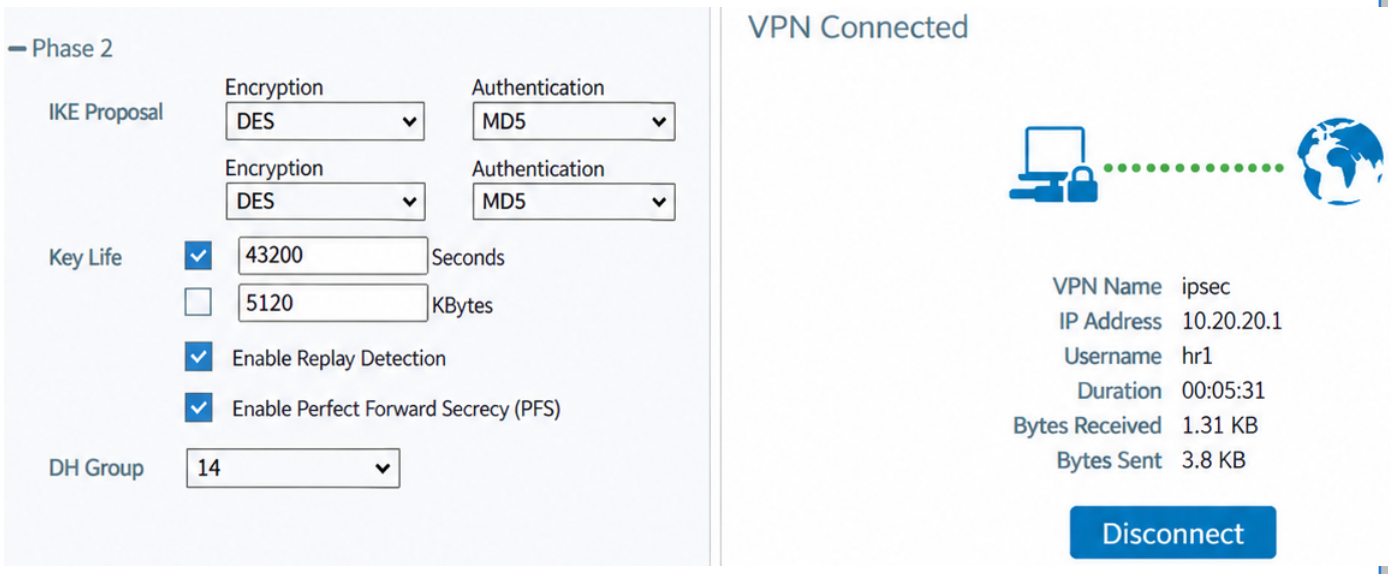


Figure: FortiClient Configuration

We can perform IPsec remote vpn verification as per below GUI and CLI Commands.

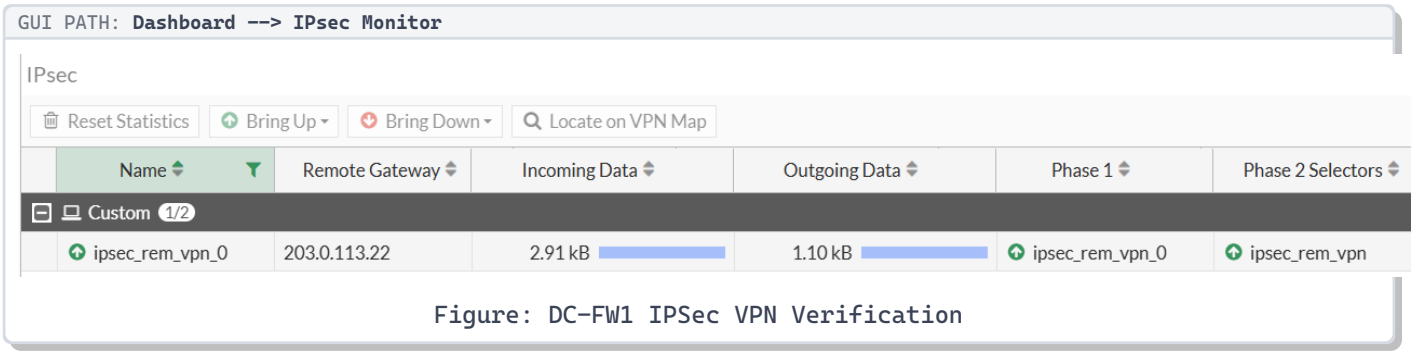


Figure: DC-FW1 IPSec VPN Verification

DC-FW1 ipsec_vpn_tunnel verification

```
DC-FW1 (FG-traffic) # get router info routing-table all
Routing table for VRF=0
S       10.20.20.1/32 [15/0] via ipsec_rem_vpn tunnel 10.20.20.1 vrf 0, [1/0]
!
DC-FW1 (FG-traffic) # get vpn ipsec tunnel summary
'ipsec_rem_vpn_0' 203.0.113.22:4500  selectors(total,up): 1/1  rx(pkt,err): 62/0  tx(pkt,err)
    : 21/0
!
DC-FW1 (FG-traffic) # diagnose sniffer packet any 'host 10.20.20.1' 4
filters=[host 10.20.20.1]
15.806903 ipsec_rem_vpn in 10.20.20.1 -> 192.168.12.1: icmp: echo request
15.807130 ipsec_rem_vpn out 192.168.12.1 -> 10.20.20.1: icmp: echo reply
```

Observation from above output

- **Routing Table:** Host route 10.20.20.1/32 is installed via the tunnel ipsec_rem_vpn. *Observation:* Confirms that the FortiGate dynamically created a route for the connected remote client.
- **Tunnel Summary:** Tunnel ipsec_rem_vpn_0 is up with selectors active (1/1). RX packets: 62, TX packets: 21, errors: 0. *Observation:* Indicates successful Phase 1 and Phase 2

negotiation, with bidirectional traffic flow and no errors.

- **Sniffer Output:** ICMP echo request from remote client 10.20.20.1 to internal host 192.168.12.1, followed by echo reply. *Observation:* End-to-end communication is functional; traffic is encrypted/decrypted correctly through the VPN tunnel.

The IPsec Remote VPN is fully operational – routing, tunnel establishment, and encrypted traffic flow are verified.

On client machine with trace we can verify the assigned ip range from IPsec remote-access vpn and then verify split-tunnel operation as per below cli output.

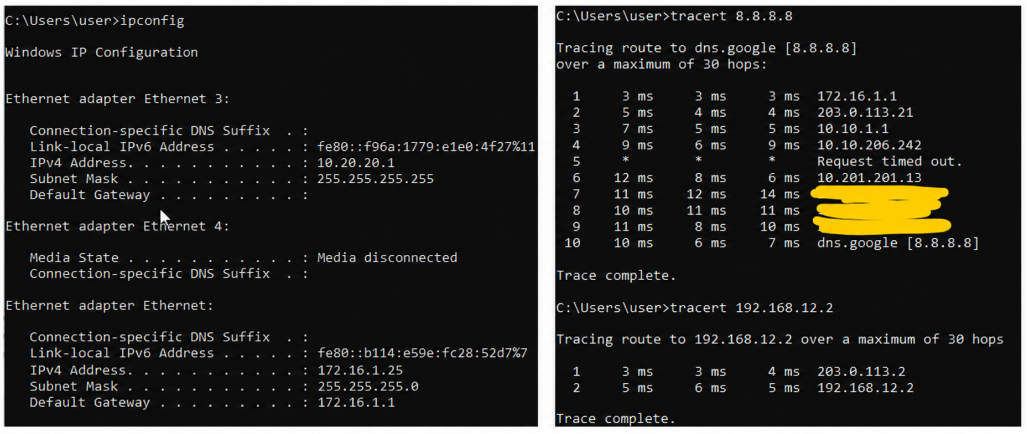


Figure: Remote Machine FortiClient Verification

15.5 VPN Troubleshooting Scenario

15.5.1 VPN Tunnel Down Phase-1 Fails

- **Symptoms:** Tunnel status shows down or VPN not visible in tunnel summary. Logs indicate repeated IKE negotiation timeout or no response from peer. No Phase-2 (Quick Mode) initiation observed.
- **Troubleshooting Focus:**
  - Verify peer IP address and interface binding are correct.
  - Check UDP 500/4500 reachability between peers.
  - Confirm Pre-Shared Key (PSK) matches on both ends.
  - Ensure IKE version and proposal parameters (encryption, authentication, DH group) align.
  - Validate NAT-Traversal settings if either peer is behind NAT.
  - Review firewall policies allowing IKE and ESP traffic.
  - Inspect system time synchronization (important for certificate-based VPNs).

> ★ **Verification Command**
>
> - execute ping <peer-ip> - check connectivity.
> - get vpn ipsec tunnel summary - status of all IPsec tunnels
> - diagnose vpn ike log-filter dst-addr4 <peer-ip> - set filter to debug

- diagnose debug application ike -1 - start debug process.

- diagnose debug enable - start debug process.

- diagnose debug reset - clears all active debug filters

- diagnose debug disable - halts all debug output

- show vpn ipsec phase1-interface - Phase1 parameters

Scenario-1

Observation

```
BR1-FW1 (FG-traffic) # execute ping-options source 100.64.100.2
BR1-FW1 (FG-traffic) # execute ping 100.64.100.1
PING 100.64.100.1 (100.64.100.1): 56 data bytes
64 bytes from 100.64.100.1: icmp_seq=0 ttl=254 time=1.6 ms
64 bytes from 100.64.100.1: icmp_seq=1 ttl=254 time=1.5 ms
64 bytes from 100.64.100.1: icmp_seq=2 ttl=254 time=1.4 ms
--- 100.64.100.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.4/1.5/1.6 ms
!
DC-FW1 (FG-traffic) # get vpn ipsec tunnel summary
'dc-br1-tunnel' 100.64.100.2:0  selectors(total,up): 3/3  rx(pkt,err): 0/0  tx(pkt,err):
    0/0
!
BR1-FW1 (FG-traffic) # diagnose vpn ike log-filter dst-addr4 100.64.100.1
BR1-FW1 (FG-traffic) # diagnose debug application ike -1
BR1-FW1 (FG-traffic) # diagnose debug enable
Debug messages will be on for 30 minutes.
BR1-FW1 (FG-traffic) # diagnose debug ike 1:br1-dc-tunnel:4: out
ike 1:br1-dc-tunnel:4: sent IKE msg (P1_RETRANSMIT): 100.64.100.2:500->100.64.100.1:500,
    len=308, vrf=0,
ike 1: comes 100.64.100.1:500->100.64.100.2:500,ifindex=7,vrf=0....
ike 1:br1-dc-tunnel:4: responder: main mode get 3rd message...
ike 1:br1-dc-tunnel:4: parse error
ike 1:br1-dc-tunnel:4: probable pre-shared secret mismatch
```

- **Root Cause:** In the above observations we can clearly see a PSK mismatch (log explicitly hints at this).

- **Fix Note:** Re-enter identical PSK on both peers.

Scenario-2

Observation

```
BR1-FW1 (FG-traffic) # diagnose vpn ike log-filter dst-addr4 100.64.100.1
BR1-FW1 (FG-traffic) # diagnose debug application ike -1
BR1-FW1 (FG-traffic) # diagnose debug enable
Debug messages will be on for 30 minutes.
ike 1:71f6735ce8c5dd47/0000000000000000:106: proposal id = 0:
ike 1:71f6735ce8c5dd47/0000000000000000:106:          type=OAKLEY_HASH\_ALG, val=SHA2\_512.
ike 1:71f6735ce8c5dd47/0000000000000000:106: my proposal, gw br1-dc-tunnel:
ike 1:71f6735ce8c5dd47/0000000000000000:106: proposal id = 1
ike 1:71f6735ce8c5dd47/0000000000000000:106:          type=OAKLEY_HASH\_ALG, val=MD5.
```

```
!
ike 1:71f6735ce8c5dd47/0000000000000000:106: negotiation failure
!
!
DC-FW1 (FG-traffic) # show vpn ipsec phase1-interface
config vpn ipsec phase1-interface
    edit "dc-br1-tunnel"
        set proposal des-sha512
!
BR1-FW1 (FG-traffic) # show vpn ipsec phase1-interface
config vpn ipsec phase1-interface
    edit "br1-dc-tunnel"
        set proposal des-md5
        set dhgrp 19 14 5
```

- **Root Cause:** The two peers are offering different hash algorithms (SHA-512 vs MD5) and DH Group Since there is no common proposal, the negotiation fails.
- **Fix Note:** Align the Phase-1 proposals on both peers so they share at least one identical encryption + hash + DH group combination.

Scenario-3

Observation

```
DC-FW1 (FG-traffic) # diagnose vpn ike log-filter dst-addr4 100.64.100.2

DC-FW1 (FG-traffic) # diagnose debug application ike -1
Debug messages will be on for 30 minutes.

DC-FW1 (FG-traffic) # diagnose debug enable
!
!
No debug message.
!
DC-FW1 (FG-traffic) # show firewall policy
config firewall policy
        set name "dc-fw1-loopback-ipsec"
        set service "ALL_ICMP" "PING"
!
BR1-FW1 (FG-traffic) # sh firewall policy
config firewall policy
        set name "br1-fw1-loopback-ipsec"
        set service "ALL_ICMP" "IKE" "PING"
```

- **Root Cause:** DC-FW1 is dropping IKE packets because the firewall policy does not include the IKE service (UDP/500 or 4500). That's why debug shows nothing – no packets are reaching the IKE daemon.
- **Fix Note:** Update the DC-FW1 policy to include IKE (UDP/500 or 4500).

15.5.2 VPN Tunnel Phase-1 Up, Phase 2 Down

- **Symptoms:** Phase-1 established but no Phase-2SAs.
- **Troubleshooting Focus:**
  - Verify Phase-2 selectors (Proxy-IDs) match exactly on both peers.

- Confirm Phase-2 proposals (encryption, authentication, DH group) are identical.
- Check firewall policies allowing traffic between protected subnets.
- Ensure routing directs packets into the VPN interface.
- Validate SA lifetimes and replay settings are consistent.
- If NAT is used, confirm NAT-Traversal is enabled on both sides.

★ **Verification Command**

- get vpn ipsec tunnel summary
- diagnose vpn tunnel list
- show vpn ipsec phase2-interface - Phase2 parameters
- diagnose vpn ike log-filter dst-addr4 <peer-ip>
- diagnose debug application ike -1
- diagnose debug enable

Observation

```
BR1-FW1 (FG-traffic) # get vpn ipsec tunnel summary
'br1-dc-tunnel' 100.64.100.1:0  selectors(total,up): 3/0  rx(pkt,err): 3070/0  tx(pkt,err)
    : 3070/68
!
BR1-FW1 (FG-traffic) # diagnose debug enableike 1: comes 100.64.100.1:500-
    >100.64.100.2:500,ifindex=7,vrf=0....
ike 1: IKEv1 exchange=Informational id=7fb1a34b25fc6a2a/75a679da1eb8ff2b:6ca415e5 len=84
    vrf=0
ike 1: in 7FB1A34B26
ike 1:br1-dc-tunnel:202: dec 7FB1A34B
ike 1:br1-dc-tunnel:202: notify msg received: R-U-THERE
ike 1:br1-dc-tunnel:202: enc 7FB1A34B2
ike 1:br1-dc-tunnel:202: out 7FB1A34B2
ike 1:br1-dc-tunnel:202: sent IKE msg (R-U-THERE-ACK): 100.64.100.2:500->100.64.100.1:500,
     len=84, vrf=0,
!
Messages like notify msg received: R-U-THERE and sent IKE msg (R-U-THERE-ACK) confirm that
     the tunnel endpoints are alive and responding. Since -Phase1 is up and DPD is working,
     the issue lies in -Phase2 negotiation - no Quick Mode SAs are being built.
!
BR1-FW1 (FG-traffic) # show vpn ipsec phase2-interface
config vpn ipsec phase2-interface
    edit "br1-dc-tunnel-hr"
        set phase1name "br1-dc-tunnel"
        set proposal des-md5
DC-FW1 (FG-traffic) # show vpn ipsec phase2-interface
config vpn ipsec phase2-interface
    edit "dc-br1-tunnel-hr"
        set phase1name "dc-br1-tunnel"
        set proposal des-sha512
        set dhgrp 19 14 5
!
BR1-FW1 (FG-traffic) # show vpn ipsec phase2-interface
config vpn ipsec phase2-interface
    edit "br1-dc-tunnel-sales"
```

```
        set phase1name "br1-dc-tunnel"
        set proposal des-md5
        set src-name "br1-tunnel_sales"
        set dst-name "dc-tunnel_sales"
DC-FW1 (FG-traffic) # show vpn ipsec phase2-interface
config vpn ipsec phase2-interface
edit "dc-br1-tunnel-sales"
        set phase1name "dc-br1-tunnel"
        set proposal des-md5
        set src-name "dc-tunnel_prod"
        set dst-name "br1-tunnel_sales"
!
BR1-FW1 (FG-traffic) # show vpn ipsec phase2-interface
config vpn ipsec phase2-interface
    edit "br1-dc-tunnel-prod"
        set phase1name "br1-dc-tunnel"
        set proposal des-md5
DC-FW1 (FG-traffic) # show vpn ipsec phase2-interface
config vpn ipsec phase2-interface
    edit "dc-br1-tunnel-prod"
        set phase1name "dc-br1-tunnel"
        set proposal null-md5
```

- **Root Cause:**
  - **HR Tunnel:** BR1 uses des-md5, DC uses des-sha512 → no common proposal.
  - **Prod Tunnel:** BR1 uses des-md5, DC uses null-md5 → mismatched/insecure.
  - **Sales Tunnel:** Both use des-md5, phase-2 selector mismatch on DC-FW1 for sales tunnel prod lan name address called.
- **Fix Note:**
  - Correct HR tunnel: proposal matching BR1 and DC.
  - Correct Prod tunnel: replace null-md5 with a proposal matching BR1.
  - Align Sales tunnel selectors on both DC and BR1 firewalls.

15.5.3 VPN Tunnel Up but No Traffic

- **Symptoms:** Phase 1 and Phase 2 are established, but protected networks cannot communicate.
- **Troubleshooting Focus:**
  - Verify Phase-2 selectors (Proxy-IDs) match exactly on both peers.
  - Check firewall policies allowing traffic between protected subnets.
  - Confirm routing directs packets into the VPN interface.
  - Ensure NAT-Traversal is enabled if peers are behind NAT.
  - Validate SA lifetimes and replay settings are consistent.
  - Check for overlapping subnets or incorrect static routes.

> ★ **Verification Command**
> - get vpn ipsec tunnel summary
> - diagnose vpn tunnel list
> - show vpn ipsec phase2-interface

- show firewall policy

- get router info routing-table static

Observation

```
DC-FW1 (FG-traffic) # get vpn ipsec tunnel summary
'dc-br1-tunnel' 100.64.100.2:0  selectors(total,up): 3/3  rx(pkt,err): 3070/0  tx(pkt,err)
    : 3070/0
!
BR1-FW1 (FG-traffic) # get vpn ipsec tunnel summary
'br1-dc-tunnel' 100.64.100.1:0  selectors(total,up): 3/3  rx(pkt,err): 3070/0  tx(pkt,err)
    : 3070/253
!
DC-FW1 (FG-traffic) # show router static
config router static
    edit 3
        set device "dc-br1-tunnel"
        set dstaddr "dc-tunnel"
    next
    edit 4
        set distance 254
        set blackhole enable
        set dstaddr "br1-tunnel"
    next
end
DC-FW1 (FG-traffic) #
!
BR1-FW1 (FG-traffic) # sh router static
config router static
    edit 3
        set device "br1-dc-tunnel"
        set dstaddr "dc-tunnel"
    next
    edit 4
        set distance 254
        set blackhole enable
        set dstaddr "dc-tunnel"
    next
end
BR1-FW1 (FG-traffic) #
!
BR1-FW1 (FG-traffic) # get router info routing-table static
Routing table for VRF=0
S*      0.0.0.0/0 [10/0] via 203.0.113.9, port5, [1/0]
S       192.168.11.0/30 [10/0] via br1-dc-tunnel tunnel 100.64.100.1, [1/0]
S       192.168.12.0/30 [10/0] via br1-dc-tunnel tunnel 100.64.100.1, [1/0]
S       192.168.13.0/30 [10/0] via br1-dc-tunnel tunnel 100.64.100.1, [1/0]
!
DC-FW1 (FG-traffic) # get router info routing-table static
Routing table for VRF=0
S*      0.0.0.0/0 [10/0] via 203.0.113.1, port5, [1/0]
S       192.168.21.0/24 [254/0] is a summary, Null, [1/0]
S       192.168.22.0/24 [254/0] is a summary, Null, [1/0]
S       192.168.23.0/24 [254/0] is a summary, Null, [1/0]
!
DC-FW1 (FG-traffic) # execute ping-options source 192.168.11.1
DC-FW1 (FG-traffic) # execute ping 192.168.21.1
```

```
PING 192.168.21.1 (192.168.21.1): 56 data bytes
sendto failed
--- 192.168.21.1 ping statistics ---
5 packets transmitted, 0 packets received, 100\% packet loss
DC-FW1 (FG-traffic) # execute ping-options source 192.168.12.1
DC-FW1 (FG-traffic) # execute ping 192.168.22.1
PING 192.168.22.1 (192.168.22.1): 56 data bytes
sendto failed
--- 192.168.22.1 ping statistics ---
5 packets transmitted, 0 packets received, 100\% packet loss
DC-FW1 (FG-traffic) # execute ping-options source 192.168.13.1
DC-FW1 (FG-traffic) # execute ping 192.168.23.1
PING 192.168.23.1 (192.168.23.1): 56 data bytes
sendto failed
--- 192.168.23.1 ping statistics ---
5 packets transmitted, 0 packets received, 100\% packet loss
```

- **Root Cause:** BR1 side has correct static routes pointing into br1-dc-tunnel, but DC side is misrouted, causing 100% packet loss.
- **Fix Note:** On DC-FW1, change static route edit 3 so that dstaddr references the br1-tunnel address group (192.168.21.0/24, 192.168.22.0/24, 192.168.23.0/24) instead of dc-tunnel.

Observation

```
DC-FW1 (FG-traffic) # get router info routing-table static
Routing table for VRF=0
S*      0.0.0.0/0 [10/0] via 203.0.113.1, port5, [1/0]
S       192.168.21.0/24 [10/0] via dc-br1-tunnel tunnel 100.64.100.2, [1/0]
S       192.168.22.0/24 [10/0] via dc-br1-tunnel tunnel 100.64.100.2, [1/0]
S       192.168.23.0/24 [10/0] via dc-br1-tunnel tunnel 100.64.100.2, [1/0]
```

16 HA (High Availability)

In this section, we will cover the following objectives.

> **Objective**
>
> - Understand FGCP architecture.
> - Configure Active-Passive and Active-Active HA Clusters.
> - Perform HA verification and failover testing.
> - Troubleshoot HA cluster formation, failover, heartbeat, and split-brain scenarios.

In this lab demonstration, we use a two-member cluster; supported cluster size depends on model and FortiOS version. Firewalls are deployed in High Availability pairs to eliminate a single point of failure, with their configurations synced to ensure seamless failover. All units must have an identical FortiGuard subscription, the same FortiOS version and build number, a matching interface layout (port names and types), identical licenses, and a uniform hardware configuration. In this lab, we will connect two firewalls in a group. The primary firewall's shared configuration syncs with the secondary firewall, except for the hostname and the priority of the HA. The default HA priority is 128.

16.1 FGCP (FortiGate Clustering Protocol)

The FortiGate Clustering Protocol (FGCP) is Fortinet's proprietary protocol for High Availability (HA). It defines how cluster members communicate, synchronize, and perform failover. FGCP discovers other firewalls and elects the primary firewall. The following are some key points about FGCP:

- **Layer 2 – EtherType:**
  - FGCP uses Layer 2 HA heartbeat frames to allow cluster members to discover each other and monitor the health and status of the cluster.
  - In NAT mode, FGCP HA heartbeat packets use EtherType 0x8890 and in Transparent mode, FGCP heartbeat packets use EtherType 0x8891.
  - EtherType 0x8893 is used for HA telnet sessions and configuration synchronization over the HA heartbeat link.
  - These EtherType-based frames are exchanged through the configured HA heartbeat interfaces.
- **Layer 3 – IP:**
  - FGCP primarily uses Layer 2 frames for HA heartbeat communication over the heartbeat interfaces.
  - FortiGate also uses Layer 3 communication for synchronization, logs, and other locally generated traffic between HA members.
  - FGCP automatically assigns link-local IPv4 addresses from the 169.254.0.0/26 range to the virtual HA heartbeat interface.
  - The heartbeat IP addresses are automatically assigned based on the serial-number priority of the cluster members and are used internally by FGCP.
- **Layer 4 – TCP/UDP:**
  - TCP/703 and UDP/703 are used for HA synchronization and related FGCP communication.

  - TCP/703 is also used for FGCP heartbeat communication in configurations where the heartbeat operates using IP-based communication.
  - The exact communication method depends on the HA mode and configuration, such as broadcast or unicast heartbeat.

- **FGCP Packet Types**

  - Hello Packets – exchanged over Ethertype 0x8890 to detect health.
  - Sync Packets – replicate configuration and session tables (TCP 703).
  - Election Packets – decide primary unit (UDP broadcast).
  - Failover Packets – trigger role switch when primary fails.

16.2 HA Link

In a FortiGate firewall HA cluster, the HA-Link (heartbeat link) is the dedicated interface (or interfaces) used for cluster communication. It's the lifeline of the HA setup, ensuring both units stay synchronized and aware of each other's state. The following are some important points regarding HA Link.

- **Role of the HA-Link**

  - **Heartbeat Exchange**: Sends hello packets to confirm cluster health and detect failures.
  - **Configuration Sync**: Keeps system settings aligned across all HA members.
  - **Session Synchronization**: Ensures active sessions continue seamlessly during failover.
  - **Election & Role Assignment**: Helps decide which unit is primary and which is secondary.

- **Configuration Guidelines**

  - **Minimum Two Interfaces**: Fortinet recommends at least two heartbeat interfaces for redundancy.
  - **Isolation**: Connect HA-Links to a dedicated switch or directly between units. Avoid mixing with production traffic.
  - **Priorities**: Assign heartbeat interface priorities so the cluster knows which link to prefer.
  - **Bandwidth**: Use interfaces with sufficient bandwidth to handle sync traffic.
  - **Security**: Keep HA-Link isolated to prevent exposure of sensitive cluster data.

- **Risks if HA-Link Fails**

  - **Split-Brain Scenario**: Both units may assume the primary role, causing duplicate IP/MAC addresses and traffic disruption.
  - **Loss of Sync**: Configurations or sessions may not replicate, leading to inconsistent firewall behavior.
  - **Failover Failure**: Without heartbeat detection, failover may not trigger correctly.

16.3 A-P HA (Active-Passive HA)

In our lab demonstration, we will configure Active-Passive HA on the data center firewalls. Both firewalls must be in the same subnet. Note that Active-Passive HA does not increase session capacity or network throughput. The following are some key concepts regarding A-P HA.

- **Active Firewall:**
  - Handles all traffic. Enforces policies and maintains session tables.
- **Passive Firewall:**
  - Remains in standby mode, monitoring the active firewall via heartbeat links. Synchronizes configuration and session data but does not forward traffic.
- **Failover:**
  - Triggered if the active firewall fails due to hardware fault, interface down, or heartbeat loss. The passive firewall automatically takes over as the active unit.
- **Recovery:**
  - When the original firewall comes back online, it can either remain passive or reclaim the active role. Behavior depends on configuration settings such as override and priority.

The following figure illustrates the Active-Passive HA workflow.

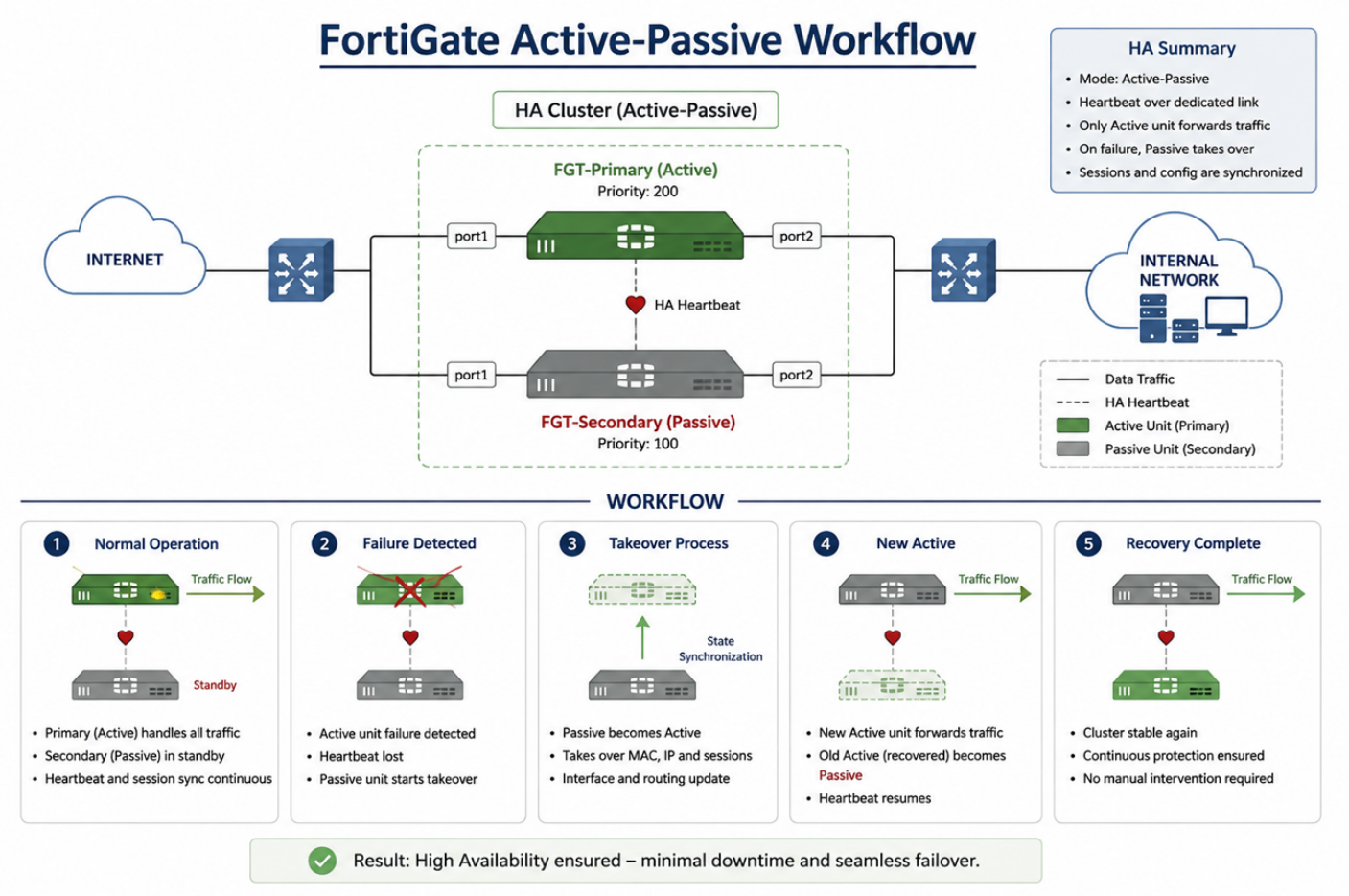


Figure 1: FortiGate Active-Passive HA Workflow

Below are some limitations of Active-Passive HA

- No Load Balancing: Only one firewall processes traffic at a time.

- Hardware Cost: Requires at least two identical firewalls.
- Configuration Discipline: Policies, routes, and licenses must be identical across all units.

Before starting the configuration, we will break down the configuration fields.

Configuration Field Breakdown

- **Mode**: Defines the HA operation type. Active-Passive or Active-Active.
- **Device Priority**: Determines which unit becomes the primary (active) firewall. Higher values take precedence during cluster formation or recovery.
- **Group Name**: Logical name for the HA cluster. Must match on all units to form a cluster.
- **Password**: Used for authentication between HA members during cluster formation and synchronization. Must be identical on all units.
- **Session Pickup**: Enables session continuity during failover. When enabled, existing sessions are preserved and users experience minimal disruption.
- **Monitor Interfaces**: Interfaces monitored for link failure. If any monitored interface goes down, failover is triggered.
- **Heartbeat Interfaces**: Dedicated interfaces used for HA communication (heartbeat and synchronization). These links exchange cluster health and configuration data.
- **Heartbeat Interface Priority**: Assigns weights to heartbeat interfaces. Higher priority interfaces are preferred for HA communication. If one fails, the next is used.
- **Management Interface Reservation**: When enabled, reserves a management interface that remains reachable even during failover. Typically disabled in simple setups.
- **Unicast Heartbeat**: Optional mode for heartbeat communication using unicast instead of broadcast. Useful in networks where broadcast is restricted.
- **VDOM Partitioning:** FG-traffic and root VDOMs mapped to Virtual Cluster 1.

Configure HA on both firewalls using the HA link, monitored links, and priority values shown in the table below.

Site	HA Mode	Device	HA-Link	Monitor Link	Priority Value
Data Center	A-P	DC-FW1	HA1:port7, HA2:port8	Lag1, port5, port6	Primary:200
		DC-FW2			Secondary:100

Table 18: FortiGate Data Center HA Configuration Details

GUI PATH: **System --> HA**

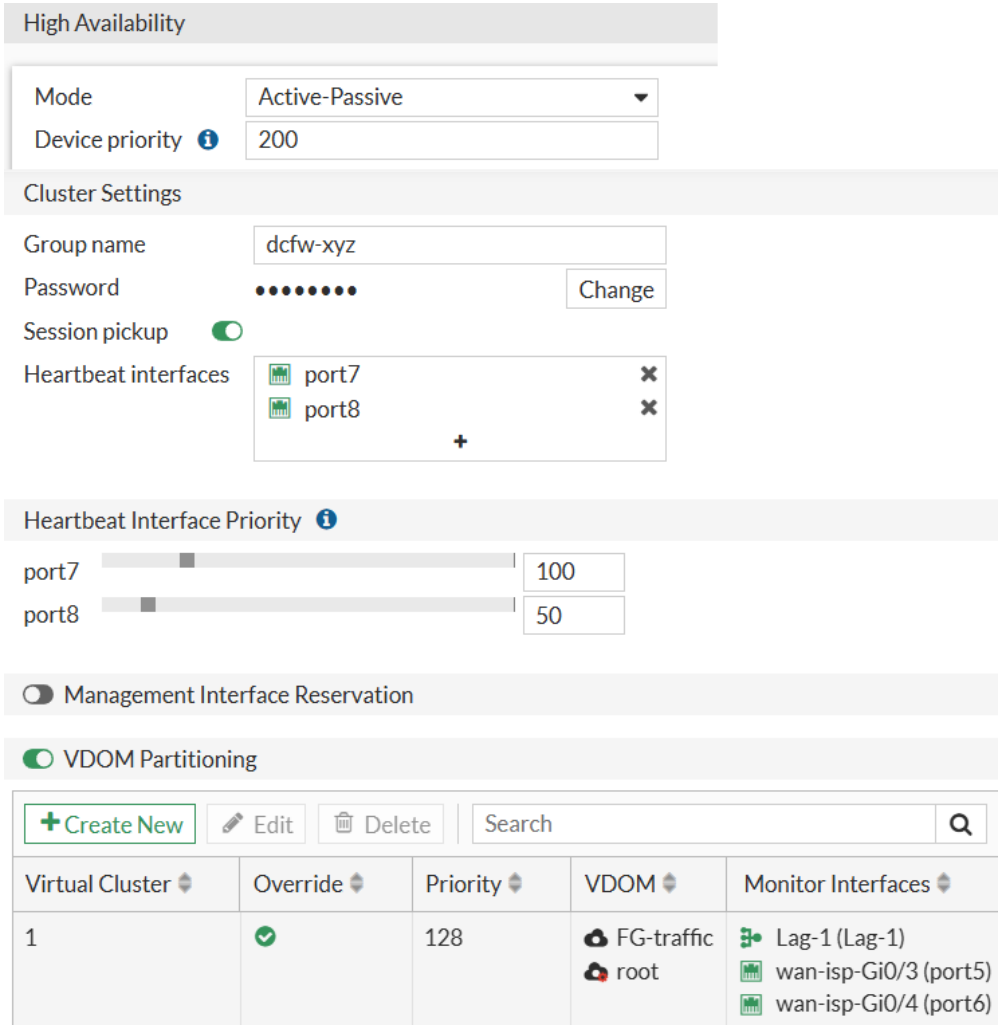
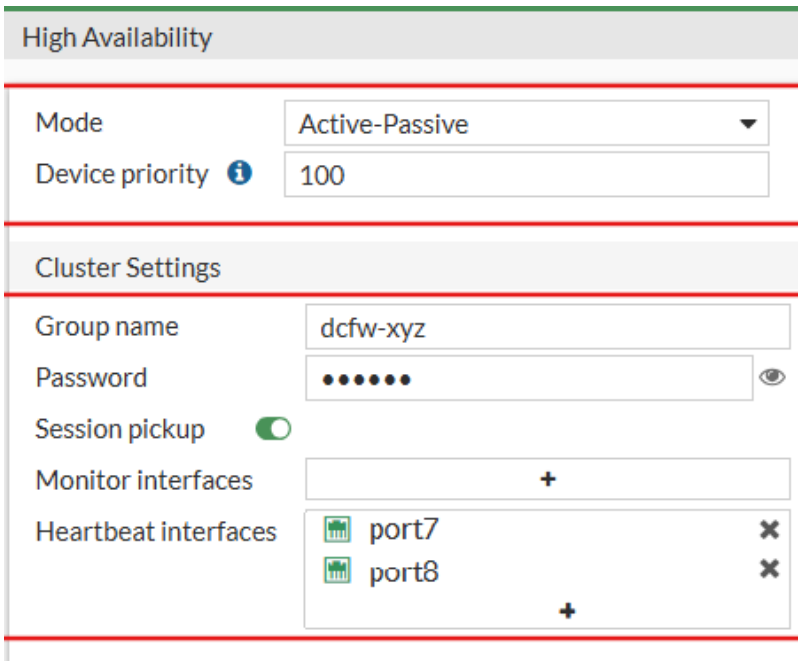


Figure: DC-FW1 FortiGate Active-Passive HA

GUI PATH: **System --> HA**



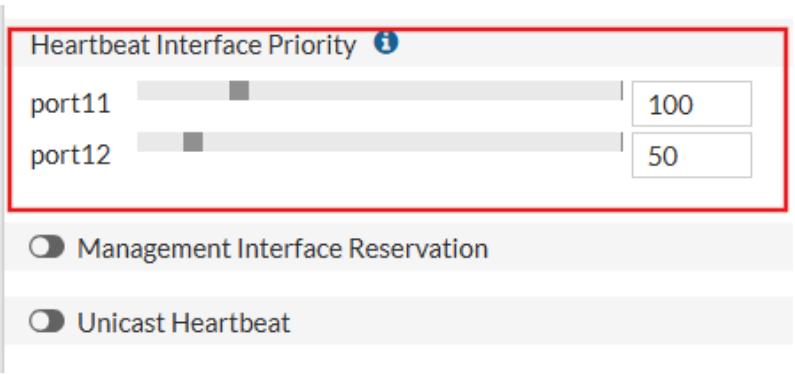


Figure: DC-FW2 FortiGate Active-Passive HA (continued)

Before start the cli configuration please note following point regarding to **set override enable**. We cannot enable override via GUI.

> ☞ **Note**
> In FortiGate HA configuration, the set override enable command is used to control primary unit election behavior when multiple devices in the cluster have different priorities. We cannot set it via GUI.

- By default, HA election is based on uptime, monitor status, and priority.
- With **override disabled**, once a unit becomes primary, it stays primary until it fails – even if another unit has higher priority.
- With **override enabled**, device priority takes precedence over uptime during primary election. Therefore, if DC-FW1 is healthy and has the higher priority of 200, it is preferred as the primary unit.

The following CLI commands configure Active-Passive HA.

➲ DC-FW1 HA Active-Passive Configuration

```
config global
 config system ha
  set group-name dcfw-xyz
    set mode a-p
  set password {password}
  set hbdev port7 100 port8 50
  set session-pickup enable
      set vcluster-status enable
      config vcluster
       edit 1
        set override enable
        set priority 200
        set monitor Lag-1 port5 port6
        set vdom FG-traffic root
       next
  end
end
```

➲ DC-FW2 HA Active-Passive Configuration

```
config global
 config system ha
    set group-name dcfw-xyz
    set mode a-p
  set password {password}
  set hbdev port7 100 port8 50
  set session-pickup enable
      set vcluster-status enable
       config vcluster
        edit 1
         set override enable
         set priority 100
         set vdom FG-traffic root
        next
       end
 end
```

We will verify the status for HA in FortiGate GUI via the below path .

GUI PATH: **System --> HA**

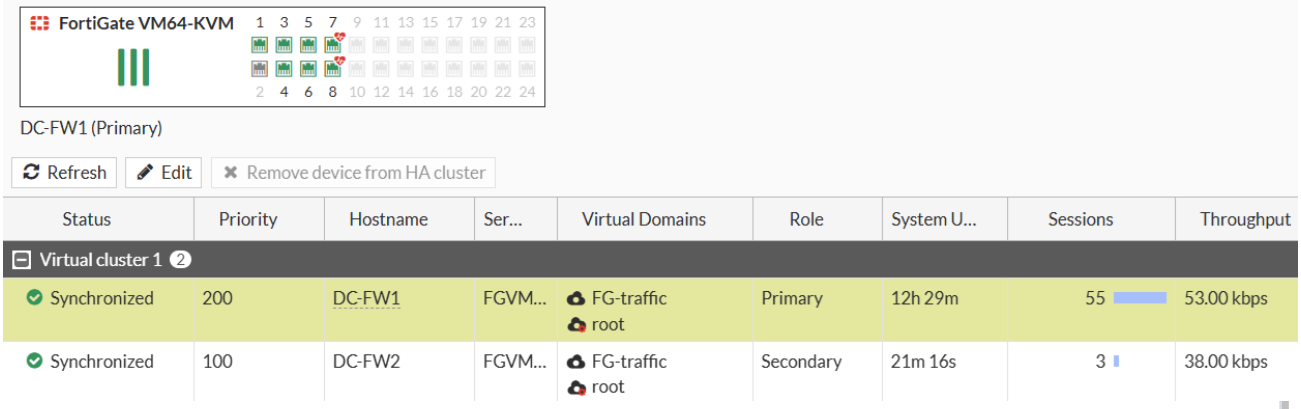


Figure: DC-FW1 FortiGate HA Status

Also with following CLI Commands we can verify or see more details about HA parameters.

HA status on DC-FW1

```
DC-FW1 (global) # get system ha status
HA Health Status: OK
Model: FortiGate-VM64-KVM
Mode: HA A-P
Cluster Uptime: 0 days 0:23:50
Cluster state change time: 2026-09-13 18:50:31
Primary selected using:
    <2026/09/13 18:50:31> vcluster-1: FGVMEVXXXXXXXX9C is selected as the primary because its
     override priority is larger than peer member FGVMEVXXXXXXXX5D.
    <2026/09/13 18:50:14> vcluster-1: FGVMEVXXXXXXXX9C is selected as the primary because it'
    s the only member in the cluster.
ses_pickup: enable, ses_pickup_delay=disable
override:
    vcluster_1 enable
Configuration Status:
    FGVMEVXXXXXXXX9C(updated 5 seconds ago): in-sync
    FGVMEVXXXXXXXX5D(updated 2 seconds ago): in-sync
HBDEV stats:
    FGVMEVXXXXXXXX9C(updated 5 seconds ago):
        port7: physical/10000full, up, rx-bytes/packets/dropped/errors=4073621/12396/0/0, tx
    =5471566/11901/0/0
        port8: physical/10000full, up, rx-bytes/packets/dropped/errors=3132689/7010/0/0, tx
    =3233992/7156/0/0
    FGVMEVXXXXXXXX5D(updated 2 seconds ago):
        port7: physical/10000full, up, rx-bytes/packets/dropped/errors=5479418/11921/0/0, tx
    =4080455/12416/0/0
        port8: physical/10000full, up, rx-bytes/packets/dropped/errors=3240320/7170/0/0, tx
    =3138513/7023/0/0
MONDEV stats:
  vcluster_1:
    FGVMEVXXXXXXXX9C(updated 5 seconds ago):
        Lag-1: aggregate/00, up, rx-bytes/packets/dropped/errors=106254545/184729/0/0, tx
    =98040079/78969/0/0
        port5: physical/10000full, up, rx-bytes/packets/dropped/errors=103928365/88049/0/0,
    tx=99845927/86119/0/0
        port6: physical/10000full, up, rx-bytes/packets/dropped/errors=1803259/12489/0/0, tx
    =2171222/29489/0/0
```

```
    FGVMEVXXXXXXXX5D(updated 2 seconds ago):
        Lag-1: aggregate/00, up, rx-bytes/packets/dropped/errors=155031/2086/0/0, tx
    =12606/103/0/0
        port5: physical/10000full, up, rx-bytes/packets/dropped/errors=48304/432/0/0, tx
    =90/1/0/0
        port6: physical/10000full, up, rx-bytes/packets/dropped/errors=48304/432/0/0, tx
    =90/1/0/0
PINGSVR stats:
  vcluster_1:
    FGVMEVXXXXXXXX9C(updated 5 seconds ago):
    FGVMEVXXXXXXXX5D(updated 2 seconds ago):
Primary     : DC-FW1          , FGVMEVXXXXXXXX9C, HA cluster index = 1
Secondary   : DC-FW2          , FGVMEVXXXXXXXX5D, HA cluster index = 0
number of vcluster: 1
vcluster 1: work 0.0.0.0
Primary: FGVMEVXXXXXXXX9C, HA operating index = 0
Secondary: FGVMEVXXXXXXXX5D, HA operating index = 1
```

HA diagnose status on DC-FW1

```
DC-FW1 (global) # diagnose sys ha status
HA information
Model=80001, Mode=2 Group=0 Debug=0
nvcluster=1, ses_pickup=1, delay=0
[Debug_Zone HA information]
HA group member information: is_manage_primary=1.
FGVMEVXXXXXXXX9C:      Primary, serialno_prio=1, usr_priority=200, hostname=DC-FW1
FGVMEVXXXXXXXX5D:    Secondary, serialno_prio=0, usr_priority=100, hostname=DC-FW2
[Kernel HA information]
vcluster 1, state=work, primary_ip=169.254.0.2, primary_id=0:
FGVMEVXXXXXXXX9C:      Primary, ha_prio/o_ha_prio=0/0
FGVMEVXXXXXXXX5D:    Secondary, ha_prio/o_ha_prio=1/1
```

HA vcluster dump on DC-FW1

```
DC-FW1 (global) # diagnose sys ha dump-by vcluster
            HA information.
vcluster_nr=1
vcluster-1: start_time=1789350594(2026-09-13 18:49:54), state/o/chg_time=2(work)/2(work)
    /1789350614(2026-09-13 18:50:14)
        pingsvr_flip_timeout/expire=3600s/1626s
        'FGVMEVXXXXXXXX9C': ha_prio/o=0/0, link_failure=0, pingsvr_failure=0, flag=0x00000001
    , mem_failover=0, uptime/reset_cnt=333/0
        'FGVMEVXXXXXXXX5D': ha_prio/o=1/1, link_failure=0, pingsvr_failure=0, flag=0x00000000
    , mem_failover=0, uptime/reset_cnt=0/1
```

16.4 A-A HA (Active-Active HA)

In Active-Active HA, multiple FortiGate units in a cluster can process traffic simultaneously, unlike Active-Passive where only one unit is active. This setup provides both redundancy and load balancing for TCP traffic only, making it ideal for high-throughput environments. In an Active-Active HA, **load balancing** is dependent on the traffic type; the primary firewall distributes TCP sessions across all firewalls in the setup. Non-session traffic (ICMP, multicast, broadcast) is always handled by the primary unit, and no load balancing occurs. Active-Active HA is most beneficial in environments with heavy TCP traffic (e.g., web, SSL, VPN), rather than ICMP/multicast/broadcast-heavy workloads. Active-Active

HA does not guarantee throughput increases proportional to cluster size.
The following are the key concepts of Active-Active HA.

- **Primary Firewall:**
  - One FortiGate remains the primary unit for cluster management, configuration synchronization, and coordination of traffic distribution.
  - The primary can distribute eligible processing or sessions to other Active-Active cluster members.
- **Secondary / Cluster Members:**
  - Unlike Active-Passive HA, secondary members in an Active-Active cluster can participate in traffic processing.
  - By default, supported proxy-based security processing can be distributed among cluster members.
  - When set load-balance-all enable is configured, TCP sessions can also be distributed according to the configured HA load-balancing schedule.
  - ICMP, multicast, and broadcast traffic is not distributed by load-balance-all and remains processed by the primary unit.
- **Failover:**
  - If a cluster member fails, the remaining members continue processing supported traffic.
  - Session synchronization helps reduce disruption during failover.
- **Recovery:**
  - When a failed unit comes back online, it rejoins the cluster and synchronizes with the other members.
  - Primary election behavior depends on HA settings such as priority and override.

  The following Figure shows the workflow of Active-Active Firewall.

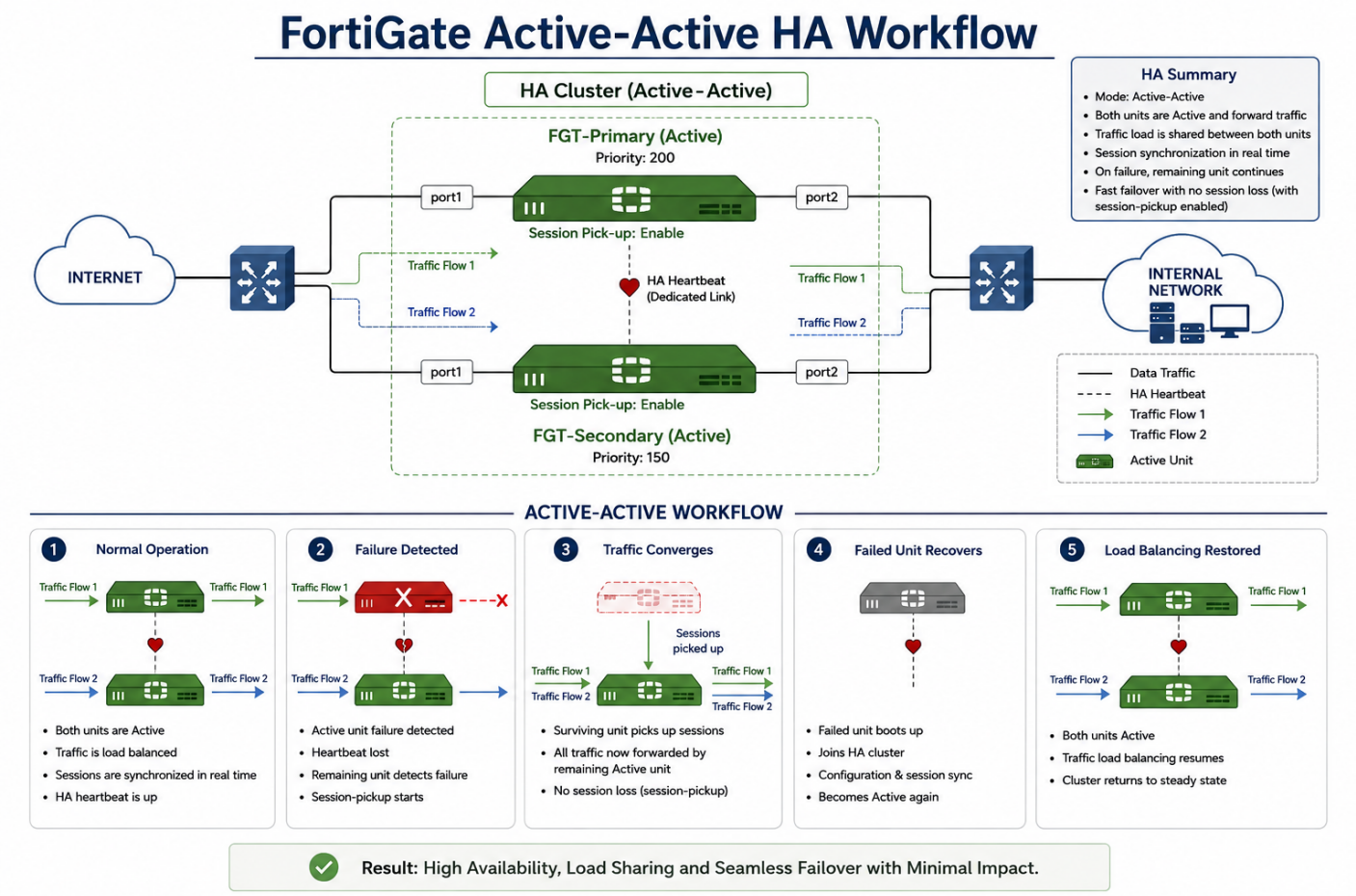


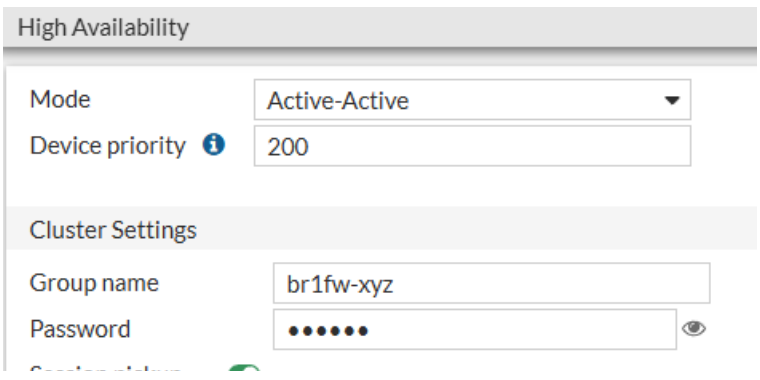
Figure 2: FortiGate Active-Active HA Workflow

Site	HA Mode	Device	HA-Link	Monitor Link	Priority
Branch 1	A-A	BR1-FW1	HA1:Port7, HA2:Port8	Lag-1, port5, port6	Primary:200
		BR1-FW2		Lag-1, port5, port6	Secondary:150

Table 19: Branch 1 HA Configuration Details

We can configure Active-Active HA using the GUI screenshots below. The configuration field breakdown already explain in Active-Passive scenario.

GUI PATH: **System --> HA**



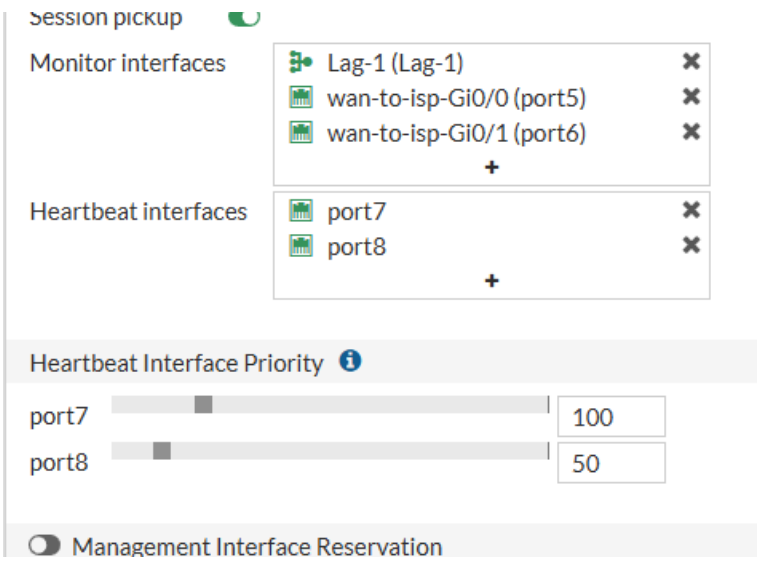


Figure: BR1-FW1 FortiGate Active-Active HA (continued)

GUI PATH: **System --> HA**

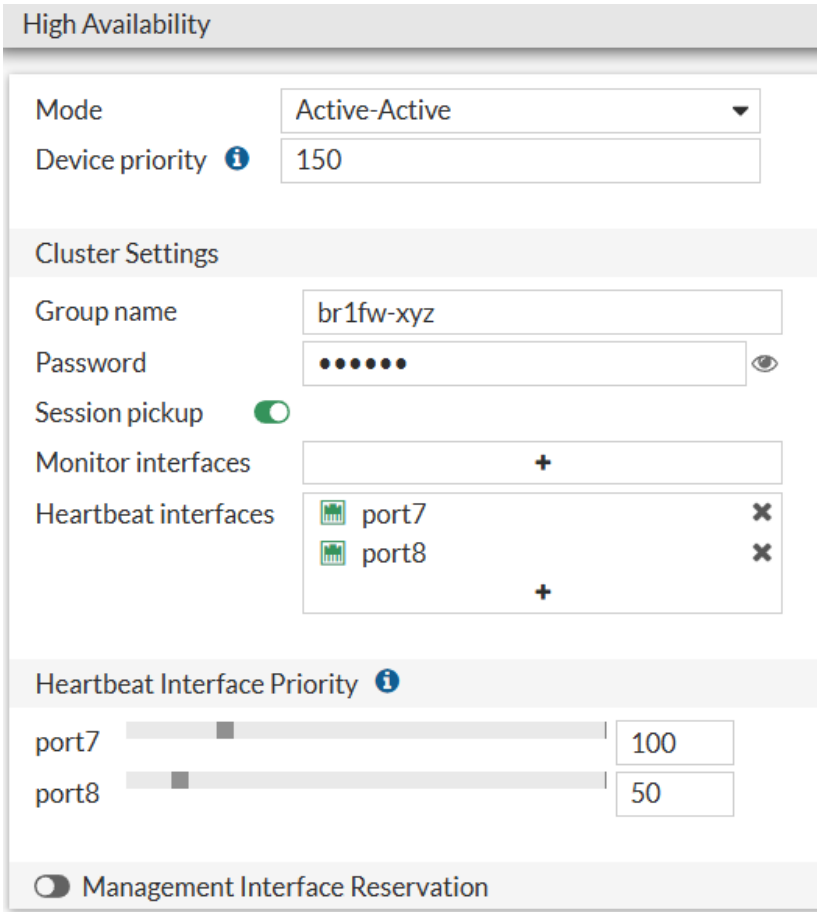


Figure: BR1-FW2 FortiGate Active-Active HA

16.4.1 Behavior of set override disable in Active-Active HA

- Active-Active HA does not use the same strict active-versus-standby forwarding model as Active-Passive HA.
- The primary unit still exists for cluster management (GUI access, config sync, logging),

but traffic distribution is shared.

- With set override disable, once a unit becomes the primary, it will remain primary until it fails, regardless of another unit's higher priority.
- This avoids unnecessary role switching in A-A HA, since throughput scaling comes from multiple units forwarding traffic, not from which one is designated primary.

16.4.2 Effect of set load-balance-all enable in Active-Active HA

When set load-balance-all enable is configured in a FortiGate Active-Active HA cluster, the primary unit can distribute TCP sessions across the available cluster members. By default, Active-Active HA primarily load balances supported proxy-based security profile processing. Enabling load-balance-all extends load balancing to TCP sessions.

- **TCP Sessions**: With load-balance-all enable, TCP sessions can be distributed across Active-Active HA cluster members according to the configured HA load-balancing schedule.
- **Proxy-Based Security Processing**: Even when load-balance-all is disabled, supported proxy-based security processing such as antivirus, web filtering, email filtering, and DLP can be distributed across cluster members.
- **ICMP, Multicast, and Broadcast**: These sessions are not load balanced by FGCP Active-Active HA and are processed by the primary unit.
- **Performance Consideration**: Enabling load balancing for all TCP sessions introduces additional processing overhead and does not necessarily increase overall firewall throughput.

The load-balance-all option specifically extends Active-Active HA load balancing to TCP sessions. It should not be interpreted as enabling load balancing for ICMP, multicast, or broadcast traffic.

We will configure Active-Active HA as per Below CLI Configuration commands. We will keep override disabled and enable load balancing.

➲ BR1-FW1 HA Active-Active Configuration

```
config global
config system ha
set group-name br1fw-xyz
set mode a-a
set password {password}
set priority 200
set override disable
set monitor Lag-1 port5 port6
set hbdev port7 100 port8 50
set session-pickup enable
set load-balance-all enable
end
end
```

➲ BR1-FW2 HA Active-Active Configuration

```
config global
config system ha
set group-name br1fw-xyz
set mode a-a
set password {password}
set priority 150
set override disable
set hbdev port7 100 port8 50
set session-pickup enable
set load-balance-all enable
end
end
```

We will verify the status for HA in FortiGate GUI via the below path .

GUI PATH: **System --> HA**

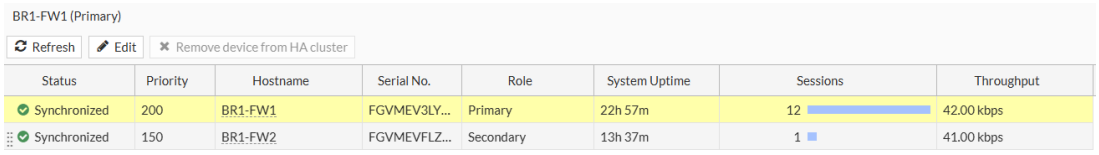


Figure: BR1-FW1 A-A HA Status

Also with following CLI Commands we can verify or see more details about HA parameters.

HA status on BR1-FW1

```
BR1-FW1 # get system ha status
HA Health Status: OK
Model: FortiGate-VM64-KVM
Mode: HA A-A
Group: 0
Debug: 0
Cluster Uptime: 0 days 0:35:23
Cluster state change time: 2026-08-11 16:33:18
Primary selected using:
<2026/08/11 16:33:18> vcluster-1: FGVMEVXXXXXXXX6F is selected as the primary because its
  uptime is larger than peer member FGVMEVXXXXXXXXCB.
<2026/08/11 16:22:32> vcluster-1: FGVMEVXXXXXXXX6F is selected as the primary because it's
  the only member in the cluster.
ses_pickup: enable, ses_pickup_delay=disable
load_balance: enable
load_balance_udp: disable
schedule: Round robin.
upgrade_mode: unset
override: disable
Configuration Status:
FGVMEVXXXXXXXX6F(updated 2 seconds ago): in-sync
FGVMEVXXXXXXXXCB(updated 4 seconds ago): in-sync
System Usage stats:
FGVMEVXXXXXXXX6F(updated 2 seconds ago):
sessions=13, average-cpu-user/nice/system/idle=0%/0%/0%/98%, memory=44%
FGVMEVXXXXXXXXCB(updated 4 seconds ago):
sessions=1, average-cpu-user/nice/system/idle=0%/0%/0%/99%, memory=44%
HBDEV stats:
FGVMEVXXXXXXXX6F(updated 2 seconds ago):
port7: physical/10000full, up, rx-bytes/packets/dropped/errors=4142681/11400/0/0, tx
  =4870156/9973/0/0
port8: physical/10000full, up, rx-bytes/packets/dropped/errors=3375175/7250/0/0, tx
  =3468916/7242/0/0
FGVMEVXXXXXXXXCB(updated 4 seconds ago):
port7: physical/10000full, up, rx-bytes/packets/dropped/errors=4865377/9963/0/0, tx
  =4137187/11386/0/0
port8: physical/10000full, up, rx-bytes/packets/dropped/errors=3464128/7232/0/0, tx
  =3370981/7241/0/0
MONDEV stats:
FGVMEVXXXXXXXX6F(updated 2 seconds ago):
Lag-1: aggregate/00, up, rx-bytes/packets/dropped/errors=2424348/27524/0/0, tx
  =729982/5247/0/0
port5: physical/10000full, up, rx-bytes/packets/dropped/errors=2202230/27476/0/0, tx
  =1084567/13446/0/0
port6: physical/10000full, up, rx-bytes/packets/dropped/errors=518884/3089/0/0, tx
```

```
  =924/16/0/0
FGVMEVXXXXXXXXCB(updated 4 seconds ago):
Lag-1: aggregate/00, up, rx-bytes/packets/dropped/errors=803503/8613/0/0, tx=24872/172/0/0
port5: physical/10000full, up, rx-bytes/packets/dropped/errors=2319732/29599/0/0, tx
  =0/0/0/0
port6: physical/10000full, up, rx-bytes/packets/dropped/errors=907563/9986/0/0, tx
  =223226/387/0/0
PINGSVR stats:
FGVMEVXXXXXXXX6F(updated 2 seconds ago):
FGVMEVXXXXXXXXCB(updated 4 seconds ago):
Primary     : BR1-FW1         , FGVMEVXXXXXXXX6F, HA cluster index = 1
Secondary   : BR1-FW2         , FGVMEVXXXXXXXXCB, HA cluster index = 0
number of vcluster: 1
vcluster 1: work 169.254.0.2
Primary: FGVMEVXXXXXXXX6F, HA operating index = 0
Secondary: FGVMEVXXXXXXXXCB, HA operating index = 1
```

HA diagnose status on BR1-FW1

```
BR1-FW1 # diagnose sys ha status
HA information
Statistics
traffic.local = s:0 p:580180 b:167172223
traffic.total = s:0 p:580193 b:167174708
activity.ha_id_changes = 4
activity.fdb  = c:0 q:0
Model=80001, Mode=1 Group=0 Debug=0
nvcluster=1, ses_pickup=1, delay=0, load_balance=1, schedule=3, ldb_udp=0, upgrade_mode=0.
[Debug_Zone HA information]
HA group member information: is_manage_primary=1.
FGVMEVXXXXXXXX6F:      Primary, serialno_prio=1, usr_priority=200, hostname=BR1-FW1
FGVMEVXXXXXXXXCB:    Secondary, serialno_prio=0, usr_priority=150, hostname=BR1-FW2
[Kernel HA information]
vcluster 1, state=work, primary_ip=169.254.0.2, primary_id=0
FGVMEVXXXXXXXX6F:      Primary, ha_prio/o_ha_prio=0/0
FGVMEVXXXXXXXXCB:    Secondary, ha_prio/o_ha_prio=1/1
```

HA vcluster dump on BR1-FW1

```
BR1-FW1 # diagnose sys ha dump-by vcluster
HA information.
vcluster_nr=1
vcluster-1: start_time=1786490532(2026-08-11 16:22:12), state/o/chg_time=2(work)/2(work)
  /1786490552(2026-08-11 16:22:32)
pingsvr_flip_timeout/expire=3600s/1434s
mondev: Lag-1(prio=50,is_aggr=1,status=1) port5(prio=50,is_aggr=0,status=1) port6(prio=50,
  is_aggr=0,status=1)
'FGVMEVXXXXXXXX6F': ha_prio/o=0/0, link_failure=0, pingsvr_failure=0, flag=0x00000001,
  mem_failover=0, uptime/reset_cnt=666/0
'FGVMEVXXXXXXXXCB': ha_prio/o=1/1, link_failure=0, pingsvr_failure=0, flag=0x00000000,
  mem_failover=0, uptime/reset_cnt=613/0
```

16.5 Troubleshooting Scenario

16.5.1 HA Cluster Won't Form / Peer Not Detected

- **Symptoms:** Units show as standalone; Cluster does not form.

- **Troubleshooting Focus:**
  - Verify HA parameters (group-name, mode, password, group-id) match exactly.
  - Check heartbeat cabling and hbdev configuration on both units. Prefer direct or isolated heartbeat connectivity when practical.
  - Confirm identical firmware version and build.
  - Ensure hardware models are supported for clustering.
  - If heartbeat interfaces pass through a switch, use a dedicated Layer-2 segment and configure appropriate edge/PortFast behavior where supported. Avoid unnecessary Layer-2 convergence delays on dedicated HA links.

> ★ **Verification Command**
>
> - diagnose sys ha status
> - get system ha status
> - diagnose sys ha dump-by group
> - show system ha

Observation

```
DC-FW1 (global) # get system ha status
Primary selected using:
<2026/08/27 10:40:29> FGVMEVXXXXXXXX82 is selected as the primary because it's the
only member in the cluster.
DC-FW2 (global) # get system ha status
Cluster state change time: 2026-08-27 10:40:26
Primary selected using:
<2026/08/27 10:40:26> FGVMEVXXXXXXXX04 is selected as the primary because it's the
only member in the cluster.
!
DC-FW1 (global) # diagnose sys ha status
[Debug_Zone HA information]
HA group member information: is_manage_primary=1.
FGVMEVXXXXXXXX82:      Primary, serialno_prio=0, usr_priority=200, hostname=DC-FW1
!
DC-FW2 (global) # diagnose sys ha status
[Debug_Zone HA information]
HA group member information: is_manage_primary=1.
FGVMEVXXXXXXXX04:      Primary, serialno_prio=0, usr_priority=100, hostname=DC-FW2
!
DC-FW1 (global) # show system ha
config system ha
set group-name "dcfw-xyz"
!
DC-FW2 (global) # sh system ha
config system ha
set group-name "dcfw-xyZ"
```

- **Root Cause:** The HA group-name does not match between the two firewalls (dcfw-xyz on DC-FW1 and dcfw-xyZ on DC-FW2). Therefore, each firewall remains standalone and elects itself as primary.
- **Fix Note:** Corrected HA group-name case mismatch so both units join the same cluster.

16.5.2 Constant Failover / Flapping

- **Symptoms:** Cluster repeatedly fails over; primary/secondary roles keep switching; logs show frequent HA events.
- **Troubleshooting Focus:**
  - Verify heartbeat link stability (cabling, switch ports, STP).
  - Check HA health-check targets for reachability and latency.
  - Monitor CPU/memory spikes that may trigger failover.
  - Ensure identical firmware/build and HA parameters.

> ★ **Verification Command**
>
> - diagnose sys ha status
> - diagnose sys ha history read
> - diagnose sys ha statistics

Observation

```
DC-FW1 (global) # diagnose sys ha history read
<2026-08-27 11:27:24> heartbeats from FGVMEVXXXXXXXX04 are lost on all hbdev
<2026-08-27 11:27:20> port port7 link status changed: 1->0
<2026-08-27 11:27:20> hbdev port7 link status changed: 1->0
<2026-08-27 11:27:20> member FGVMEVXXXXXXXX04 lost heartbeat on hbdev port8
<2026-08-27 11:27:16> port port8 link status changed: 1->0
<2026-08-27 11:27:16> hbdev port8 link status changed: 1->0
<2026-08-27 11:27:03> port port7 link status changed: 0->1
<2026-08-27 11:27:03> hbdev port7 link status changed: 0->1
<2026-08-27 11:26:57> port port8 link status changed: 0->1
<2026-08-27 11:26:57> hbdev port8 link status changed: 0->1
<2026-08-27 11:26:57> new member 'FGVMEVXXXXXXXX04' joins the cluster
<2026-08-27 11:26:57> FGVMEVXXXXXXXX82 is elected as the cluster primary of 2 member
<2026-08-27 11:25:27> FGVMEVXXXXXXXX82 is elected as the cluster primary of 1 member
<2026-08-27 11:25:27> member FGVMEVXXXXXXXX04 lost heartbeat on hbdev port7
```

- **Root Cause:** Unstable heartbeat connectivity.
- **Fix Note:** Stabilized heartbeat links and Replaced flapping cabling on the heartbeat links and re-enabled the member.

16.5.3 Failover Not Happening When It Should

- **Symptoms:** Primary unit's LAG-1 interface goes down, but secondary does not take over; traffic continues through failed primary.
- **Troubleshooting Focus:**
  - Verify HA monitor-port configuration includes LAG-1.
  - Check heartbeat link status and HA health-check targets.
  - Review override/priority settings that may block takeover.

> ★ **Verification Command**
>
> - diagnose sys ha status

- get system ha status
- show system ha

Observation

```
DC-FW1 (global) # diagnose sys ha status
[Debug_Zone HA information]
HA group member information: is_manage_primary=1.
FGVMEVXXXXXXXX82:      Primary, serialno_prio=1, usr_priority=200, hostname=DC-FW1
FGVMEVXXXXXXXX04:    Secondary, serialno_prio=0, usr_priority=100, hostname=DC-FW2
!
DC-FW1 (global) # sh system ha
set override enable
set priority 200
set monitor "port5" "port6"
end
!
DC-FW2 (global) # sh system ha
set override enable
set priority 100
set monitor "port5" "port6"
end
!
SW2-DC#sh etherchannel summary
Group  Port-channel  Protocol    Ports
1      Po1(SD)         LACP      Gi1/1(s)    Gi1/2(s)
```

- **Root Cause:** LAG-1 interface not configured as a monitor port in HA settings.
- **Fix Note:** Added LAG-1 to HA monitor-port configuration to ensure failover triggers correctly.

16.5.4 Split-Brain (Both Units Think They're Primary)

Split-brain in FortiGate high availability (HA) occurs when the firewalls in a cluster lose communication on their heartbeat interfaces, causing both units to assume the primary role simultaneously. Because both firewalls think they are the active member, they both attempt to process traffic using the exact same virtual IP and MAC addresses. This creates MAC flapping on connected switches, drops traffic, resets active sessions, and triggers intermittent network outages.

- **Symptoms:** Both FortiGate units report themselves as primary; duplicate virtual IP/MAC addresses are advertised; traffic drops or intermittent connectivity.
- **Troubleshooting Focus:**
  - Verify heartbeat cabling and redundancy (direct vs switch).
  - Check HA group parameters (group-name, group-id, password).
  - Review firmware/build consistency across units.
  - Monitor latency or congestion on heartbeat interfaces.

★ **Verification Command**

- get system ha status – shows current HA role.
- diagnose sys ha status – confirms heartbeat state.

> - diagnose sys ha dump-by vcluster - detailed cluster info.

Observation

```
DC-FW1 (global) # diagnose sys ha status
[Debug_Zone HA information]
HA group member information: is_manage_primary=1.
FGVMEVXXXXXXXX82:      Primary, serialno_prio=0, usr_priority=200, hostname=DC-FW1
!
DC-FW2 (global) # diagnose sys ha status
[Debug_Zone HA information]
HA group member information: is_manage_primary=1.
FGVMEVXXXXXXXX04:      Primary, serialno_prio=0, usr_priority=100, hostname=DC-FW2
```

- **Root Cause:** Loss of heartbeat communication or misconfigured HA parameters leading to split-brain condition.
- **Fix Note:** Restored heartbeat connectivity and corrected HA parameters to eliminate split-brain and ensure that a single primary is elected.

17 Daily FortiGate Firewall Commands

- **System / Interface**
  - get system status
  - get system interface physical <interface>
  - show system interface
  - show system interface <interface>
  - show system vdom-property
  - show system zone
  - get system performance status
- **Routing / BGP**
  - get router info routing-table all
  - get router info routing-table all vrf <vrf-id>
  - get router info routing-table static
  - get router info routing-table bgp
  - get router info bgp summary
  - get router info bgp neighbor
  - get router info bgp network
  - show router static
- **Firewall Policy / Objects**
  - show firewall policy
  - show firewall policy <policy-id>
  - show firewall address
  - show firewall addrgrp
  - show firewall vip
- **Central SNAT**
  - show system settings | grep central
  - show firewall central-snat-map
- **DHCP**
  - show system dhcp server
  - execute dhcp lease-list <interface>
  - execute dhcp lease-clear <leased-ip>
- **Users**
  - get user local
  - show user local
  - get user group
  - show user group

- **Connectivity**
  - execute ping <ip>
  - execute ping-options source <source-ip>
  - execute traceroute <ip>
- **Packet Flow Debug**
  - diagnose debug reset
  - diagnose debug flow filter addr <ip>
  - diagnose debug flow trace start 50
  - diagnose debug enable
  - diagnose debug disable
- **IPSec VPN**
  - get vpn ike gateway
  - get vpn ipsec tunnel summary
  - show vpn ipsec phase1-interface
  - show vpn ipsec phase2-interface
  - diagnose debug application ike -1
  - diagnose debug enable
  - diagnose debug disable
  - diagnose debug reset
- **SSL VPN**
  - get vpn ssl monitor
  - get vpn ssl settings
- **High Availability**
  - get system ha status
  - show system ha

18 Conclusion

This concludes **FortiGate Firewall Lab – Volume 1**. Throughout this guide, we built and validated a practical FortiGate lab environment using step-by-step configuration, verification, and troubleshooting exercises.

The objective of this volume was not only to demonstrate how FortiGate features are configured, but also to understand how those features behave during normal operation and how to troubleshoot them when something goes wrong.

What We Covered

- FortiGate lab prerequisites and deployment in EVE-NG and FortiGate hardware architecture.
- Virtual Domains (VDOMs) and security zones.
- Initial firewall configuration, management access, and interface configuration.
- Physical, Aggregate, VLAN, and Loopback interfaces. FortiGate DHCP server configuration and lease verification.
- WAN connectivity using Static Routes, BGP, routing objects, prefix lists, and route maps.
- Firewall address objects, address groups, policies, inspection modes, and policy verification.
- Source NAT using interface IPs, IP Pools, One-to-One NAT, Fixed Port Range, Port Block Allocation, and Central SNAT.
- Destination NAT using Virtual IPs (VIPs) and port forwarding.
- Local user authentication and user groups.
- Site-to-Site and Remote Access VPN technologies using IPsec and SSL VPN.
- FortiGate High Availability using Active-Passive and Active-Active FGCP clusters.

Continue Learning

The journey continues in **FortiGate Firewall Lab – Volume 2**, which builds on the foundation established in this volume.
Volume 2 focus on security inspection, centralized management, logging, automation, LDAP Authentication and advanced troubleshooting.

End of Volume 1

FortiGate Firewall Lab
A Practical Hands-On Guide

Prepared By
Asif Patel